

Quantitative-Analytic Report on Chess
Carlsen vs Caruana
2018 World Championship, London

Generated by Stockfish 17.1

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Contents

1	Carlsen, Magnus vs Caruana, Fabiano	5
1.1	Game Information	5
1.2	Player Statistics	5
1.2.1	Carlsen, Magnus (White)	5
1.2.2	Caruana, Fabiano (Black)	5
1.3	Positional Analysis	6
1.3.1	Position Evaluation Over Time	6
1.3.2	Space Control	6
1.3.3	Mobility (Piece Activity)	7
1.3.4	King Safety	7
1.3.5	Threats	7
1.4	Annotated Game	8
1.5	Critical Positions	9
1.6	Fireteam Index Prediction	12
1.7	Analysis Information	14
2	Carlsen, Magnus vs Caruana, Fabiano	15
2.1	Game Information	15
2.2	Player Statistics	15
2.2.1	Carlsen, Magnus (White)	15
2.2.2	Caruana, Fabiano (Black)	15
2.3	Positional Analysis	16
2.3.1	Position Evaluation Over Time	16
2.3.2	Space Control	16
2.3.3	Mobility (Piece Activity)	17
2.3.4	King Safety	17
2.3.5	Threats	18
2.4	Annotated Game	18
2.5	Critical Positions	19
2.6	Fireteam Index Prediction	20
2.7	Analysis Information	22
3	Carlsen, Magnus vs Caruana, Fabiano	25
3.1	Game Information	25
3.2	Player Statistics	25
3.2.1	Carlsen, Magnus (White)	25
3.2.2	Caruana, Fabiano (Black)	25

3.3	Positional Analysis	26
3.3.1	Position Evaluation Over Time	26
3.3.2	Space Control	26
3.3.3	Mobility (Piece Activity)	27
3.3.4	King Safety	27
3.3.5	Threats	27
3.4	Annotated Game	28
3.5	Critical Positions	28
3.6	Fireteam Index Prediction	30
3.7	Analysis Information	32
4	Carlsen, Magnus vs Caruana, Fabiano	33
4.1	Game Information	33
4.2	Player Statistics	33
4.2.1	Carlsen, Magnus (White)	33
4.2.2	Caruana, Fabiano (Black)	33
4.3	Positional Analysis	34
4.3.1	Position Evaluation Over Time	34
4.3.2	Space Control	34
4.3.3	Mobility (Piece Activity)	35
4.3.4	King Safety	35
4.3.5	Threats	36
4.4	Annotated Game	36
4.5	Critical Positions	36
4.6	Fireteam Index Prediction	38
4.7	Analysis Information	40
5	Carlsen, Magnus vs Caruana, Fabiano	43
5.1	Game Information	43
5.2	Player Statistics	43
5.2.1	Carlsen, Magnus (White)	43
5.2.2	Caruana, Fabiano (Black)	43
5.3	Positional Analysis	44
5.3.1	Position Evaluation Over Time	44
5.3.2	Space Control	44
5.3.3	Mobility (Piece Activity)	45
5.3.4	King Safety	45
5.3.5	Threats	46
5.4	Annotated Game	46
5.5	Critical Positions	46
5.6	Fireteam Index Prediction	48
5.7	Analysis Information	50
6	Carlsen, Magnus vs Caruana, Fabiano	53
6.1	Game Information	53
6.2	Player Statistics	53
6.2.1	Carlsen, Magnus (White)	53
6.2.2	Caruana, Fabiano (Black)	53

6.3	Positional Analysis	54
6.3.1	Position Evaluation Over Time	54
6.3.2	Space Control	54
6.3.3	Mobility (Piece Activity)	55
6.3.4	King Safety	55
6.3.5	Threats	56
6.4	Annotated Game	56
6.5	Critical Positions	57
6.6	Fireteam Index Prediction	59
6.7	Analysis Information	61
7	Carlsen, Magnus vs Caruana, Fabiano	63
7.1	Game Information	63
7.2	Player Statistics	63
7.2.1	Carlsen, Magnus (White)	63
7.2.2	Caruana, Fabiano (Black)	63
7.3	Positional Analysis	64
7.3.1	Position Evaluation Over Time	64
7.3.2	Space Control	64
7.3.3	Mobility (Piece Activity)	65
7.3.4	King Safety	65
7.3.5	Threats	66
7.4	Annotated Game	66
7.5	Critical Positions	67
7.6	Fireteam Index Prediction	68
7.7	Analysis Information	70
8	Carlsen, Magnus vs Caruana, Fabiano	73
8.1	Game Information	73
8.2	Player Statistics	73
8.2.1	Carlsen, Magnus (White)	73
8.2.2	Caruana, Fabiano (Black)	73
8.3	Positional Analysis	74
8.3.1	Position Evaluation Over Time	74
8.3.2	Space Control	74
8.3.3	Mobility (Piece Activity)	75
8.3.4	King Safety	75
8.3.5	Threats	76
8.4	Annotated Game	76
8.5	Critical Positions	77
8.6	Fireteam Index Prediction	79
8.7	Analysis Information	81
9	Carlsen, Magnus vs Caruana, Fabiano	83
9.1	Game Information	83
9.2	Player Statistics	83
9.2.1	Carlsen, Magnus (White)	83
9.2.2	Caruana, Fabiano (Black)	83

9.3	Positional Analysis	84
9.3.1	Position Evaluation Over Time	84
9.3.2	Space Control	84
9.3.3	Mobility (Piece Activity)	85
9.3.4	King Safety	85
9.3.5	Threats	86
9.4	Annotated Game	86
9.5	Critical Positions	87
9.6	Fireteam Index Prediction	88
9.7	Analysis Information	90
10	Carlsen, Magnus vs Caruana, Fabiano	93
10.1	Game Information	93
10.2	Player Statistics	93
10.2.1	Carlsen, Magnus (White)	93
10.2.2	Caruana, Fabiano (Black)	93
10.3	Positional Analysis	94
10.3.1	Position Evaluation Over Time	94
10.3.2	Space Control	94
10.3.3	Mobility (Piece Activity)	95
10.3.4	King Safety	95
10.3.5	Threats	95
10.4	Annotated Game	96
10.5	Critical Positions	96
10.6	Fireteam Index Prediction	98
10.7	Analysis Information	101
11	Carlsen, Magnus vs Caruana, Fabiano	103
11.1	Game Information	103
11.2	Player Statistics	103
11.2.1	Carlsen, Magnus (White)	103
11.2.2	Caruana, Fabiano (Black)	103
11.3	Positional Analysis	104
11.3.1	Position Evaluation Over Time	104
11.3.2	Space Control	104
11.3.3	Mobility (Piece Activity)	105
11.3.4	King Safety	105
11.3.5	Threats	106
11.4	Annotated Game	106
11.5	Critical Positions	107
11.6	Fireteam Index Prediction	108
11.7	Analysis Information	110
12	Carlsen, Magnus vs Caruana, Fabiano	113
12.1	Game Information	113
12.2	Player Statistics	113
12.2.1	Carlsen, Magnus (White)	113
12.2.2	Caruana, Fabiano (Black)	113

12.3	Positional Analysis	114
12.3.1	Position Evaluation Over Time	114
12.3.2	Space Control	114
12.3.3	Mobility (Piece Activity)	115
12.3.4	King Safety	115
12.3.5	Threats	116
12.4	Annotated Game	116
12.5	Critical Positions	117
12.6	Fireteam Index Prediction	119
12.7	Analysis Information	121
13	Fireteam Index Prediction Summary	123
13.1	Overall Results	123
13.1.1	FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)	123
13.1.2	FTI2: Tactical (0.6, 0.1, 0, 0.3)	123
13.1.3	FTI3: Strategic (0.7, 0.1, 0.1, 0.1)	123
13.1.4	FTI4: Dynamic (0.2, 0.6, 0, 0.2)	124
13.1.5	FTI5: Solid (0.3, 0.2, 0.5, 0)	124
13.1.6	Comparison	124
13.1.7	Empirically Optimal Weights	124
13.2	Interpretation	125
A	Methodology: How Metrics Are Computed	127
A.1	Space	127
A.2	Mobility	127
A.3	King Safety	128
A.4	Threats	128
A.5	Middlegame vs Endgame (MG/EG)	128
A.6	Game Character Classification	128
A.7	Fireteam Index (FTI) and Win Prediction	129

Chapter 1

Carlsen, Magnus vs Caruana, Fabiano

1.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.09
Round	1
White	Carlsen, Magnus (2847)
Black	Caruana, Fabiano (2823)
Result	1/2-1/2
Opening	B31 – Sicilian Defense, Rossolimo Variation

1.2 Player Statistics

1.2.1 Carlsen, Magnus (White)

- Total moves: 115
- Accuracy: 95.3%
- Average centipawn loss: 9.6
- Best/Excellent moves: 81 / 14
- Good moves: 12
- Inaccuracies: 4
- Mistakes: 1
- Blunders: 3

1.2.2 Caruana, Fabiano (Black)

- Total moves: 114
- Accuracy: 94.7%
- Average centipawn loss: 10.9

- Best/Excellent moves: 73 / 25
- Good moves: 9
- Inaccuracies: 1
- Mistakes: 2
- Blunders: 4

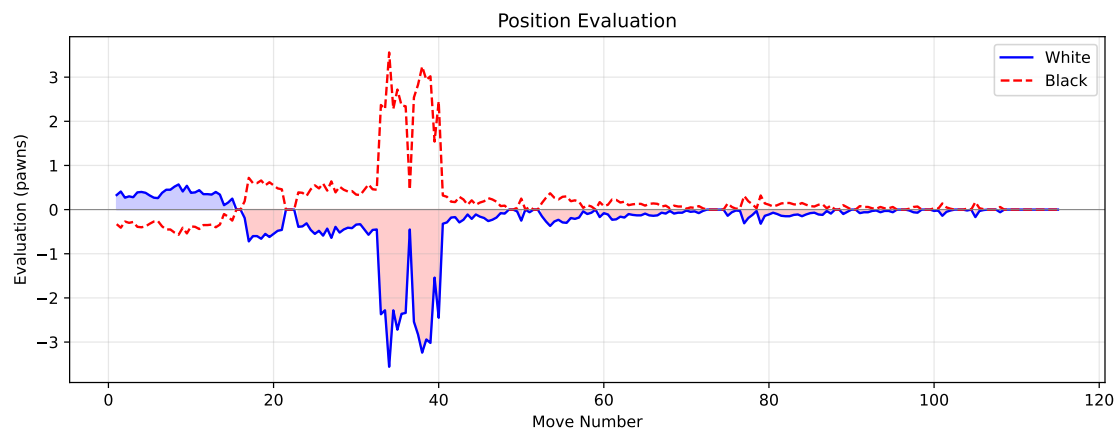
1.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

1.3.1 Position Evaluation Over Time

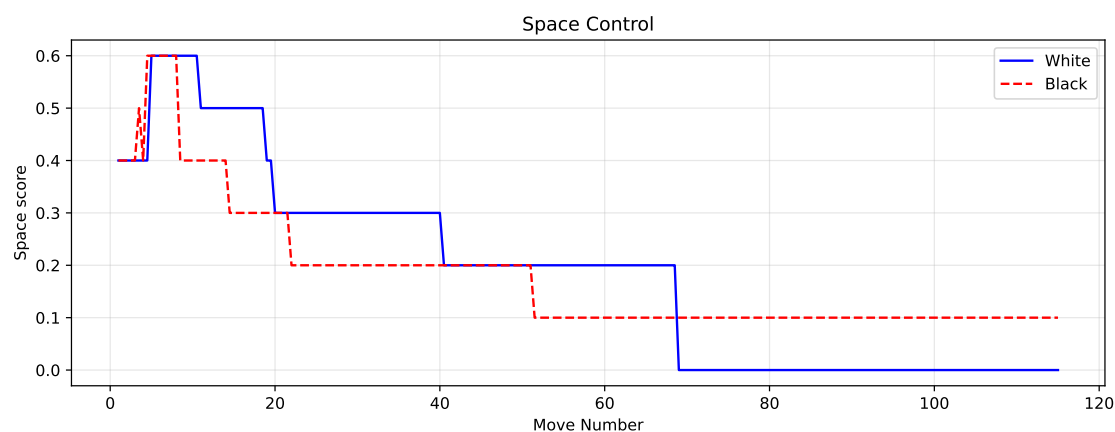
Game Character Classification. Evaluation range: minimum $m_1 = -3.56$, maximum $m_2 = +0.57$, spread $d = m_2 - m_1 = 4.13$.

This game is classified as **tactical** (normal). A tactical game with moderate swings.



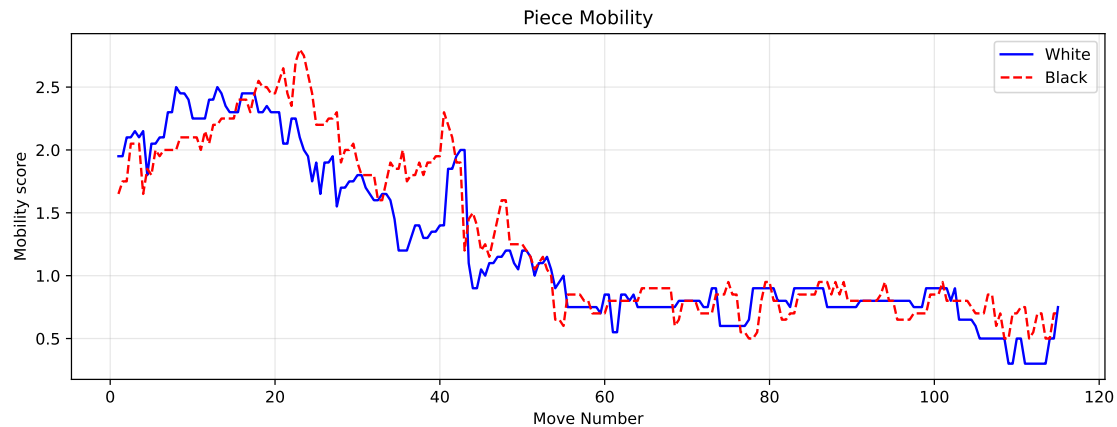
1.3.2 Space Control

- White average: 0.19 (range: 0.00 to 0.60)
- Black average: 0.18 (range: 0.10 to 0.60)



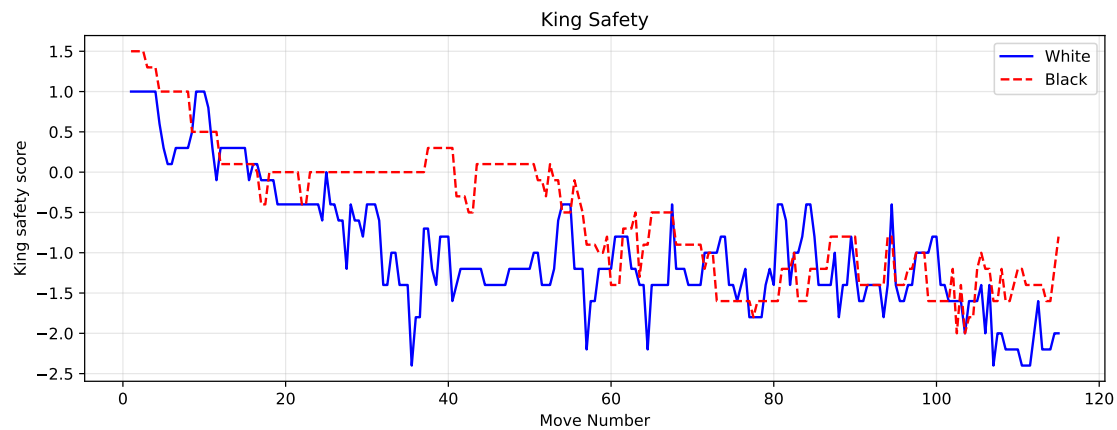
1.3.3 Mobility (Piece Activity)

- White average: 1.22 (range: 0.30 to 2.50)
- Black average: 1.30 (range: 0.50 to 2.80)



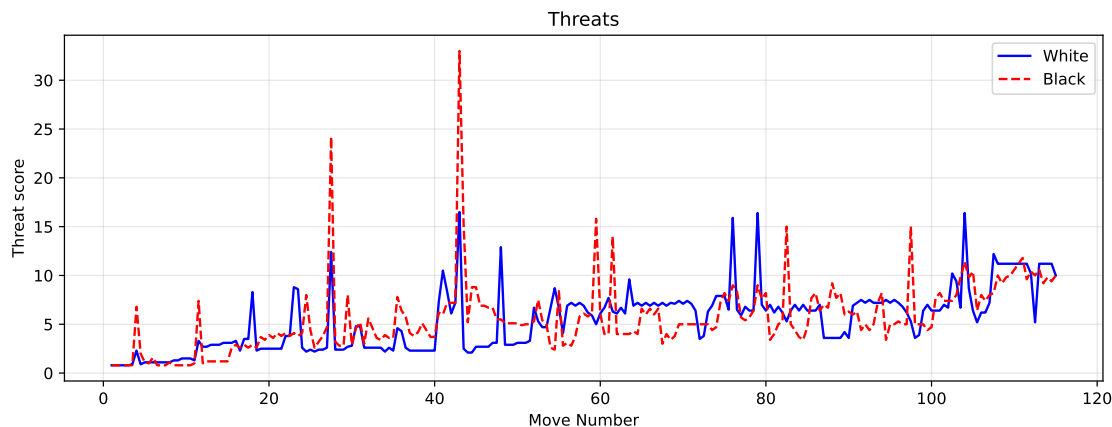
1.3.4 King Safety

- White average: -0.97
- Black average: -0.54



1.3.5 Threats

- White average: 5.45 (range: 0.80 to 16.50)
- Black average: 5.61 (range: 0.80 to 33.00)
- Average advantage: -0.16 (Black maintained more threats)



1.4 Annotated Game

1. e4 c5 2. Nf3 Nc6 3. Bb5 g6 4. Bxc6 dxc6 5. d3 Bg7 6. h3 Nf6 7. Nc3 Nd7 8. Be3! e5 9. O-O b6 10. Nh2 Nf8! 11. f4! exf4! 12. Rxf4 Be6 13. Rf2! h6! 14. Qd2?! g5 15. Raf1! Qd6! 16. Ng4 O-O-O! 17. Nf6?! Nd7 18. Nh5! Be5! 19. g4! f6! 20. b3! Bf7 21. Nd1! Nf8?! 22. Nxf6! Ne6! 23. Nh5?! Bxh5! 24. gxh5! Nf4 25. Bxf4! gxf4 26. Rg2 Rhg8 27. Qe2 Rxc2+ 28. Qxc2 Qe6 29. Nf2! Rg8 30. Ng4! Qe8 31. Qf3 Qxh5! 32. Kf2! Bc7! 33. Ke2?? *White loses 184cp. Consider instead: e5.*

33...Qg5 34. Nh2?? *White loses 135cp. Consider instead: d4, Qf2, e5.*

34...h5?? *Black loses 140cp. Consider instead: Qe5.*

35. Rf2 Qg1? *Black loses 62cp. Consider instead: Qg7, b5.*

36. Nf1! h4?? *Black loses 223cp. Consider instead: Kb7, Qg7.*

37. Kd2?? *White loses 204cp. Consider instead: e5.*

37...Kb7! 38. c3?! Be5? *Black loses 64cp. Consider instead: Rg3.*

39. Kc2 Qg7?? *Black loses 124cp. Consider instead: b5.*

40. Nh2? *White loses 105cp. Consider instead: Qh5, Nd2, a4.*

40...Bxc3?? *Black loses 229cp. Consider instead: Qg1.*

41. Qxf4! Bd4! 42. Qf7+! Ka6! 43. Qxc7! Rxc7 44. Re2! Rg3 45. Ng4! Rxh3 46. e5! Rf3 47. e6 Rf8 48. e7 Re8! 49. Nh6! h3 50. Nf5 Bf6! 51. a3! b5 52. b4! cxb4 53. axb4 Bxe7! 54. Nxe7 h2 55. Rxh2! Rxe7! 56. Rh6 Kb6 57. Kc3 Rd7 58. Rg6 Kc7 59. Rh6 Rd6 60. Rh8! Rg6 61. Ra8 Kb7 62. Rh8! Rg5 63. Rh7+! Kb6! 64. Rh6! Rg1 65. Kc2 Rf1 66. Rg6 Rh1 67. Rf6 Rh8 68. Kc3 Ra8 69. d4 Rd8! 70. Rh6 Rd7 71. Rg6 Kc7 72. Rg5 Rd6! 73. Rg8 Rh6 74. Ra8! Rh3+ 75. Kc2 Ra3 76. Kb2! Ra4 77. Kc3 a6 78. Rh8 Ra3+ 79. Kb2 Rg3 80. Kc2 Rg5 81. Rh6! Rd5 82. Kc3! Rd6 83. Rh8! Rg6 84. Kc2 Kb7 85. Kc3 Rg3+ 86. Kc2! Rg1 87. Rh5! Rg2+ 88. Kc3! Rg3+! 89. Kc2! Rg4! 90. Kc3! Kb6 91. Rh6 Rg5! 92. Rf6 Rh5 93. Rg6! Rh3+! 94. Kc2 Rh5 95. Kc3! Rd5 96. Rh6! Kc7 97. Rh7+ Rd7! 98. Rh5 Rd6! 99. Rh8! Rg6! 100. Rf8 Rg3+ 101. Kc2 Ra3 102. Rf7+ Kd6! 103. Ra7 Kd5! 104. Kb2! Rd3! 105. Rxa6! Rxd4 106. Kb3 Re4! 107. Kc3 Rc4+ 108. Kb3 Kd4 109. Rb6! Kd3! 110. Ra6! Rc2 111. Rb6 Rc3+ 112. Kb2! Rc4! 113. Kb3! Kd4 114. Ra6! Kd5 115. Ra8

1/2-1/2

1.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

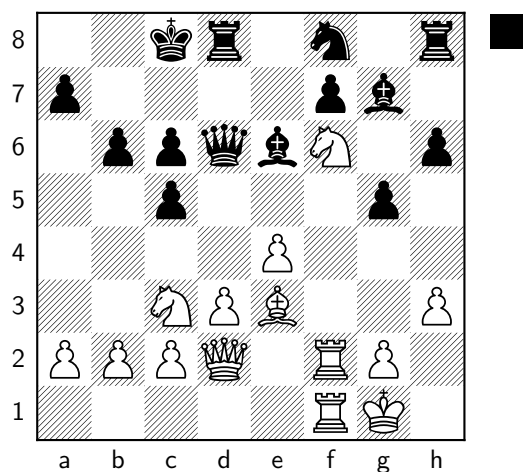
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 17. Nf6 ★

Biggest evaluation swing(s) (53cp)

Evaluation: -0.72



Metric	White	Black
Space	0.50	0.30
Mobility (MG)	4.90	4.60
King Safety	-0.10	-0.40
Threats	3.50	2.90

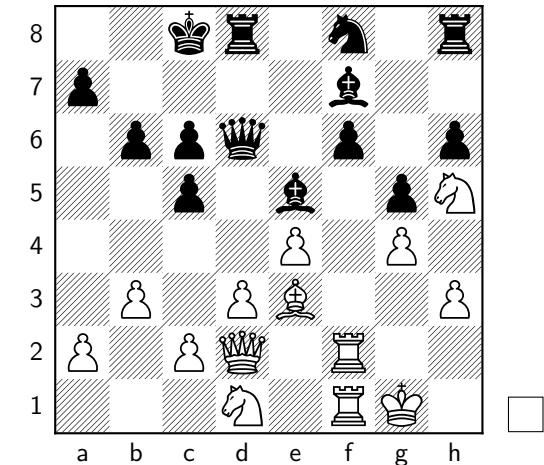
Instead of Nf6: e5, Qe2, a4

Best continuation: Nd7 Nh5 Be5

Position 2: After 21...Nf8 ★

Biggest evaluation swing(s) (46cp)

Evaluation: +0.00



Metric	White	Black
Space	0.30	0.30
Mobility (MG)	4.10	4.90
King Safety	-0.40	0.00
Threats	2.50	3.80

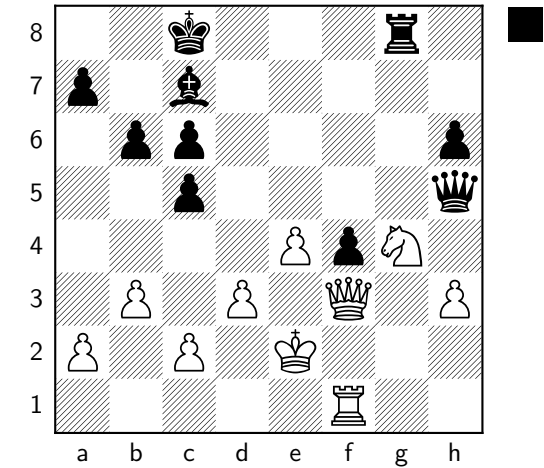
Instead of Nf8: Kb7, Qe6, Rhg8

Best continuation: Nxf6 Bg6 h4

Position 3: After 33. Ke2

White blunders (-192cp)

Evaluation: -2.37



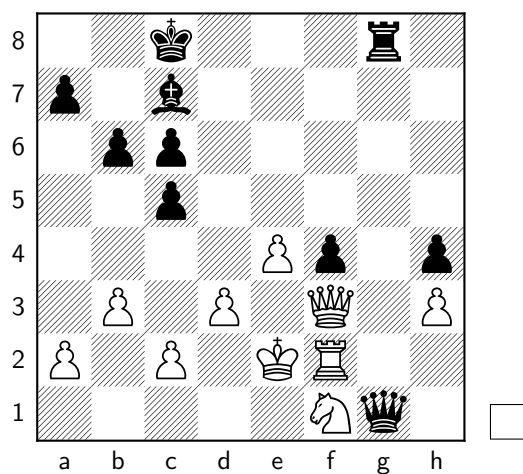
Metric	White	Black
Space	0.30	0.20
Mobility (MG)	3.30	3.20
King Safety	-1.00	0.00
Threats	2.60	3.60

Instead of Ke2: e5

Best continuation: Qg5 Qf2 h5

Position 4: After 36...h4*Black blunders (+189cp)*

Evaluation: -0.45



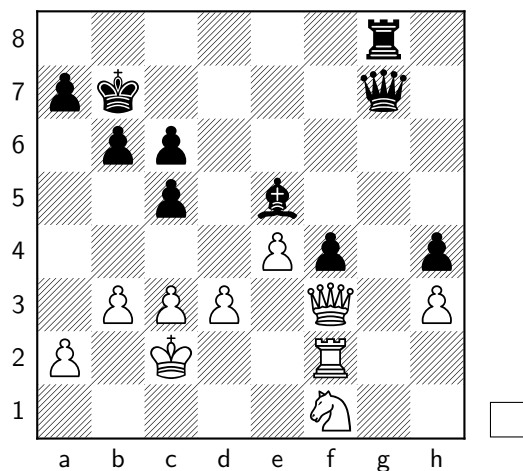
Metric	White	Black
Space	0.30	0.20
Mobility (MG)	2.60	3.60
King Safety	-1.80	0.00
Threats	2.60	5.80

Instead of h4: Kb7, Qg7

Best continuation: e5 Rg5 Nd2

Position 5: After 39...Qg7*Black blunders (+148cp)*

Evaluation: -1.54



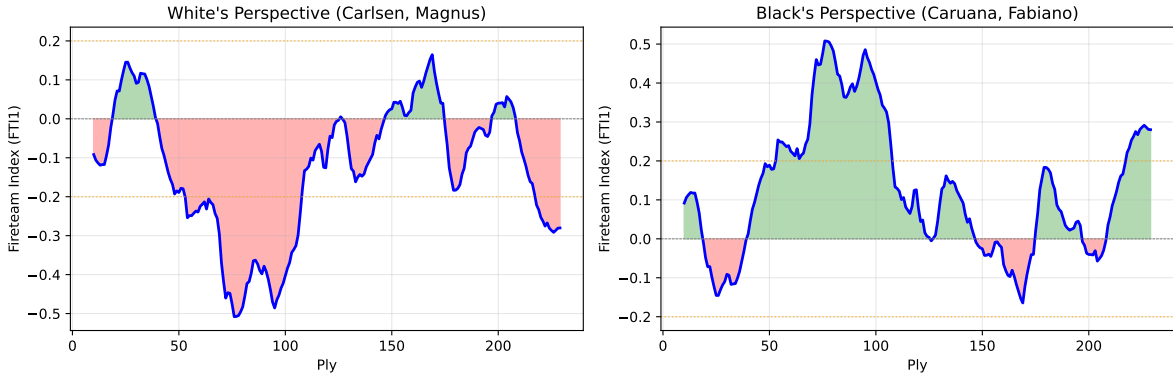
Metric	White	Black
Space	0.30	0.20
Mobility (MG)	2.70	3.90
King Safety	-0.80	0.30
Threats	2.30	3.70
Instead of Qg7:	b5	Best continuation: Qh5 Ka6 Qe2

1.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

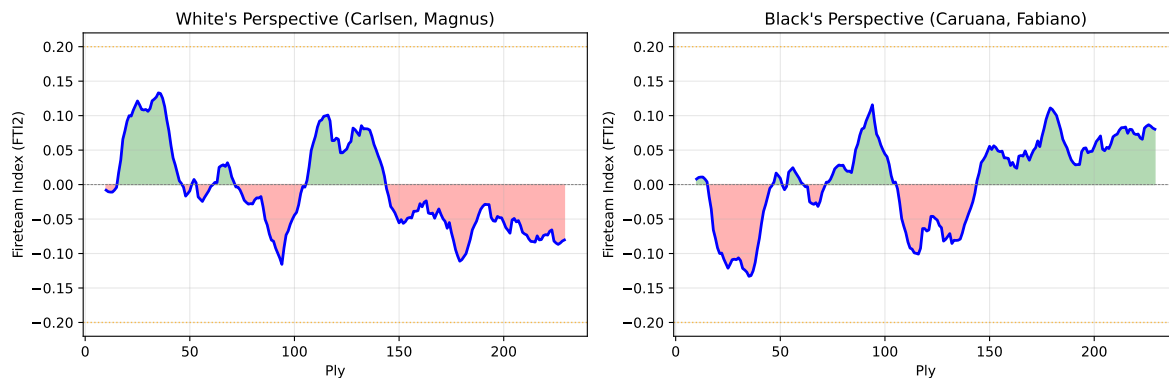
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	WIN	21 plies
Caruana, Fabiano	Windowed	WIN	55 plies

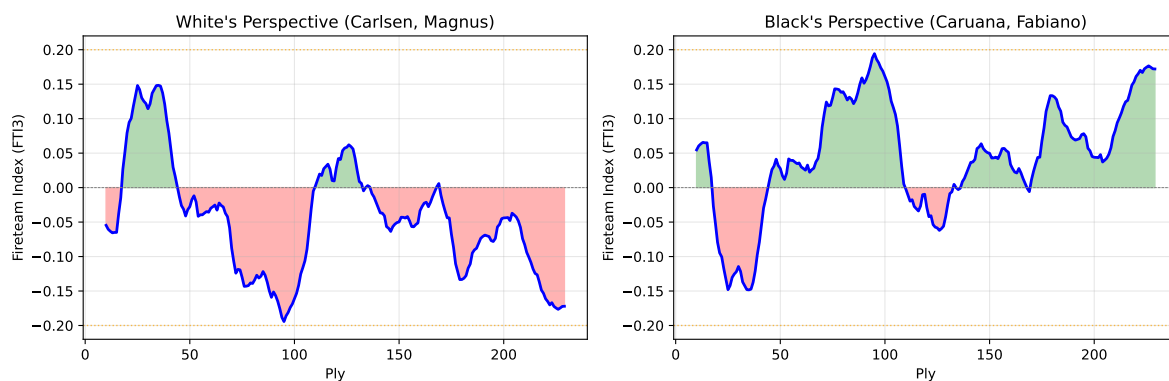


FTI2: Tactical (0.6, 0.1, 0, 0.3)

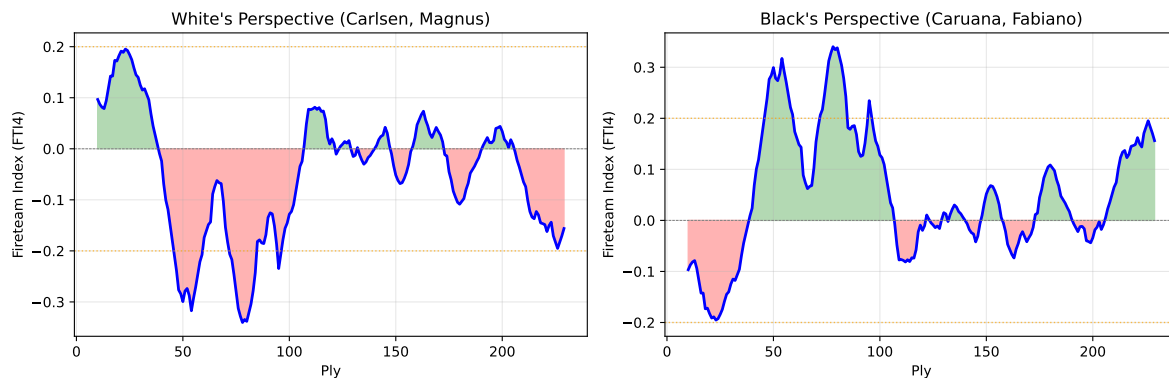
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI3: Strategic (0.7, 0.1, 0.1, 0.1)**

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	3 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

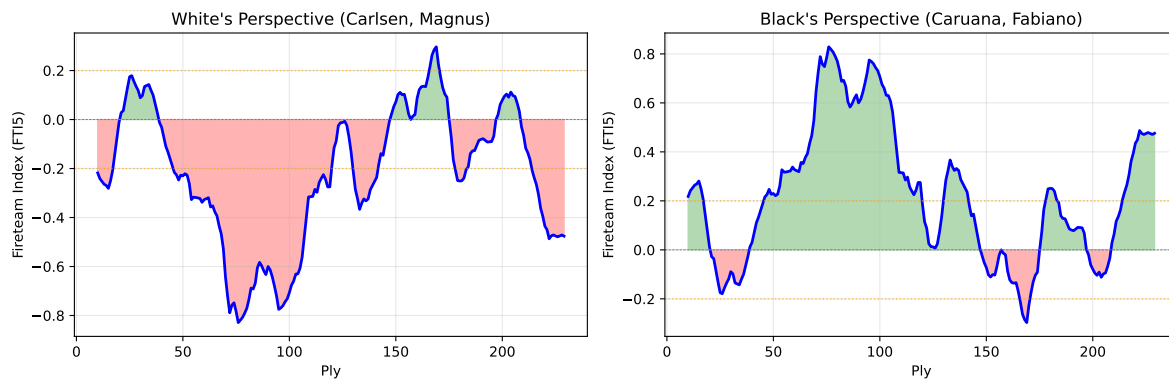
**FTI4: Dynamic (0.2, 0.6, 0, 0.2)**

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	WIN	13 plies
Caruana, Fabiano	Windowed	WIN	14 plies



FTI5: Solid (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	4 plies
Carlsen, Magnus	Windowed	DRAW	5 plies
Caruana, Fabiano	Per-ply	WIN	23 plies
Caruana, Fabiano	Windowed	WIN	74 plies



Interpretation. Black (Caruana, Fabiano) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting White successfully defended or Black failed to convert the advantage.

1.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 724.7 seconds

Chapter 2

Carlsen, Magnus vs Caruana, Fabiano

2.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.10
Round	2
White	Carlsen, Magnus (2847)
Black	Caruana, Fabiano (2823)
Result	1/2-1/2
Opening	D37 – Queen’s Gambit Declined, Harrwitz Attack

2.2 Player Statistics

2.2.1 Carlsen, Magnus (White)

- Total moves: 49
- Accuracy: 98.3%
- Average centipawn loss: 3.4
- Best/Excellent moves: 33 / 7
- Good moves: 9
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

2.2.2 Caruana, Fabiano (Black)

- Total moves: 48
- Accuracy: 98.6%
- Average centipawn loss: 2.9

- Best/Excellent moves: 33 / 6
- Good moves: 9
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

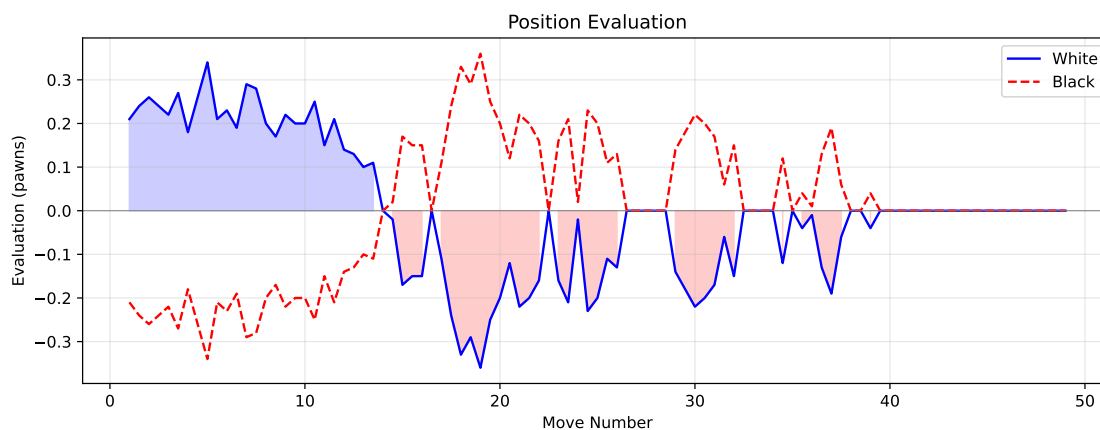
2.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

2.3.1 Position Evaluation Over Time

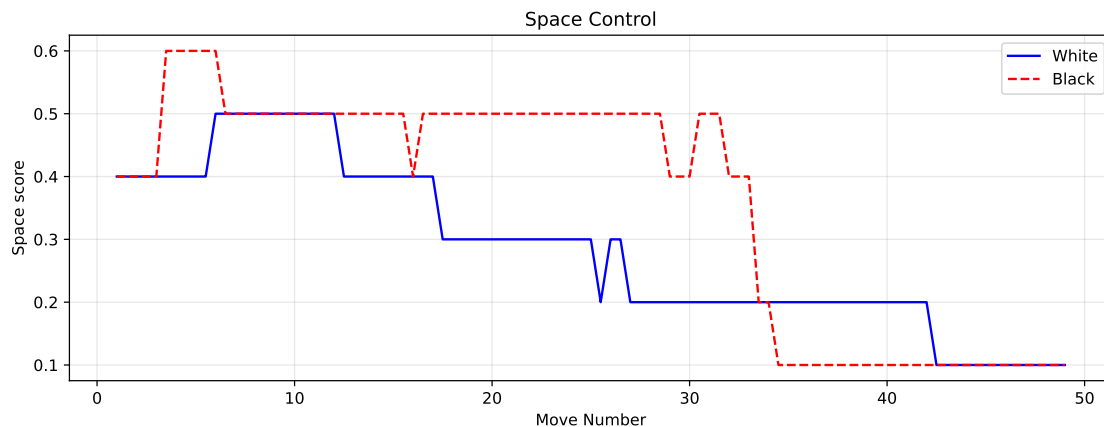
Game Character Classification. Evaluation range: minimum $m_1 = -0.36$, maximum $m_2 = +0.34$, spread $d = m_2 - m_1 = 0.70$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



2.3.2 Space Control

- White average: 0.29 (range: 0.10 to 0.50)
- Black average: 0.36 (range: 0.10 to 0.60)



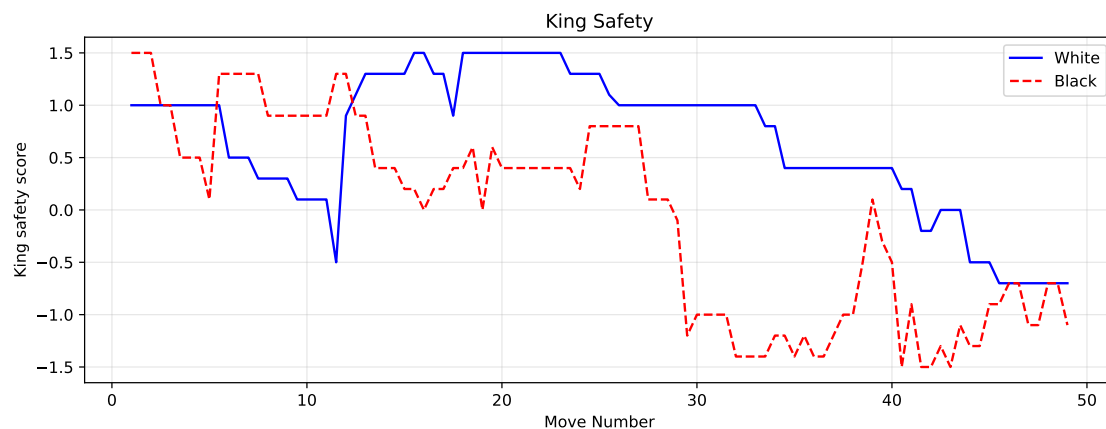
2.3.3 Mobility (Piece Activity)

- White average: 1.62 (range: 0.60 to 2.95)
- Black average: 1.42 (range: 0.65 to 2.55)



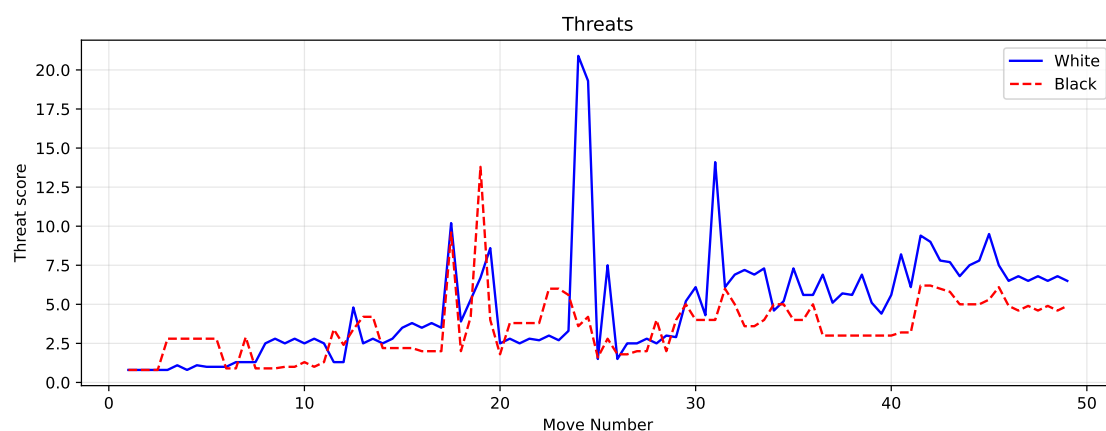
2.3.4 King Safety

- White average: 0.67
- Black average: -0.05



2.3.5 Threats

- White average: 4.78 (range: 0.80 to 20.90)
- Black average: 3.51 (range: 0.80 to 13.80)
- Average advantage: 1.28 (White maintained more threats)



2.4 Annotated Game

1. d4 Nf6 2. Nf3 d5 3. c4 e6 4. Nc3 Be7 5. Bf4 O-O 6. e3 c5 7. dxc5! Bxc5! 8. Qc2 Nc6! 9. a3 Qa5 10. Rd1! Rd8! 11. Be2 Ne4 12. O-O! Nxc3! 13. bxc3! h6! 14. a4 Ne7! 15. Ne5 Bd6 16. cxd5! Nxd5 17. Bf3 Nxf4! 18. exf4! Bxe5! 19. Rxd8+ Qxd8! 20. fxe5! Qc7 21. Rb1 Rb8! 22. Qd3 Bd7 23. a5 Bc6 24. Qd6! Qxd6! 25. exd6! Bxf3 26. gxf3! Kf8! 27. c4! Ke8! 28. a6 b6 29. c5! Kd7 30. cxb6! axb6! 31. a7! Ra8! 32. Rxb6 Rxa7! 33. Kg2 e5 34. Rb4 f5 35. Rb6 Ke6! 36. d7+ Kxd7! 37. Rb5 Ke6! 38. Rb6+! Kf7 39. Rb5 Kf6! 40. Rb6+ Kg5! 41. Rb5! Kf4! 42. Rb4+! e4 43. fxe4! fxe4! 44. h3 Ra5 45. Rb7 Rg5+ 46. Kf1! Rg6! 47. Rb4! Rg5 48. Rb7 Rg6 49. Rb4

1/2-1/2

2.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

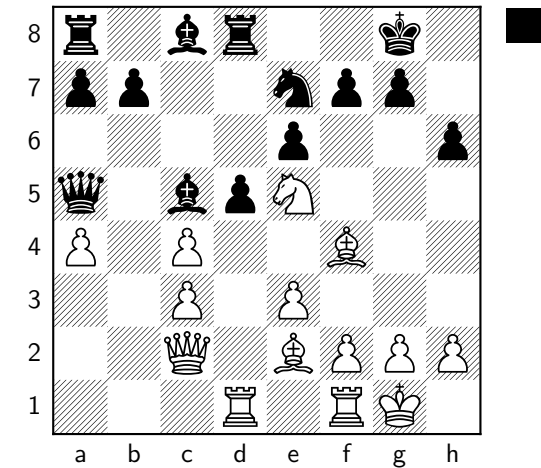
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 15. Ne5 ★

Biggest evaluation swing(s) (15cp)

Evaluation: -0.17



Metric	White	Black
Space	0.40	0.50
Mobility (MG)	5.00	4.10
King Safety	1.30	0.20
Threats	3.50	2.20

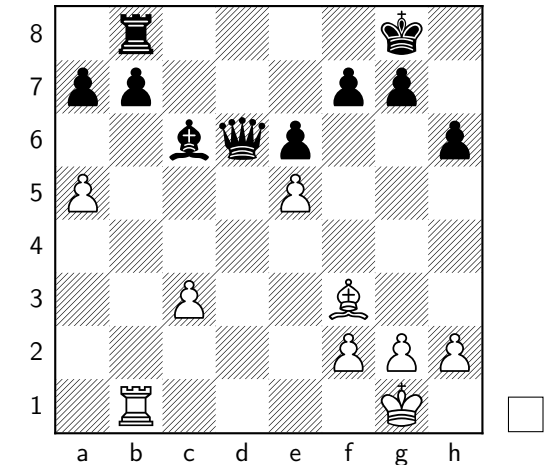
Instead of Ne5: Nd2, Rb1, Qb3

Best continuation: f6 Ng6 e5

Position 2: After 24...Qxd6 ★

Biggest evaluation swing(s) (21cp)

Evaluation: -0.23



Metric	White	Black
Space	0.30	0.50
Mobility (MG)	2.50	3.70
King Safety	1.30	0.80
Threats	19.30	4.20

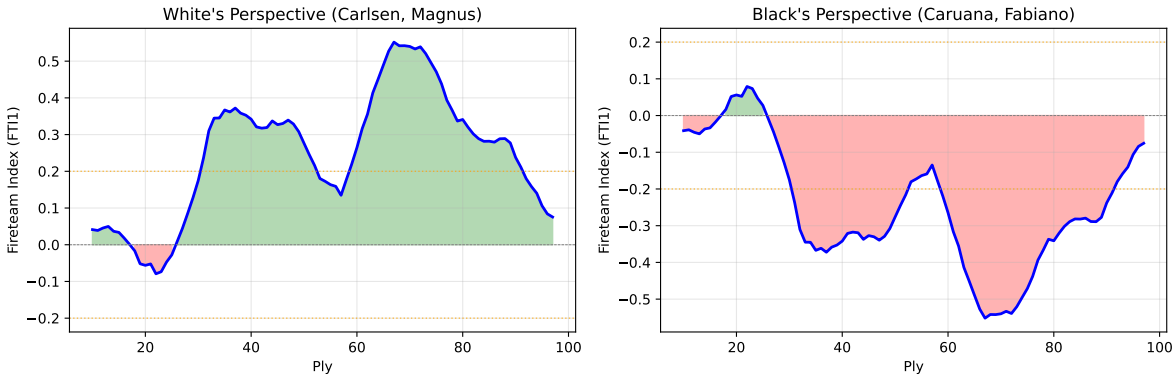
Instead of Qxd6: Rc8 Best continuation: exd6 Bxf3 gxf3

2.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

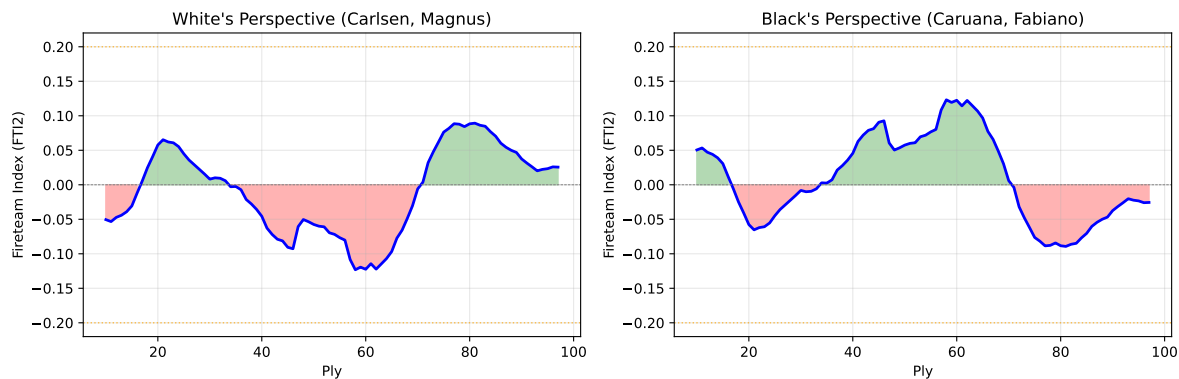
FTI1: Harmonious $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	23 plies
Carlsen, Magnus	Windowed	WIN	33 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

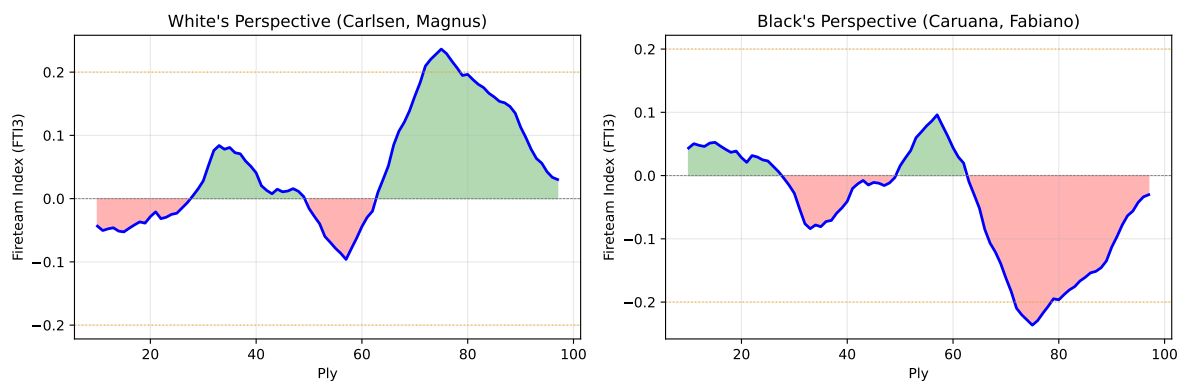


FTI2: Tactical (0.6, 0.1, 0, 0.3)

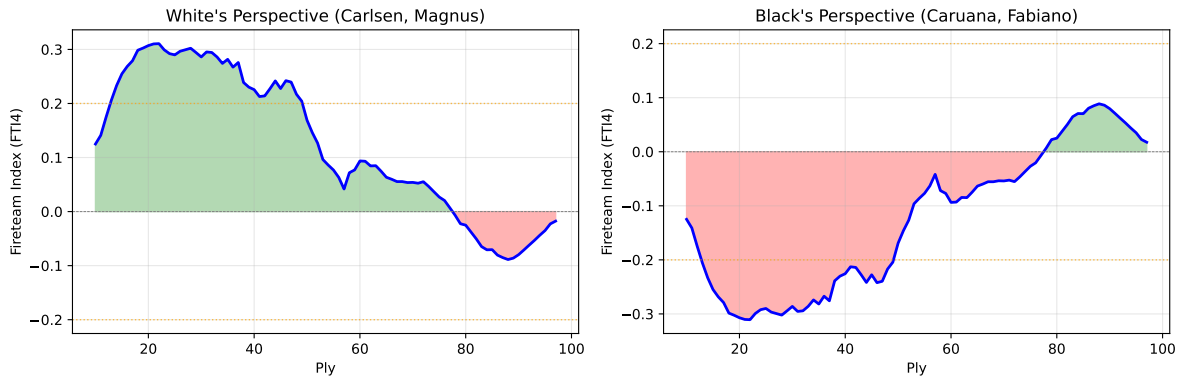
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI3: Strategic** (0.7, 0.1, 0.1, 0.1)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	10 plies
Carlsen, Magnus	Windowed	DRAW	7 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

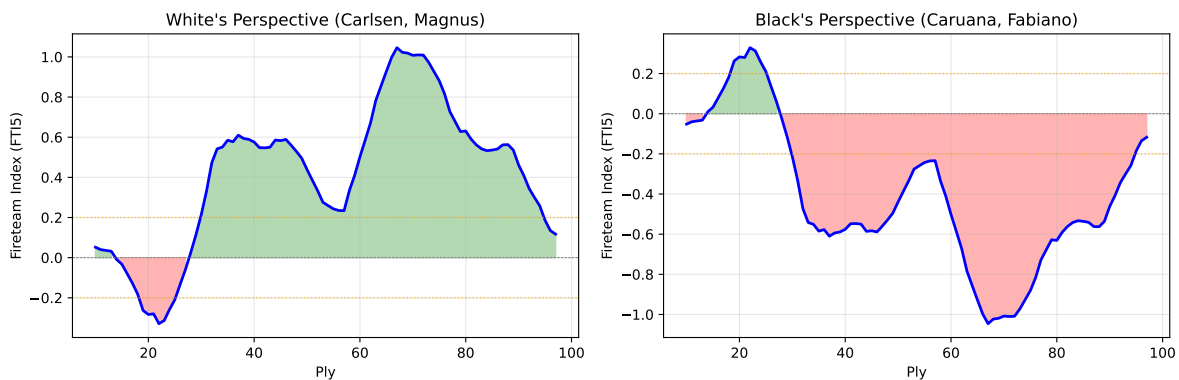
**FTI4: Dynamic** (0.2, 0.6, 0, 0.2)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	17 plies
Carlsen, Magnus	Windowed	WIN	33 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



FTI5: Solid (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	23 plies
Carlsen, Magnus	Windowed	WIN	65 plies
Caruana, Fabiano	Per-ply	DRAW	6 plies
Caruana, Fabiano	Windowed	DRAW	7 plies



Interpretation. White (Carlsen, Magnus) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting Black successfully defended or White failed to convert the advantage.

2.7 Analysis Information

- Engine: Stockfish 17.1

- Depth: 23
- Analysis time: 351.2 seconds

Chapter 3

Carlsen, Magnus vs Caruana, Fabiano

3.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.12
Round	3
White	Carlsen, Magnus (2847)
Black	Caruana, Fabiano (2823)
Result	1/2-1/2
Opening	B31 – Sicilian Defence, Rossolimo Variation

3.2 Player Statistics

3.2.1 Carlsen, Magnus (White)

- Total moves: 49
- Accuracy: 95.9%
- Average centipawn loss: 8.4
- Best/Excellent moves: 21 / 13
- Good moves: 12
- Inaccuracies: 3
- Mistakes: 0
- Blunders: 0

3.2.2 Caruana, Fabiano (Black)

- Total moves: 48
- Accuracy: 97.1%
- Average centipawn loss: 6.0

- Best/Excellent moves: 29 / 8
- Good moves: 8
- Inaccuracies: 3
- Mistakes: 0
- Blunders: 0

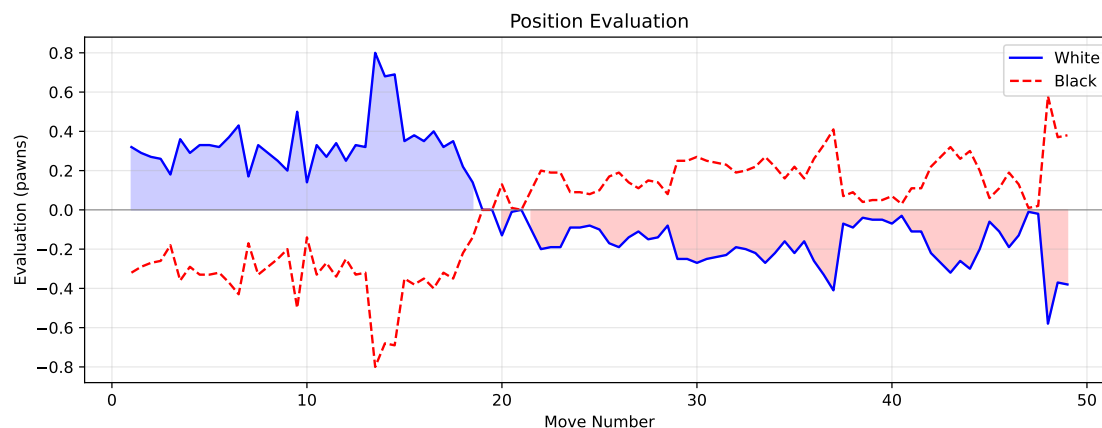
3.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

3.3.1 Position Evaluation Over Time

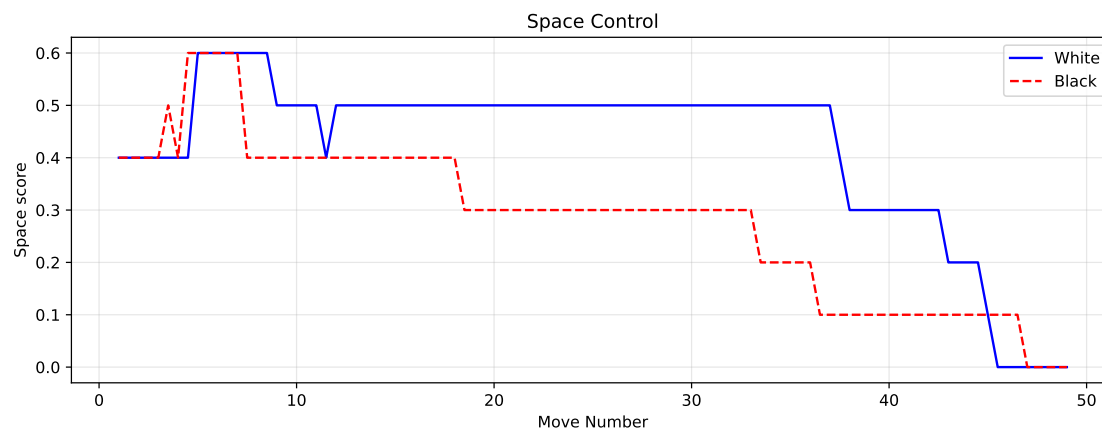
Game Character Classification. Evaluation range: minimum $m_1 = -0.58$, maximum $m_2 = +0.80$, spread $d = m_2 - m_1 = 1.38$.

This game is classified as **tense** (normal). A tense game with moderate swings.



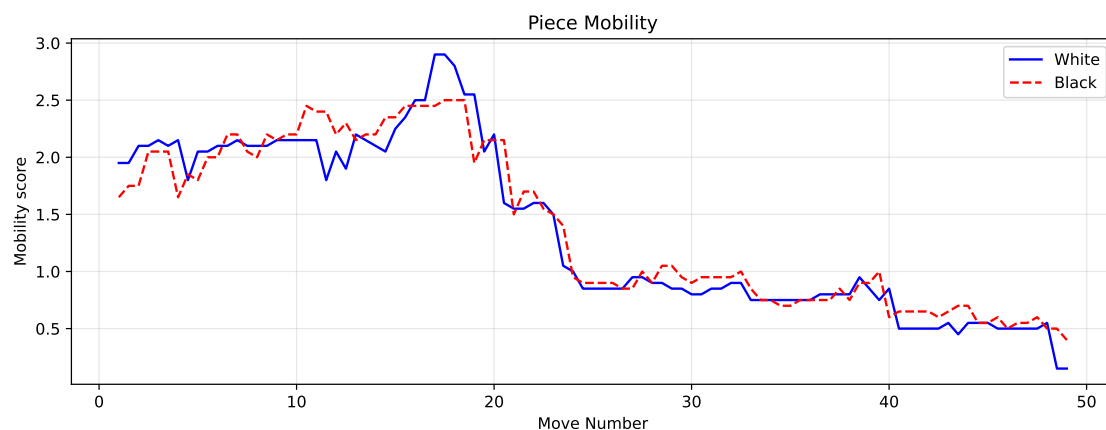
3.3.2 Space Control

- White average: 0.42 (range: 0.00 to 0.60)
- Black average: 0.28 (range: 0.00 to 0.60)



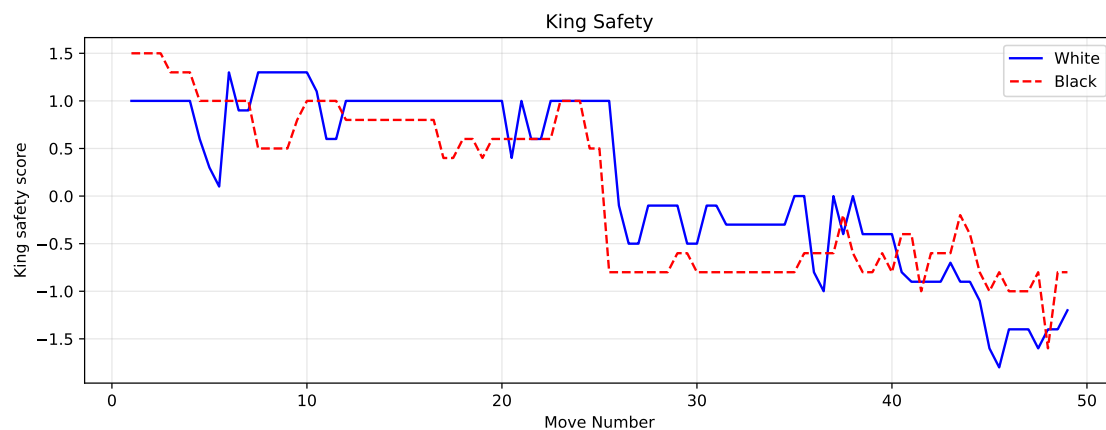
3.3.3 Mobility (Piece Activity)

- White average: 1.36 (range: 0.15 to 2.90)
- Black average: 1.39 (range: 0.40 to 2.50)



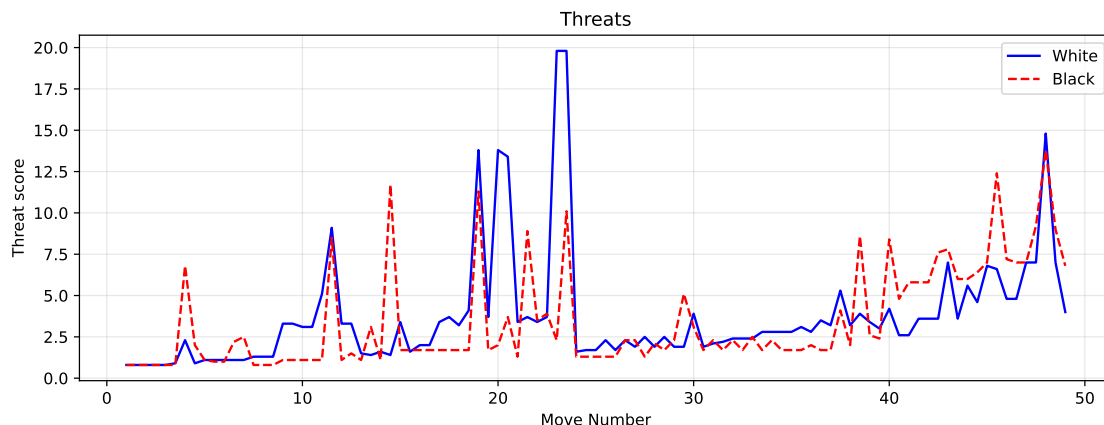
3.3.4 King Safety

- White average: 0.19
- Black average: 0.06



3.3.5 Threats

- White average: 3.78 (range: 0.80 to 19.80)
- Black average: 3.47 (range: 0.80 to 13.80)
- Average advantage: 0.31 (White maintained more threats)



3.4 Annotated Game

1. e4 c5 2. Nf3 Nc6 3. Bb5 g6 4. Bxc6 dxc6 5. d3 Bg7 6. O-O Qc7 7. Re1?! e5! 8. a3 Nf6! 9. b4 O-O 10. Nbd2 Bg4 11. h3 Bxf3 12. Nxf3 cxb4 13. axb4! a5?! 14. bxa5! Rxa5! 15. Bd2?! Raa8 16. Qb1 Nd7 17. Qb4 Rfe8 18. Bc3 b5! 19. Rxa8 Rxa8! 20. Ra1 Rxa1+! 21. Bxa1! Qa7! 22. Bc3 Qa2! 23. Qb2 Qxb2! 24. Bxb2! f6 25. Kf1 Kf7! 26. Ke2! Nc5 27. Bc3 Ne6! 28. g3 Bf8 29. Nd2 Ng5! 30. h4! Ne6! 31. Nb3 h5! 32. Bd2 Bd6 33. c3 c5! 34. Be3 Ke7 35. Kd1 Kd7 36. Kc2 f5! 37. Kd1 fxe4?! 38. dxe4! c4 39. Nd2 Nc5 40. Bxc5! Bxc5! 41. Ke2 Kc6 42. Nf1 b4! 43. cxb4! Bxb4! 44. Ne3! Kc5 45. f4! exf4! 46. gxf4 Ba5 47. f5! gxf5! 48. Nxc4?! Kxc4?! 49. exf5!

1/2-1/2

3.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

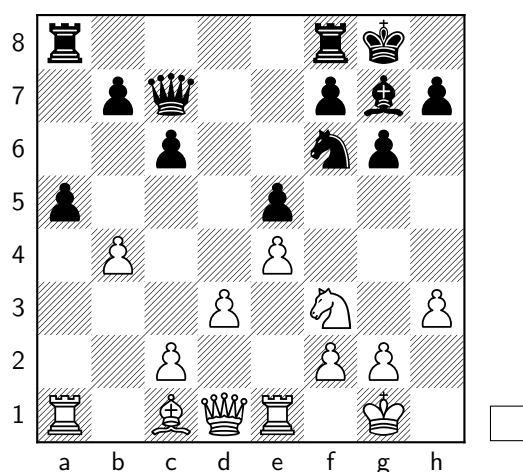
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 13...a5 ★

Biggest evaluation swing(s) (48cp)

Evaluation: +0.80



Metric	White	Black
Space	0.50	0.40
Mobility (MG)	4.30	4.40
King Safety	1.00	0.80
Threats	1.40	3.10

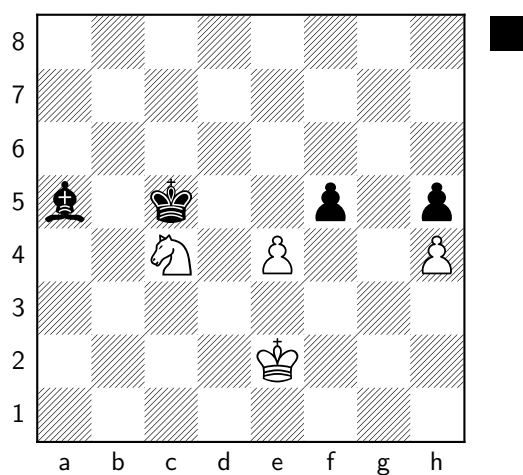
Instead of a5: Nd7, Rfd8, Rfe8

Best continuation: bxa5 Rxa5 Rxa5

Position 2: After 48. Nxc4 ★

Biggest evaluation swing(s) (56cp)

Evaluation: -0.58



Metric	White	Black
Space	0.00	0.00
Mobility (MG)	1.10	1.00
King Safety	-1.40	-1.60
Threats	14.80	13.80

Instead of Nxc4: exf5, Nxf5, e5

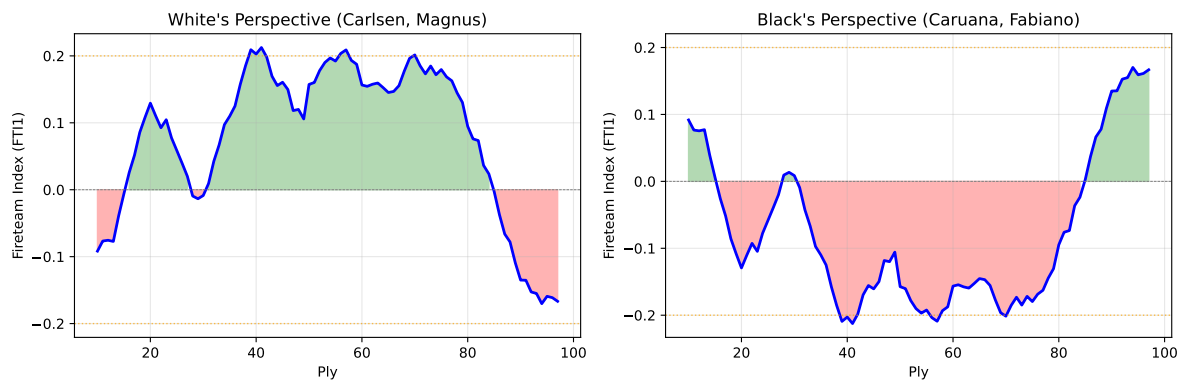
Best continuation: Kxc4 exf5 Bd8

3.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

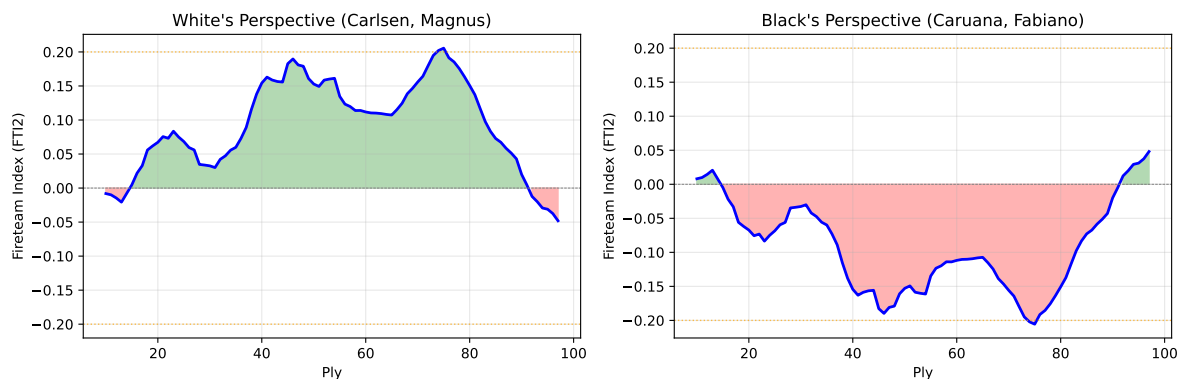
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	5 plies
Carlsen, Magnus	Windowed	DRAW	3 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



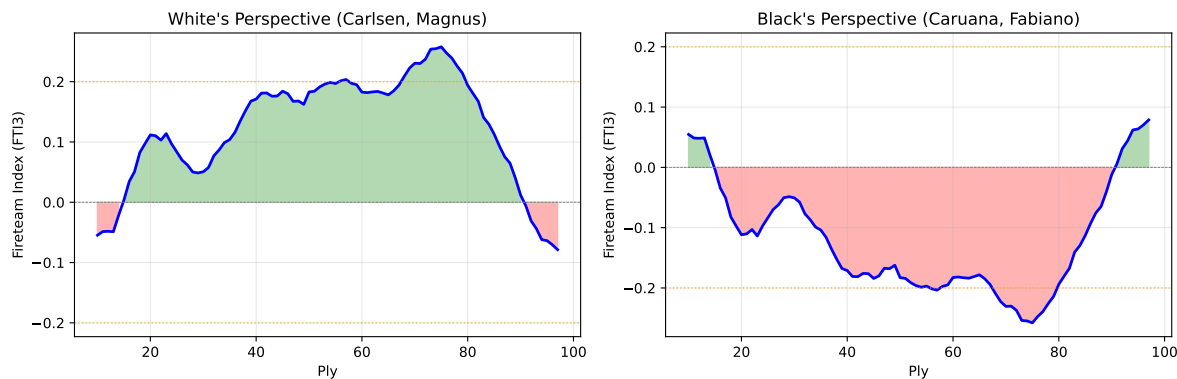
FTI2: Tactical (0.6, 0.1, 0, 0.3)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	2 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

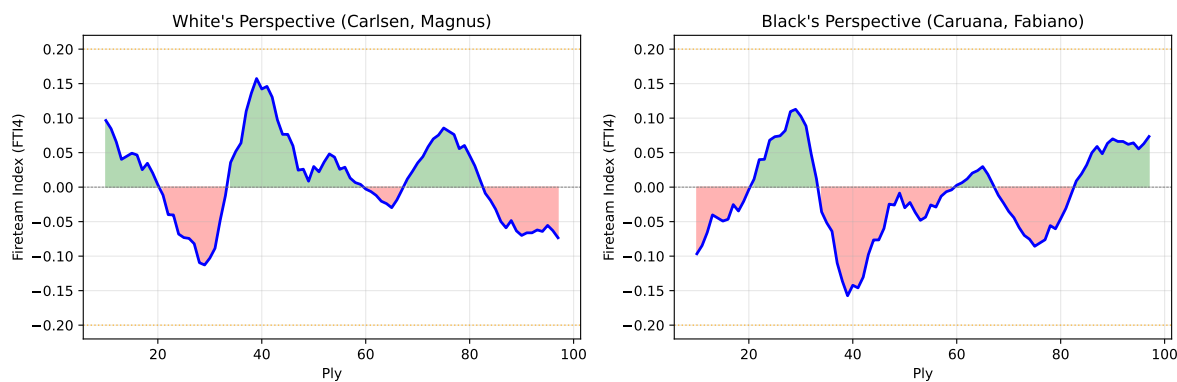


FTI3: Strategic (0.7, 0.1, 0.1, 0.1)

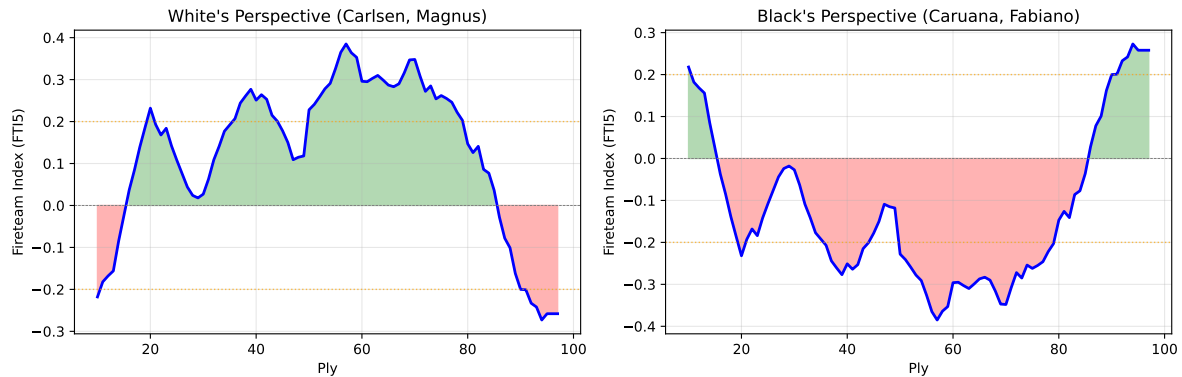
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	5 plies
Carlsen, Magnus	Windowed	WIN	12 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI4: Dynamic** (0.2, 0.6, 0, 0.2)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI5: Solid** (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	11 plies
Carlsen, Magnus	Windowed	WIN	30 plies
Caruana, Fabiano	Per-ply	DRAW	6 plies
Caruana, Fabiano	Windowed	DRAW	7 plies



Interpretation. Neither player achieved sustained positional dominance according to FTI1. Both algorithms predicted DRAW for both sides, which matches the actual result. This suggests a balanced game where neither side established lasting control.

3.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 395.8 seconds

Chapter 4

Carlsen, Magnus vs Caruana, Fabiano

4.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.13
Round	4
White	Carlsen, Magnus (2847)
Black	Caruana, Fabiano (2823)
Result	1/2-1/2
Opening	A29 – English Opening, Four Knights, Kingside Fianchetto

4.2 Player Statistics

4.2.1 Carlsen, Magnus (White)

- Total moves: 34
- Accuracy: 96.5%
- Average centipawn loss: 7.0
- Best/Excellent moves: 12 / 12
- Good moves: 10
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

4.2.2 Caruana, Fabiano (Black)

- Total moves: 33
- Accuracy: 97.4%
- Average centipawn loss: 5.3

- Best/Excellent moves: 16 / 8
- Good moves: 8
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 0

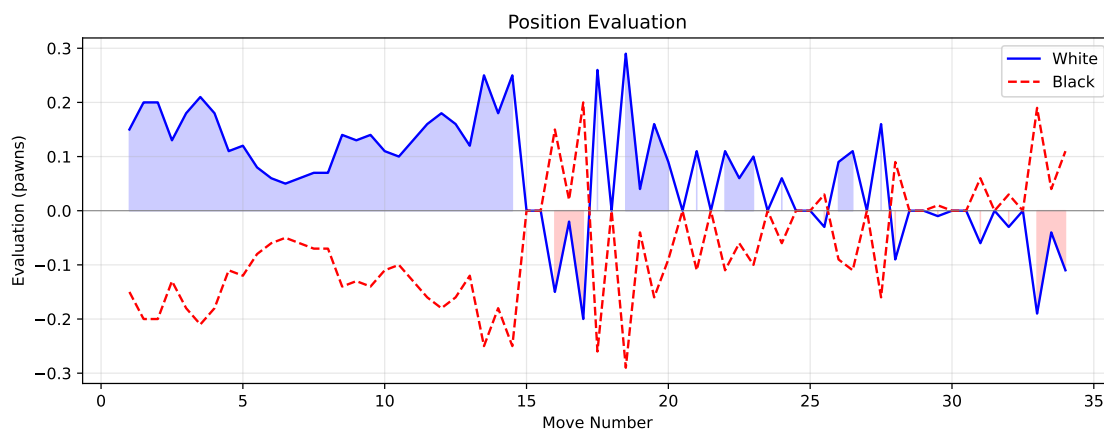
4.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

4.3.1 Position Evaluation Over Time

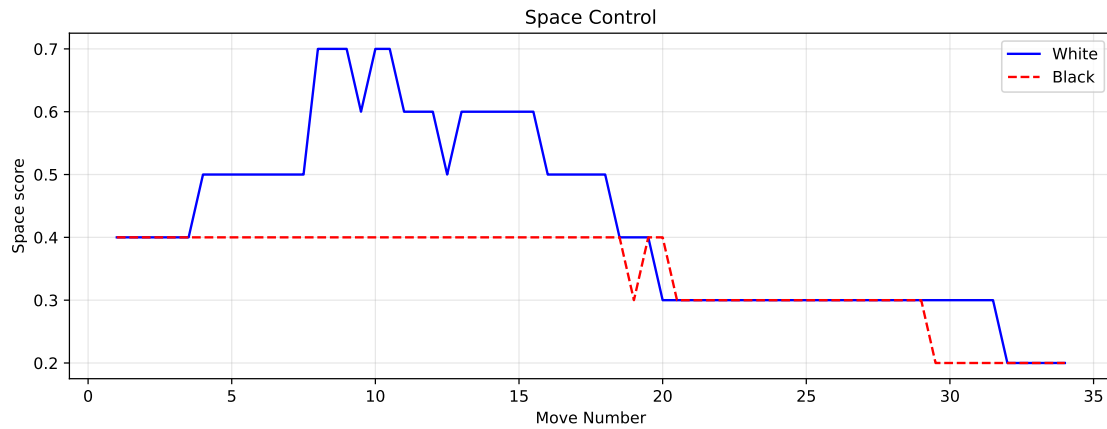
Game Character Classification. Evaluation range: minimum $m_1 = -0.20$, maximum $m_2 = +0.29$, spread $d = m_2 - m_1 = 0.49$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



4.3.2 Space Control

- White average: 0.42 (range: 0.20 to 0.70)
- Black average: 0.34 (range: 0.20 to 0.40)



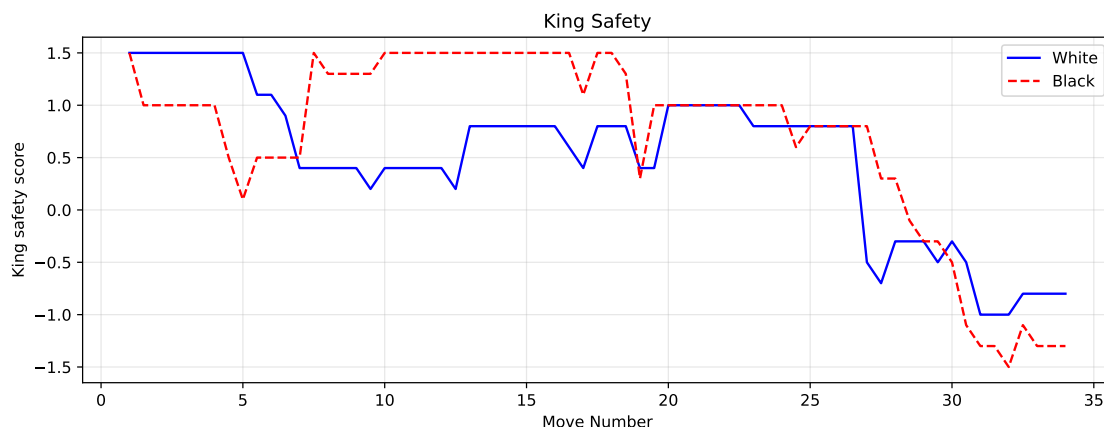
4.3.3 Mobility (Piece Activity)

- White average: 1.80 (range: 1.15 to 2.65)
- Black average: 1.91 (range: 1.15 to 2.95)



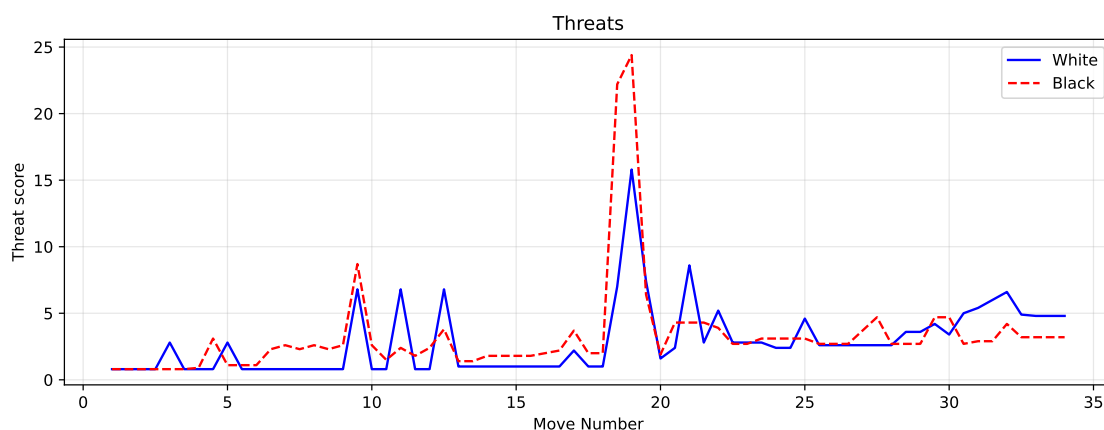
4.3.4 King Safety

- White average: 0.50
- Black average: 0.71



4.3.5 Threats

- White average: 2.91 (range: 0.80 to 15.80)
- Black average: 3.30 (range: 0.80 to 24.40)
- Average advantage: -0.39 (Black maintained more threats)



4.4 Annotated Game

1. c4 e5 2. Nc3 Nf6 3. Nf3 Nc6 4. g3 d5 5. cxd5 Nxd5 6. Bg2 Bc5 7. O-O O-O! 8. d3! Re8 9. Bd2! Nxc3 10. Bxc3 Nd4! 11. b4! Bd6 12. Rb1! Nxf3+! 13. Bxf3 a6 14. a4! c6! 15. Re1 Bd7! 16. e3 Qf6 17. Be4 Bf5?! 18. Qf3 Bxe4 19. Qxf6 gxf6! 20. dxe4 b5! 21. Red1 Bf8! 22. axb5 axb5 23. Kg2 Red8! 24. Rdc1! Kg7 25. Be1 Rdc8 26. Rc2 Ra4 27. Kf3 h5 28. Ke2 Kg6! 29. h3! f5! 30. exf5+ Kxf5! 31. f3 Be7 32. e4+ Ke6! 33. Bd2 Bd6 34. Rbc1

1/2-1/2

4.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

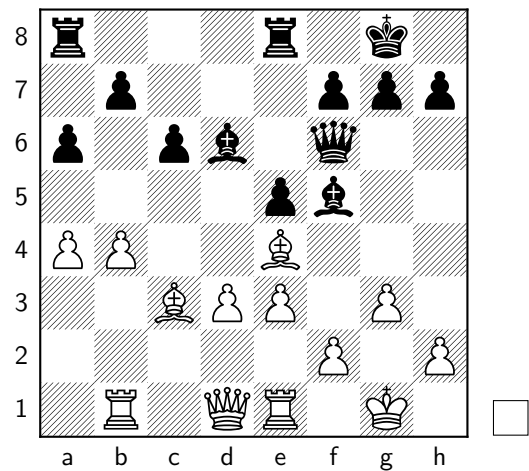
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 17...Bf5 ★

Biggest evaluation swing(s) (46cp)

Evaluation: +0.26



Metric	White	Black
Space	0.50	0.40
Mobility (MG)	4.20	4.70
King Safety	0.80	1.50
Threats	1.00	2.00

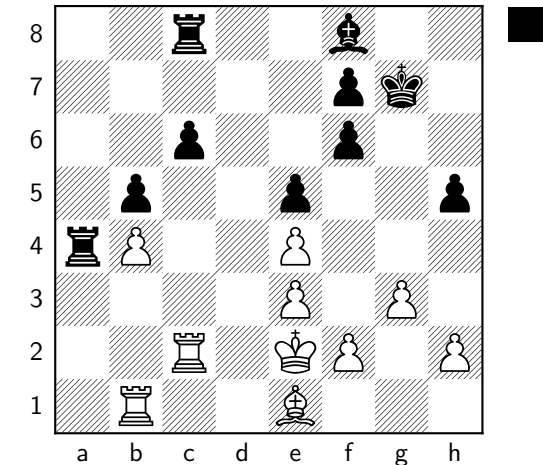
Instead of Bf5: Rac8, Rad8, h6

Best continuation: Qf3 Bg6 Qxf6

Position 2: After 28. Ke2 ★

Biggest evaluation swing(s) (25cp)

Evaluation: -0.09



Metric	White	Black
Space	0.30	0.30
Mobility (MG)	2.60	2.60
King Safety	-0.30	0.30
Threats	2.60	2.70

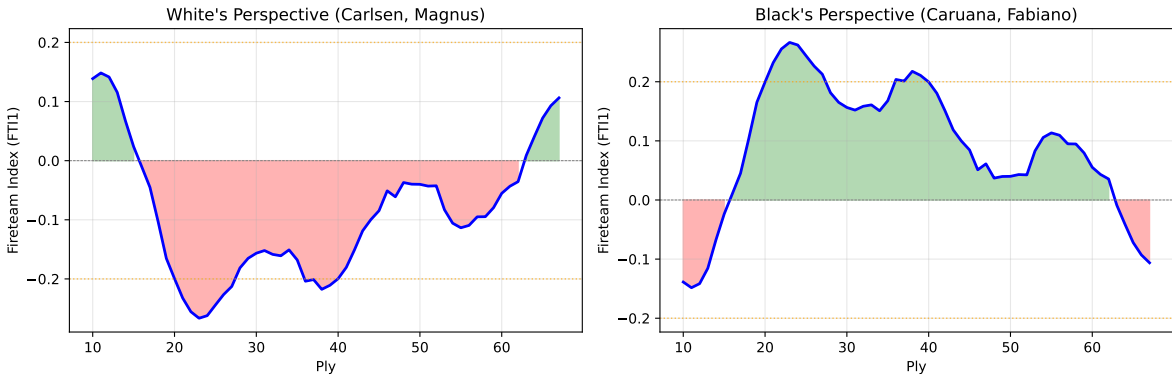
Instead of Ke2: h4, Bd2, g4 Best continuation: Kg6 f3 f5

4.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

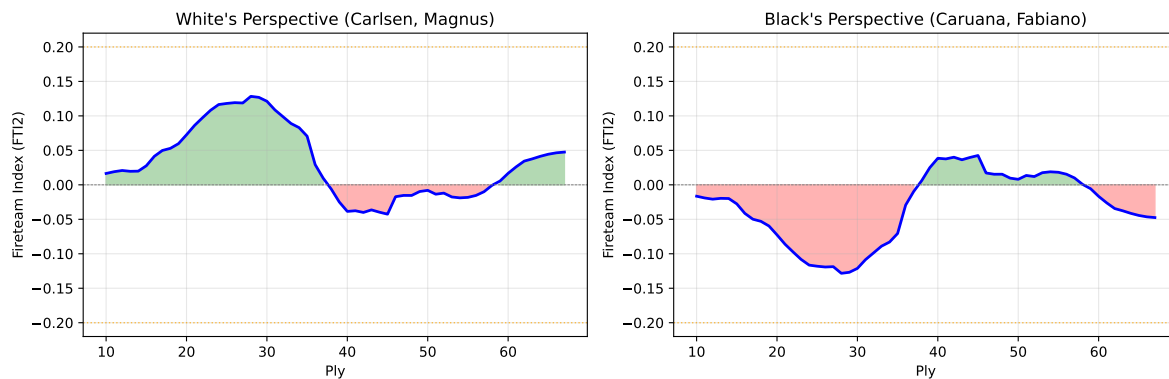
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	8 plies
Caruana, Fabiano	Windowed	DRAW	7 plies

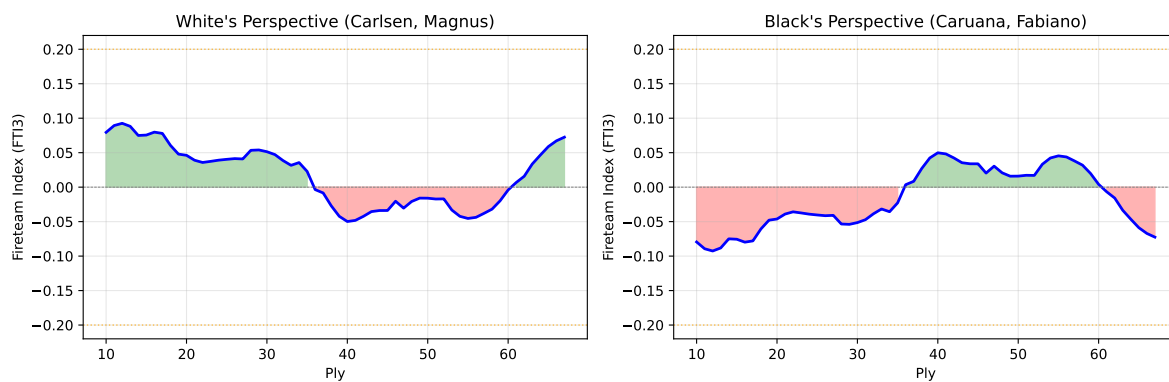


FTI2: Tactical (0.6, 0.1, 0, 0.3)

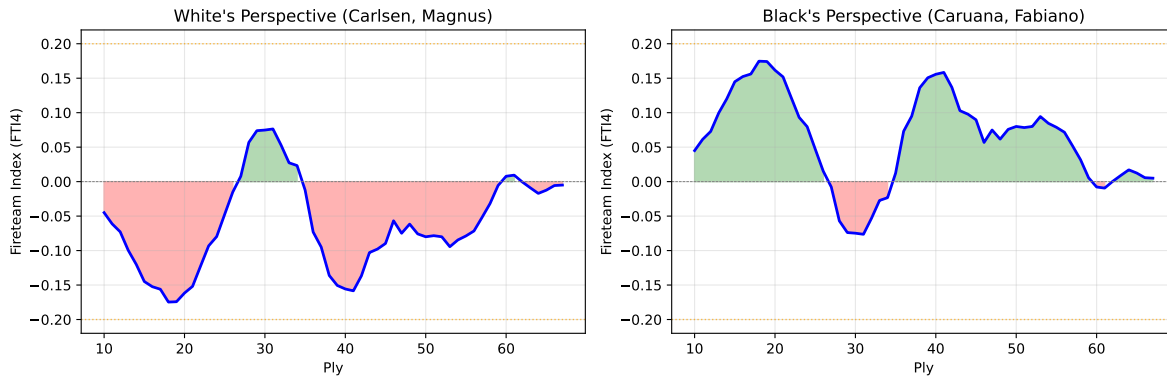
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI3: Strategic** (0.7, 0.1, 0.1, 0.1)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

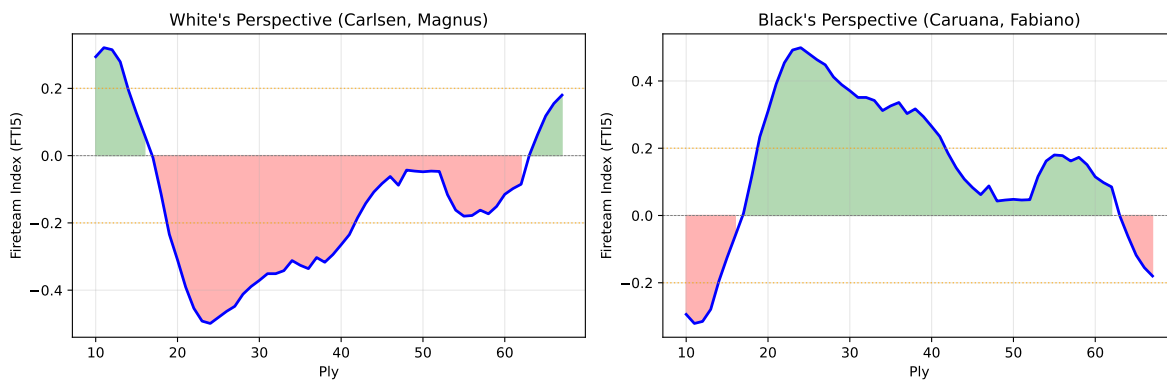
**FTI4: Dynamic** (0.2, 0.6, 0, 0.2)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



FTI5: Solid (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	WIN	20 plies
Caruana, Fabiano	Windowed	WIN	23 plies



Interpretation. Neither player achieved sustained positional dominance according to FTI1. Both algorithms predicted DRAW for both sides, which matches the actual result. This suggests a balanced game where neither side established lasting control.

4.7 Analysis Information

- Engine: Stockfish 17.1

- Depth: 23
- Analysis time: 317.3 seconds

Chapter 5

Carlsen, Magnus vs Caruana, Fabiano

5.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.15
Round	5
White	Carlsen, Magnus (2847)
Black	Caruana, Fabiano (2823)
Result	1/2-1/2
Opening	B31 – Sicilian Defence, Rossolimo Variation

5.2 Player Statistics

5.2.1 Carlsen, Magnus (White)

- Total moves: 34
- Accuracy: 98.1%
- Average centipawn loss: 3.8
- Best/Excellent moves: 21 / 6
- Good moves: 6
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 0

5.2.2 Caruana, Fabiano (Black)

- Total moves: 33
- Accuracy: 98.5%
- Average centipawn loss: 3.0

- Best/Excellent moves: 23 / 3
- Good moves: 6
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 0

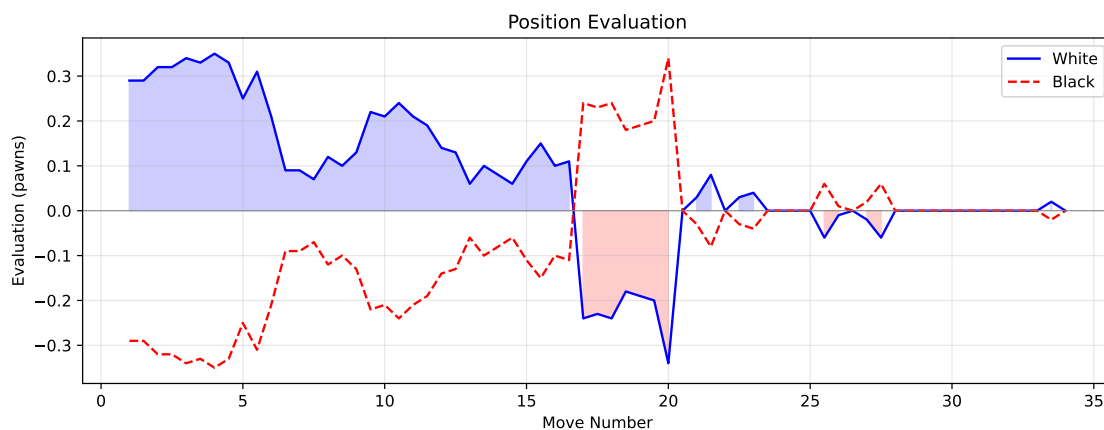
5.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

5.3.1 Position Evaluation Over Time

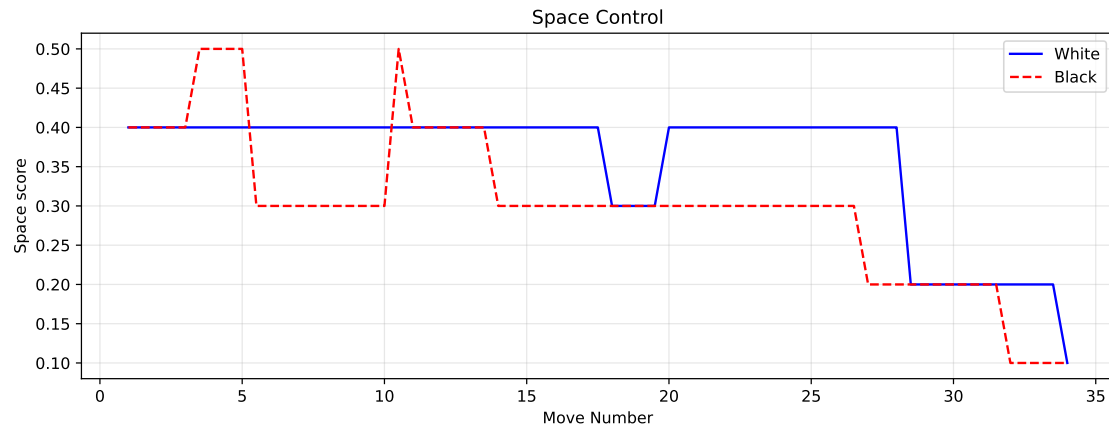
Game Character Classification. Evaluation range: minimum $m_1 = -0.34$, maximum $m_2 = +0.35$, spread $d = m_2 - m_1 = 0.69$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



5.3.2 Space Control

- White average: 0.36 (range: 0.10 to 0.40)
- Black average: 0.30 (range: 0.10 to 0.50)



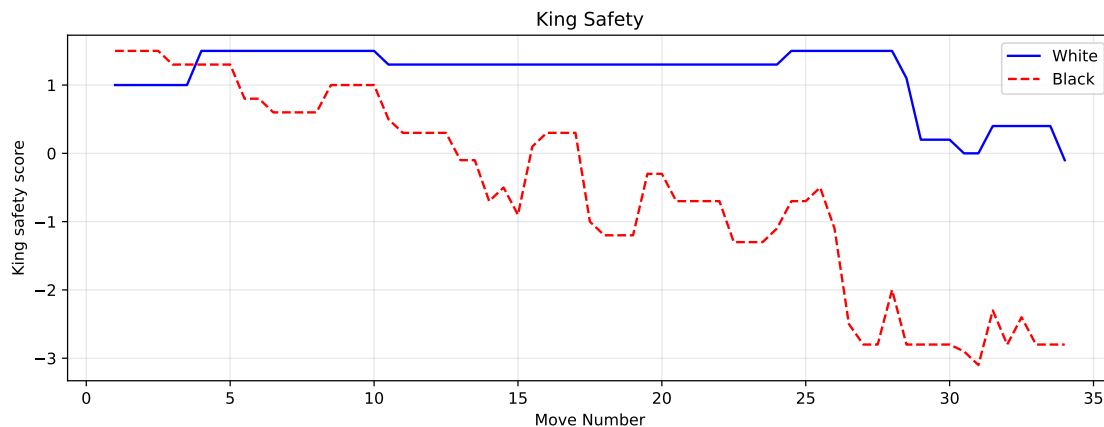
5.3.3 Mobility (Piece Activity)

- White average: 1.78 (range: 0.95 to 2.35)
- Black average: 1.72 (range: 1.00 to 2.30)



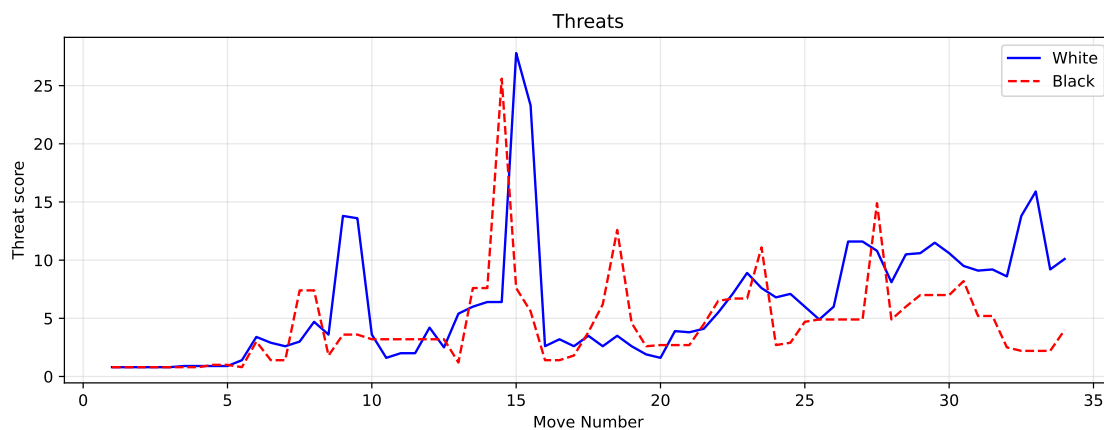
5.3.4 King Safety

- White average: 1.16
- Black average: -0.57



5.3.5 Threats

- White average: 6.18 (range: 0.80 to 27.80)
- Black average: 4.46 (range: 0.80 to 25.60)
- Average advantage: 1.72 (White maintained more threats)



5.4 Annotated Game

1. e4 c5 2. Nf3 Nc6 3. Bb5 g6 4. O-O Bg7 5. Re1 e5 6. b4 Nxb4 7. Bb2 a6! 8. a3! axb5! 9. axb4! Rxa1! 10. Bxa1! d6 11. bxc5! Ne7! 12. Qe2 b4 13. Qc4 Qa5 14. cxd6! Be6! 15. Qc7 Qxc7! 16. dxc7 Nc6! 17. c3?! Kd7! 18. cxb4! Ra8! 19. Bc3! Kxc7! 20. d3 Kb6?! 21. Bd2! Rd8 22. Be3+ Kb5! 23. Nc3+! Kxb4! 24. Nd5+! Bxd5! 25. exd5! Rxd5! 26. Rb1+! Kc3 27. Rxb7! Nd8! 28. Rc7+! Kxd3! 29. Kf1! h5! 30. h3! Ke4 31. Ng5+! Kf5! 32. Nxf7! Nxf7! 33. Rxf7+! Bf6! 34. g4+ 1/2-1/2

5.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

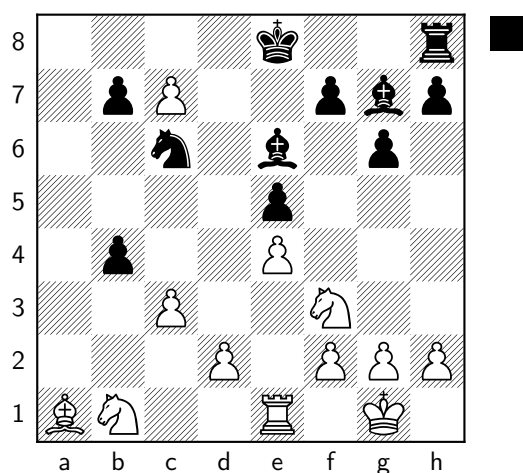
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 17. c3 ★

Biggest evaluation swing(s) (35cp)

Evaluation: -0.24



Metric	White	Black
Space	0.40	0.30
Mobility (MG)	2.70	2.80
King Safety	1.30	0.30
Threats	2.60	1.80

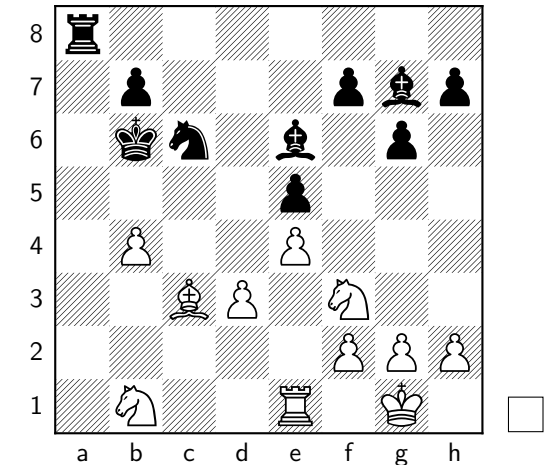
Instead of c3: Rd1, d4

Best continuation: Kd7 cxb4 Ra8

Position 2: After 20...Kb6 ★

Biggest evaluation swing(s) (34cp)

Evaluation: +0.00



Metric	White	Black
Space	0.40	0.30
Mobility (MG)	3.40	3.90
King Safety	1.30	-0.70
Threats	3.90	2.70

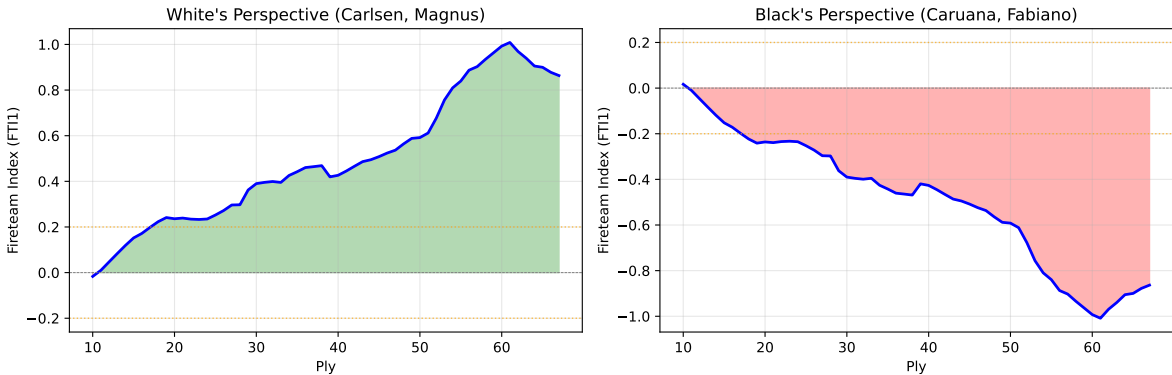
Instead of Kb6: b5, Kd7, Ra4 Best continuation: Bd2 Kc7 Rc1

5.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

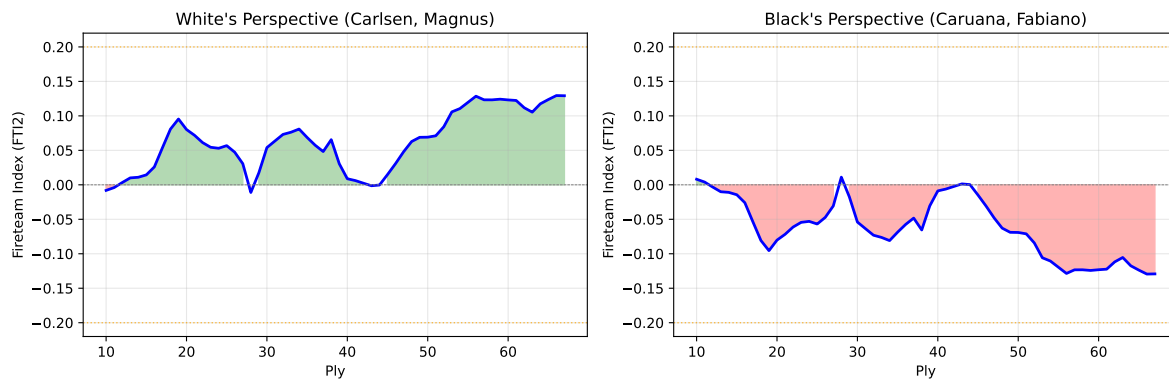
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	47 plies
Carlsen, Magnus	Windowed	WIN	50 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

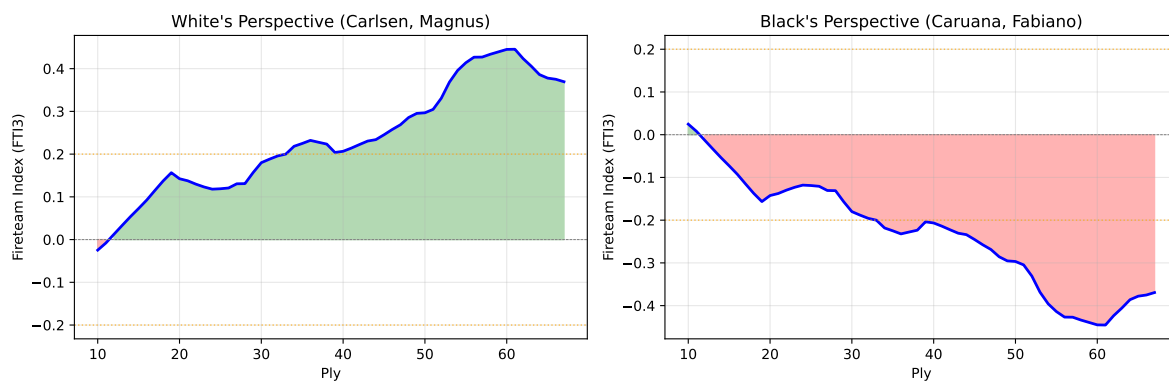


FTI2: Tactical (0.6, 0.1, 0, 0.3)

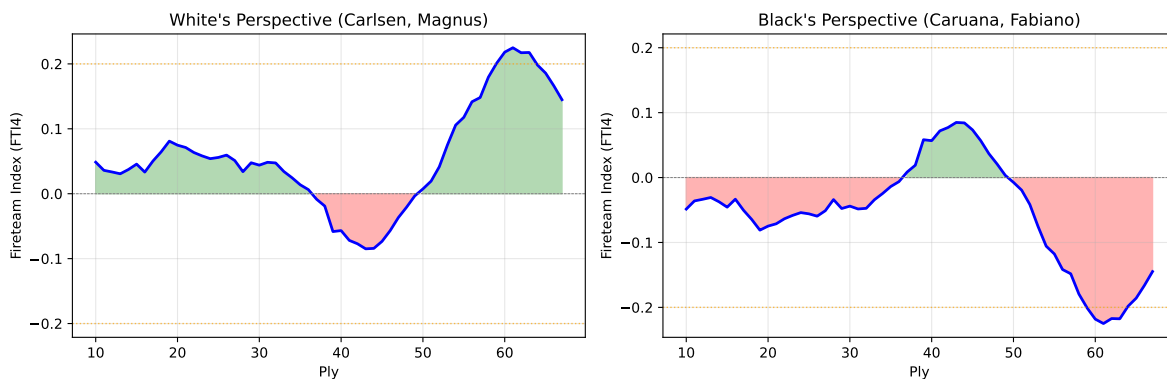
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	2 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI3: Strategic** (0.7, 0.1, 0.1, 0.1)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	28 plies
Carlsen, Magnus	Windowed	WIN	35 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

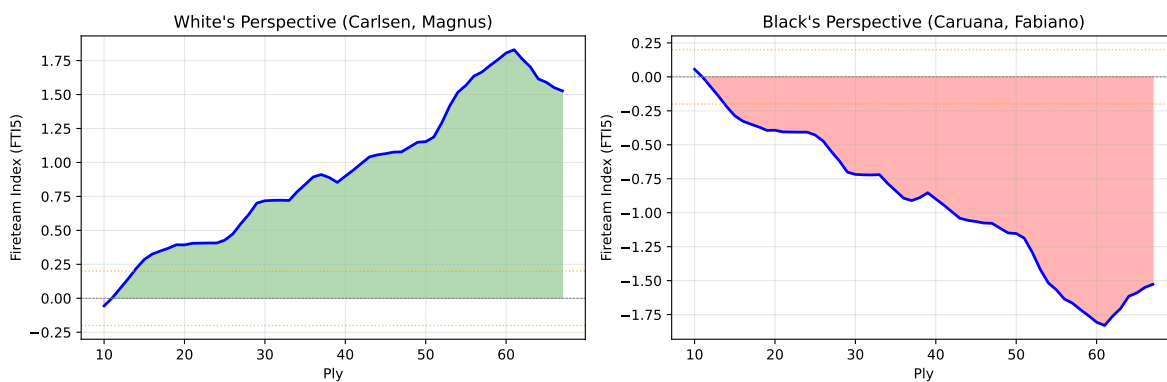
**FTI4: Dynamic** (0.2, 0.6, 0, 0.2)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	7 plies
Carlsen, Magnus	Windowed	DRAW	5 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



FTI5: Solid (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	51 plies
Carlsen, Magnus	Windowed	WIN	51 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



Interpretation. White (Carlsen, Magnus) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting Black successfully defended or White failed to convert the advantage.

5.7 Analysis Information

- Engine: Stockfish 17.1

- Depth: 23
- Analysis time: 338.6 seconds

Chapter 6

Carlsen, Magnus vs Caruana, Fabiano

6.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.16
Round	6
White	Carlsen, Magnus (2847)
Black	Caruana, Fabiano (2823)
Result	1/2-1/2
Opening	C42 – Petrov’s Defence, Karklins-Martinovsky Variation

6.2 Player Statistics

6.2.1 Carlsen, Magnus (White)

- Total moves: 80
- Accuracy: 95.6%
- Average centipawn loss: 9.0
- Best/Excellent moves: 45 / 19
- Good moves: 12
- Inaccuracies: 2
- Mistakes: 1
- Blunders: 1

6.2.2 Caruana, Fabiano (Black)

- Total moves: 80
- Accuracy: 96.2%
- Average centipawn loss: 7.7

- Best/Excellent moves: 42 / 24
- Good moves: 12
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 1

6.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

6.3.1 Position Evaluation Over Time

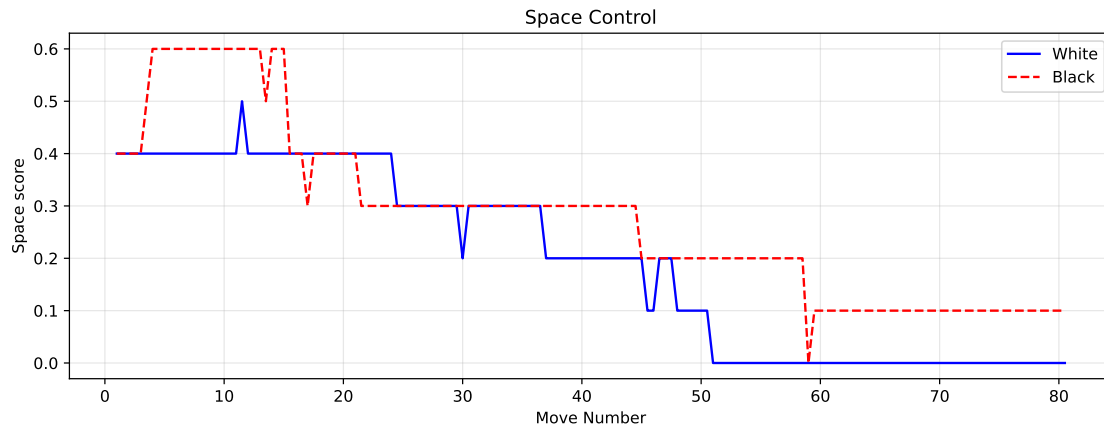
Game Character Classification. Evaluation range: minimum $m_1 = -2.48$, maximum $m_2 = +0.42$, spread $d = m_2 - m_1 = 2.90$.

This game is classified as **tense** (one-sided). A tense, one-sided game. Black maintained the advantage throughout.



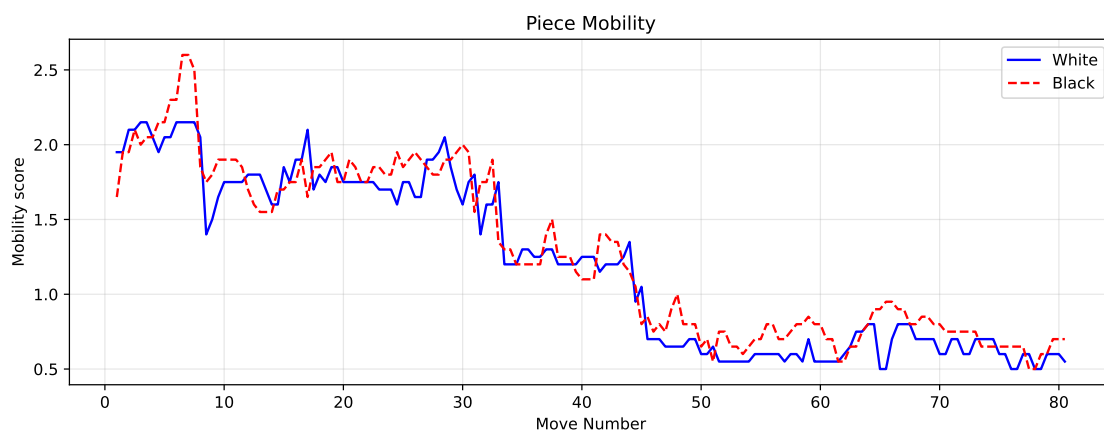
6.3.2 Space Control

- White average: 0.19 (range: 0.00 to 0.50)
- Black average: 0.28 (range: 0.00 to 0.60)



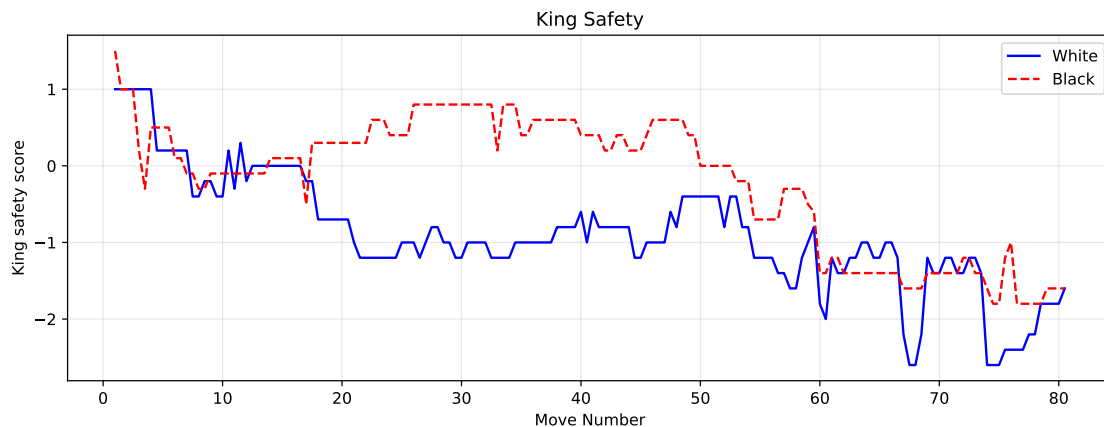
6.3.3 Mobility (Piece Activity)

- White average: 1.20 (range: 0.50 to 2.15)
- Black average: 1.27 (range: 0.50 to 2.60)



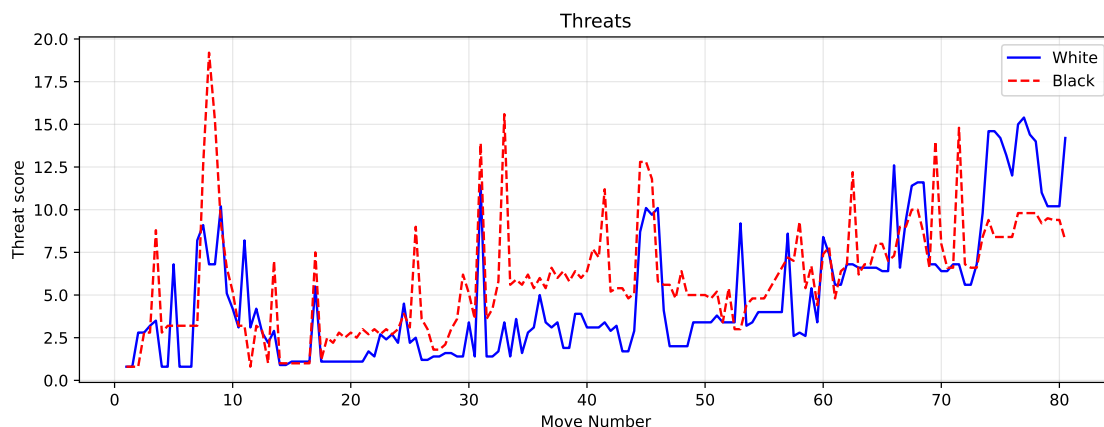
6.3.4 King Safety

- White average: -0.90
- Black average: -0.18



6.3.5 Threats

- White average: 4.86 (range: 0.80 to 15.40)
- Black average: 5.93 (range: 0.80 to 19.20)
- Average advantage: -1.07 (Black maintained more threats)



6.4 Annotated Game

1. e4 e5 2. Nf3 Nf6 3. Nxe5 d6 4. Nd3?! Nxe4 5. Qe2 Qe7 6. Nf4 Nc6 7. Nd5 Nd4! 8. Nxe7! Nxe2! 9. Nd5! Nd4! 10. Na3 Ne6 11. f3! N4c5! 12. d4 Nd7 13. c3 c6! 14. Nf4! Nb6! 15. Bd3 d5 16. Nc2! Bd6 17. Nxe6 Bxe6 18. Kf2 h5 19. h4 Nc8 20. Ne3 Ne7 21. g3 c5! 22. Bc2 O-O 23. Rd1 Rfd8 24. Ng2 cxd4! 25. cxd4! Rac8 26. Bb3! Nc6 27. Bf4! Na5 28. Rdc1! Bb4 29. Bd1 Nc4! 30. b3! Na3 31. Rxc8! Rxc8! 32. Rc1 Nb5 33. Rxc8+! Bxc8! 34. Ne3 Nc3! 35. Bc2 Ba3 36. Bb8 a6 37. f4 Bd7 38. f5 Bc6 39. Bd1! Bb2 40. Bxh5 Ne4+ 41. Kg2 Bxd4! 42. Bf4 Bc5 43. Bf3! Nd2! 44. Bxd5? *White loses 67cp. Consider instead: Nd1, Nf1, Nc2.*

44...Bxe3! 45. Bxc6! Bxf4! 46. Bxb7! Bd6! 47. Bxa6 Ne4! 48. g4! Ba3! 49. Bc4 Kf8 50. g5 Nc3! 51. b4! Bxb4 52. Kf3 Na4! 53. Bb5 Nc5?! 54. a4?! f6 55. Kg4 Ne4 56. Kh5 Be1! 57. Bd3 Nd6 58. a5 Bxa5 59. gxf6! gxf6! 60. Kg6! Bd8 61. Kh7! Nf7 62.

Bc4 Ne5! 63. Bd5 Ba5 64. h5! Bd2 65. Ba2 Nf3 66. Bd5 Nd4! 67. Kg6?? *White loses 200cp. Consider instead: Be4.*

67...Bg5! 68. Bc4 Nf3?? *Black loses 181cp. Consider instead: Bh4.*

69. Kh7! Ne5 70. Bb3 Ng4 71. Bc4 Ne3! 72. Bd3! Ng4! 73. Bc4! Nh6 74. Kg6! Ke7 75. Bb3! Kd6 76. Bc2! Ke5 77. Bd3 Kf4 78. Bc2! Ng4! 79. Bb3 Ne3 80. h6! Bxh6!

1/2-1/2

6.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

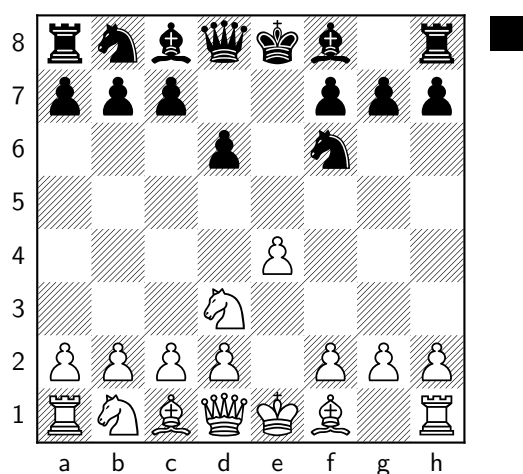
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 4. Nd3 ★

Biggest evaluation swing(s) (39cp)

Evaluation: +0.00



Metric	White	Black
Space	0.40	0.60
Mobility (MG)	4.10	4.10
King Safety	1.00	0.50
Threats	0.80	2.80

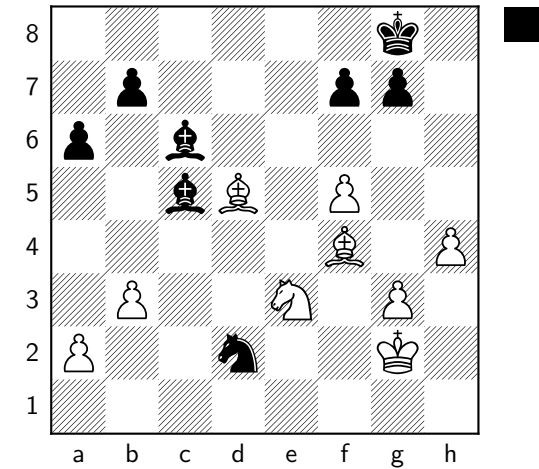
Instead of Nd3: Nf3, Nc4

Best continuation: Nxe4 Qe2 Qe7

Position 2: After 44. Bxd5 ★

Biggest evaluation swing(s) (65cp)

Evaluation: -0.81



Metric	White	Black
Space	0.20	0.30
Mobility (MG)	2.70	2.30
King Safety	-0.80	0.20
Threats	2.90	5.10

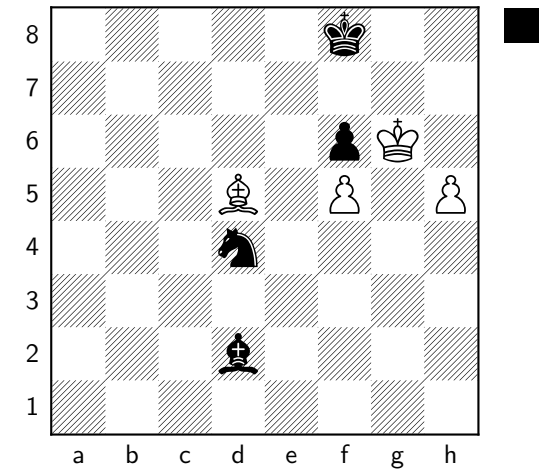
Instead of Bxd5: Nd1, Nf1, Nc2

Best continuation: Bxe3 Bxc6 Bxf4

Position 3: After 67. Kg6

White blunders (-214cp)

Evaluation: -2.47



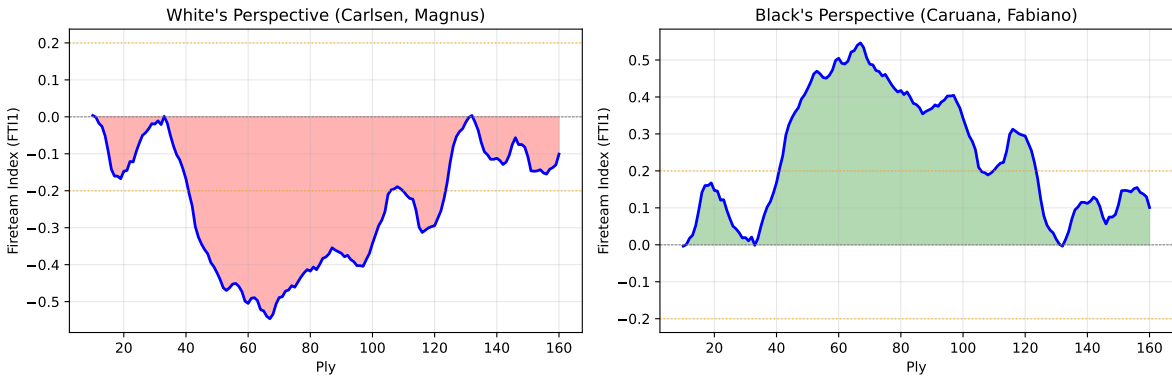
Metric	White	Black
Space	0.00	0.10
Mobility (MG)	1.60	1.80
King Safety	-2.20	-1.60
Threats	9.40	9.00
Instead of Kg6:	Be4	Best continuation: Bg5 Bc4 Bh4

6.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

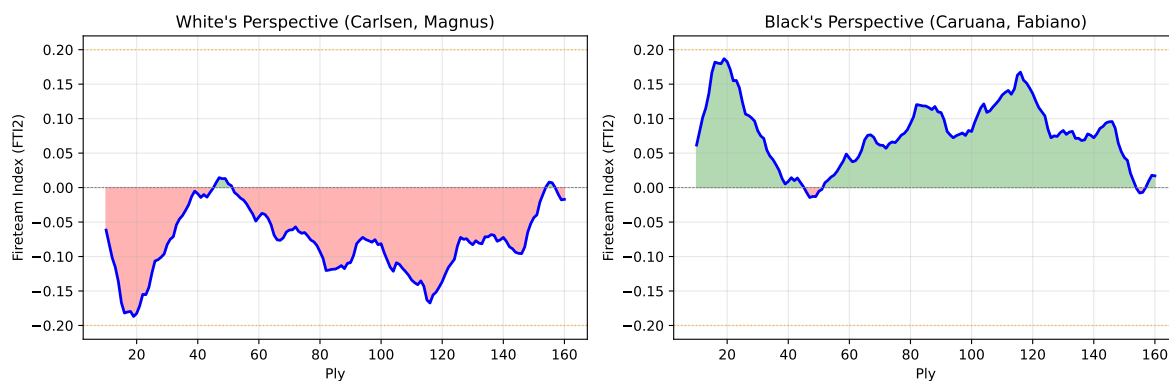
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	WIN	64 plies
Caruana, Fabiano	Windowed	WIN	65 plies

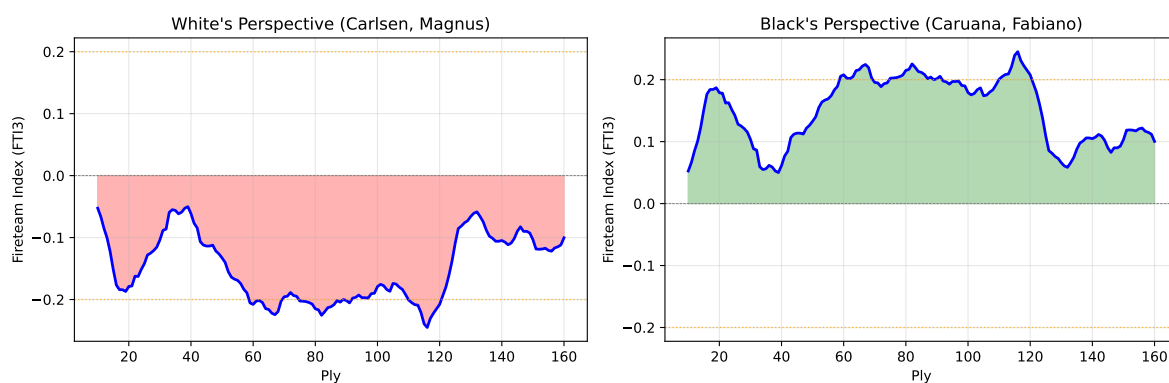


FTI2: Tactical (0.6, 0.1, 0, 0.3)

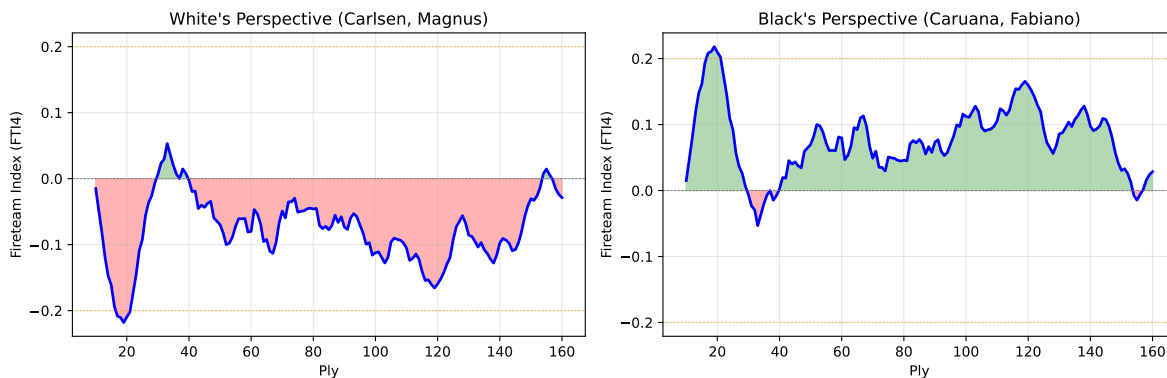
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	2 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI3: Strategic (0.7, 0.1, 0.1, 0.1)**

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	WIN	11 plies
Caruana, Fabiano	Windowed	WIN	14 plies

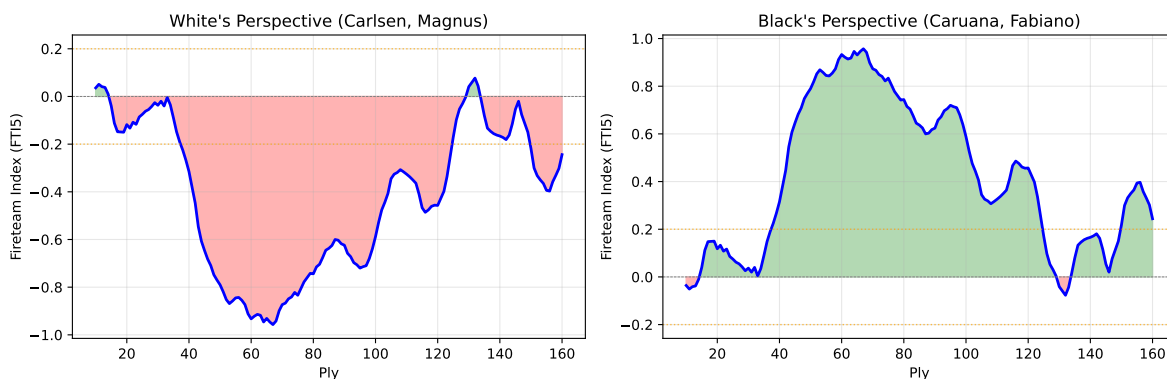
**FTI4: Dynamic (0.2, 0.6, 0, 0.2)**

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	2 plies
Caruana, Fabiano	Windowed	DRAW	5 plies



FTI5: Solid (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	WIN	71 plies
Caruana, Fabiano	Windowed	WIN	87 plies



Interpretation. Black (Caruana, Fabiano) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting White successfully defended or Black failed to convert the advantage.

6.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 621.0 seconds

Chapter 7

Carlsen, Magnus vs Caruana, Fabiano

7.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.18
Round	7
White	Carlsen, Magnus (2847)
Black	Caruana, Fabiano (2823)
Result	1/2-1/2
Opening	D37 – Queen’s Gambit Declined, Harrwitz Attack

7.2 Player Statistics

7.2.1 Carlsen, Magnus (White)

- Total moves: 40
- Accuracy: 98.4%
- Average centipawn loss: 3.2
- Best/Excellent moves: 27 / 5
- Good moves: 8
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

7.2.2 Caruana, Fabiano (Black)

- Total moves: 39
- Accuracy: 98.7%
- Average centipawn loss: 2.7

- Best/Excellent moves: 28 / 3
- Good moves: 8
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

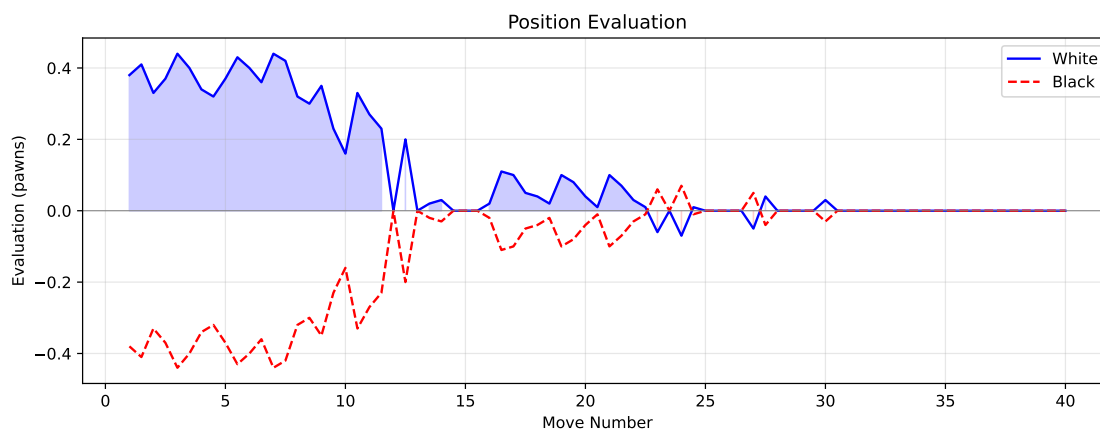
7.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

7.3.1 Position Evaluation Over Time

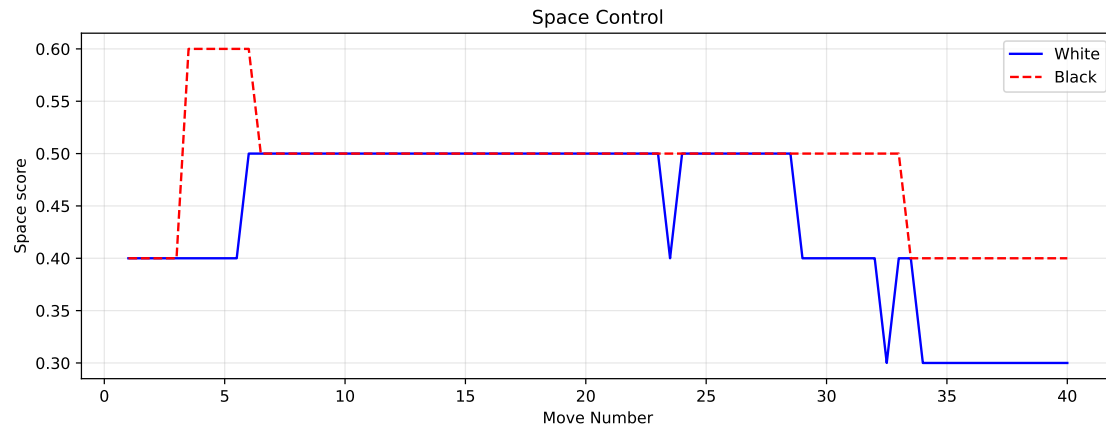
Game Character Classification. Evaluation range: minimum $m_1 = -0.07$, maximum $m_2 = +0.44$, spread $d = m_2 - m_1 = 0.51$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



7.3.2 Space Control

- White average: 0.44 (range: 0.30 to 0.50)
- Black average: 0.48 (range: 0.40 to 0.60)



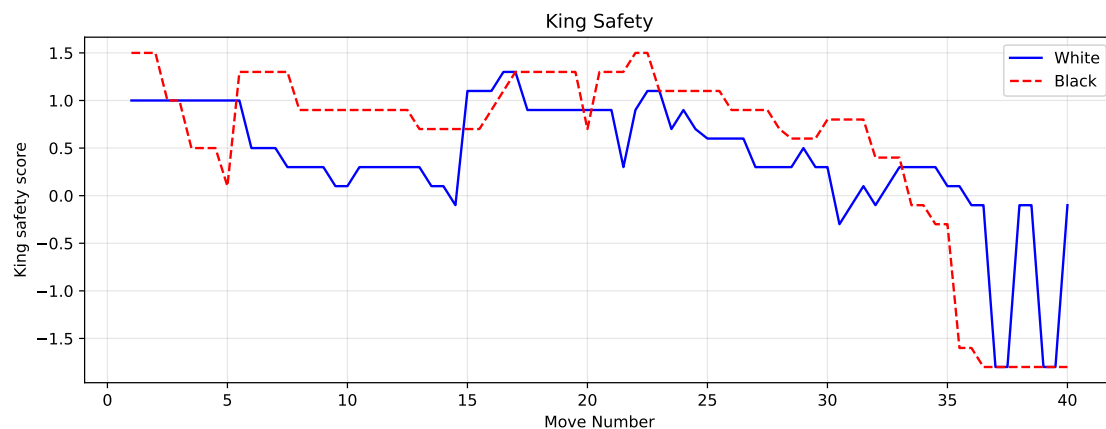
7.3.3 Mobility (Piece Activity)

- White average: 2.10 (range: 0.75 to 3.45)
- Black average: 1.96 (range: 0.70 to 3.15)



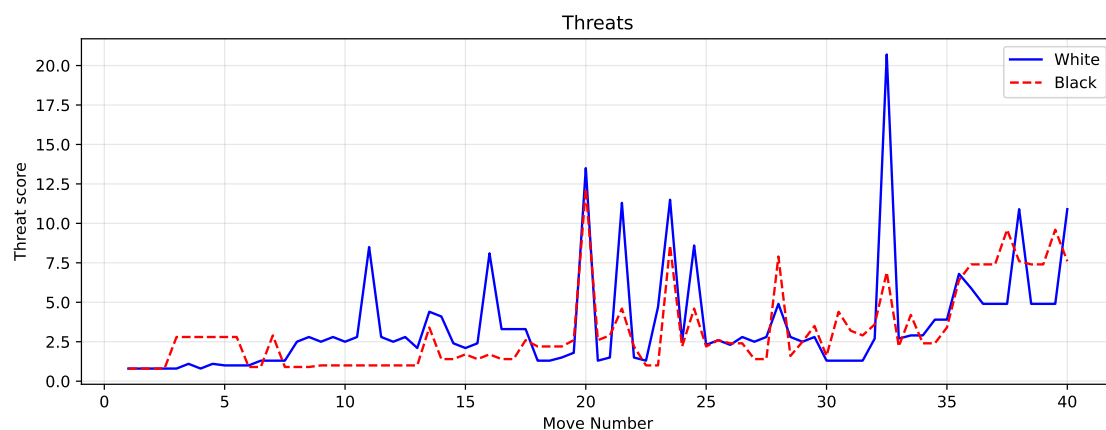
7.3.4 King Safety

- White average: 0.40
- Black average: 0.56



7.3.5 Threats

- White average: 3.58 (range: 0.80 to 20.70)
- Black average: 3.12 (range: 0.80 to 12.20)
- Average advantage: 0.45 (White maintained more threats)



7.4 Annotated Game

1. d4 Nf6 2. Nf3 d5 3. c4 e6 4. Nc3 Be7 5. Bf4 O-O 6. e3 c5 7. dxc5! Bxc5 8. Qc2 Nc6! 9. a3! Qa5! 10. Nd2 Qd8 11. Nb3 Bb6! 12. Be2 Qe7 13. Bg5 dxc4! 14. Nd2! Ne5! 15. O-O! Bd7! 16. Bf4! Ng6! 17. Bg3! Bc6! 18. Nxc4! Bc7! 19. Rfd1! Rfd8 20. Rxd8+! Rxd8! 21. Rd1 Rxd1+! 22. Qxd1! Nd5! 23. Qd4 Nxc3 24. Qxc3 Bxg3 25. hxc3! Qd7! 26. Bd3 b6 27. f3 Bb7 28. Bxg6 hxc6! 29. e4 Qc7! 30. e5 Qc5+ 31. Kh2! Ba6! 32. Nd6! Qxc3! 33. bxc3! f6 34. f4! Kf8! 35. Kg1! Ke7! 36. Kf2! Kd7! 37. Ke3! Bf1 38. Kf2! Ba6! 39. Ke3! Bf1! 40. Kf2

1/2-1/2

7.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

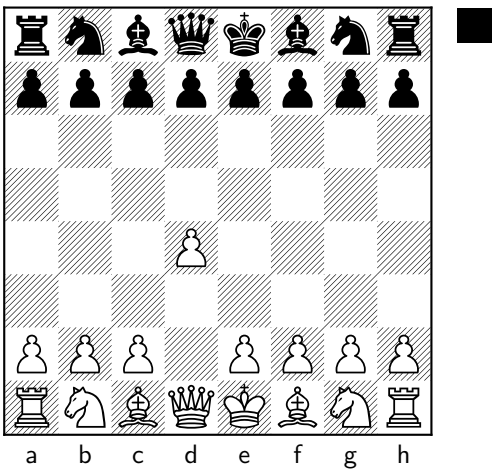
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 1. d4 ★

Biggest evaluation swing(s) (10cp)

Evaluation: +0.38



Metric	White	Black
Space	0.40	0.40
Mobility (MG)	3.80	3.30
King Safety	1.00	1.50
Threats	0.80	0.80

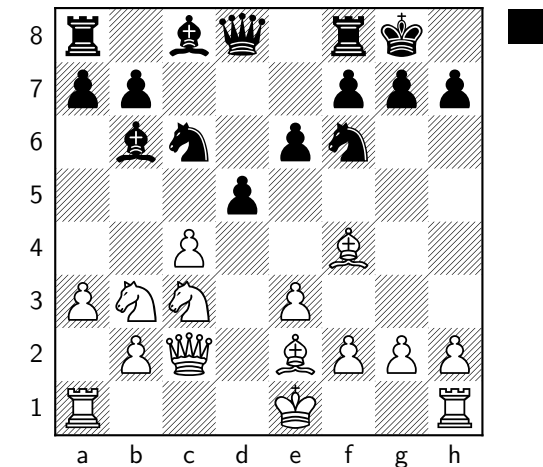
Instead of d4: e4, Nf3

Best continuation: Nf6 c4 e6

Position 2: After 12. Be2 ★

Biggest evaluation swing(s) (23cp)

Evaluation: +0.00



Metric	White	Black
Space	0.50	0.50
Mobility (MG)	5.70	4.80
King Safety	0.30	0.90
Threats	2.50	1.00

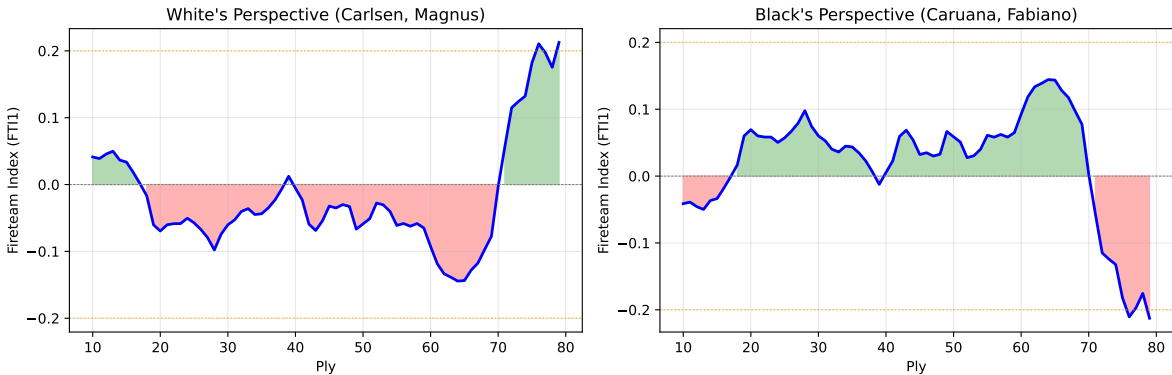
Instead of Be2: O-O-O, Rd1 Best continuation: dxc4 Bxc4 Bd7

7.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

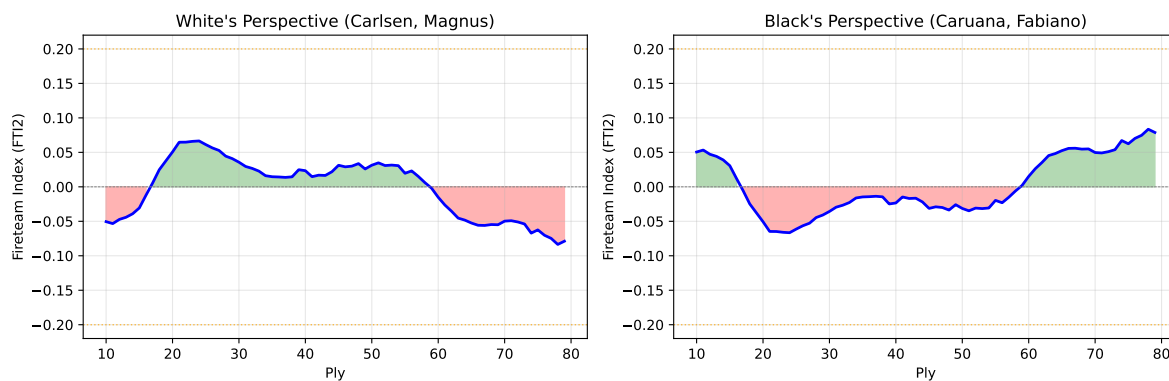
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	1 plies
Caruana, Fabiano	Per-ply	DRAW	3 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

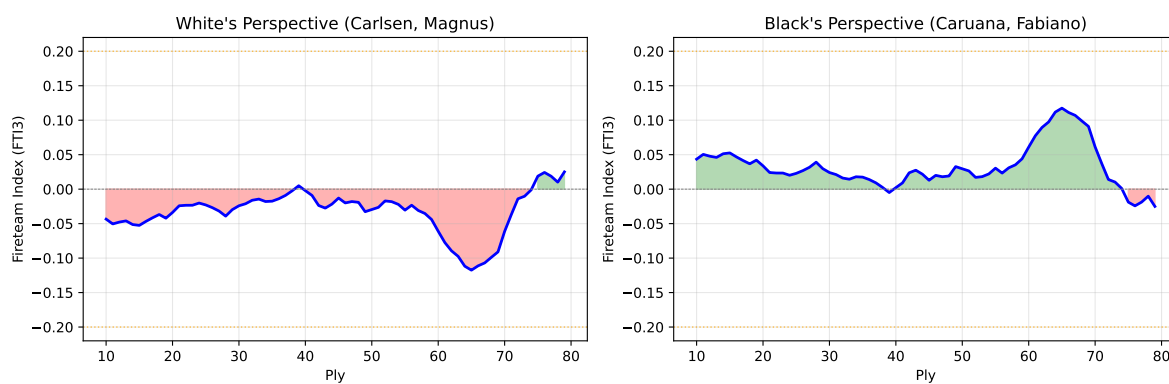


FTI2: Tactical (0.6, 0.1, 0, 0.3)

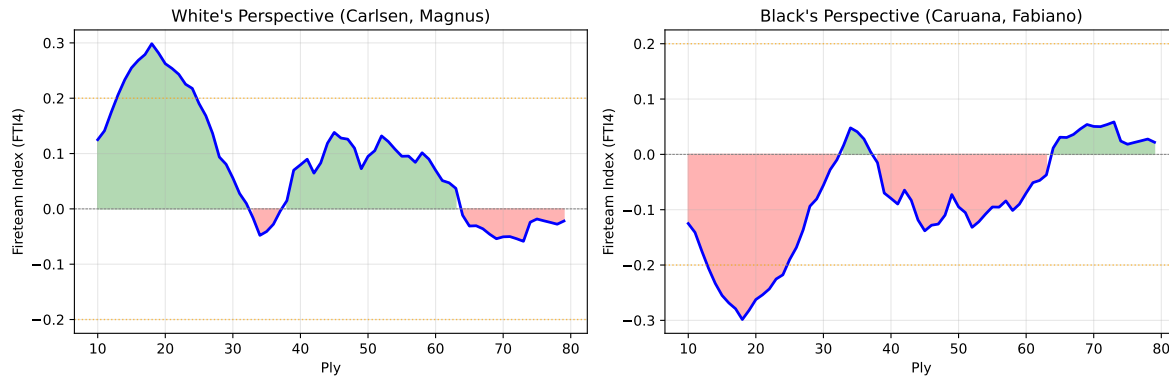
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI3: Strategic** (0.7, 0.1, 0.1, 0.1)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

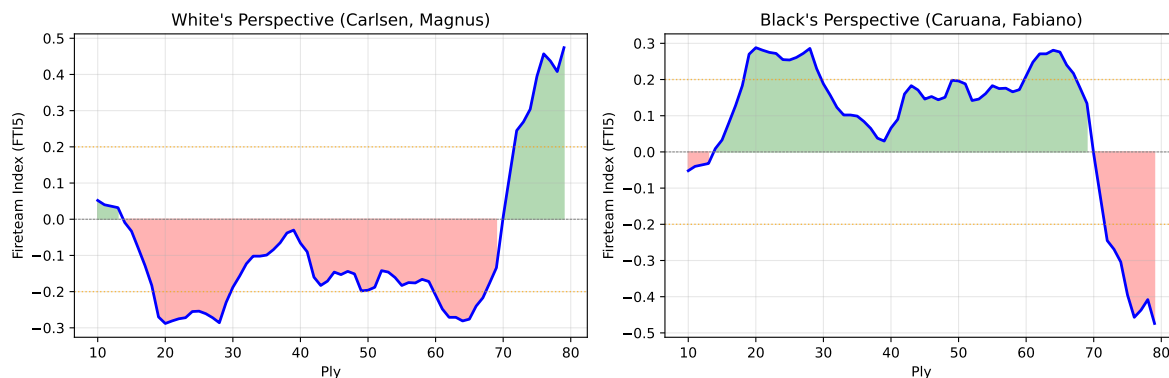
**FTI4: Dynamic** (0.2, 0.6, 0, 0.2)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	8 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



FTI5: Solid (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	8 plies
Caruana, Fabiano	Per-ply	DRAW	8 plies
Caruana, Fabiano	Windowed	WIN	11 plies



Interpretation. Neither player achieved sustained positional dominance according to FTI1. Both algorithms predicted DRAW for both sides, which matches the actual result. This suggests a balanced game where neither side established lasting control.

7.7 Analysis Information

- Engine: Stockfish 17.1

- Depth: 23
- Analysis time: 370.1 seconds

Chapter 8

Carlsen, Magnus vs Caruana, Fabiano

8.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.19
Round	8
White	Carlsen, Magnus (2847)
Black	Caruana, Fabiano (2823)
Result	1/2-1/2
Opening	B33 – Sicilian Defence, Sveshnikov Variation

8.2 Player Statistics

8.2.1 Carlsen, Magnus (White)

- Total moves: 38
- Accuracy: 96.3%
- Average centipawn loss: 7.5
- Best/Excellent moves: 25 / 4
- Good moves: 8
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 1

8.2.2 Caruana, Fabiano (Black)

- Total moves: 37
- Accuracy: 96.0%
- Average centipawn loss: 8.2

- Best/Excellent moves: 20 / 6
- Good moves: 8
- Inaccuracies: 2
- Mistakes: 1
- Blunders: 0

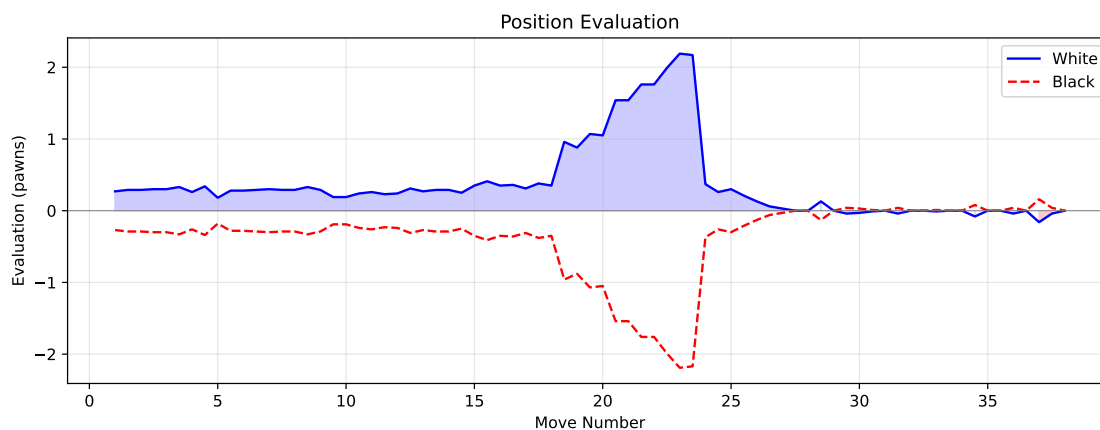
8.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

8.3.1 Position Evaluation Over Time

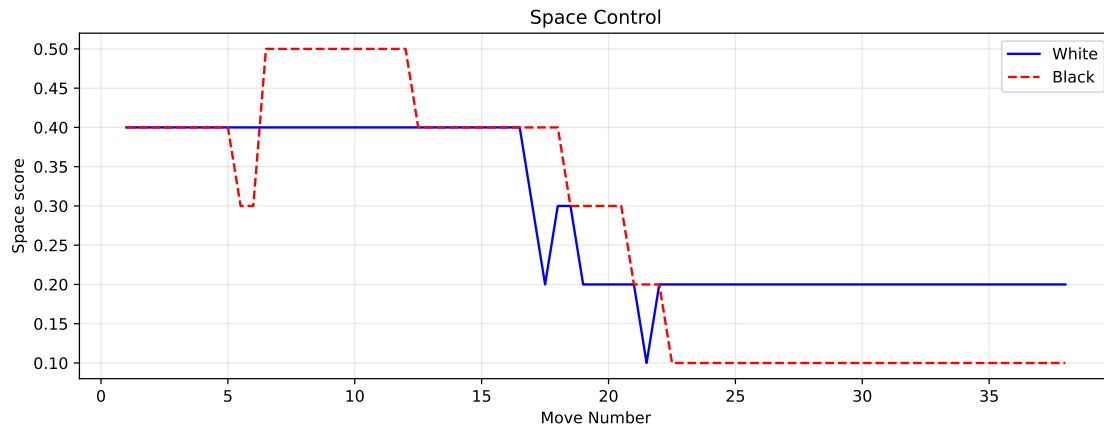
Game Character Classification. Evaluation range: minimum $m_1 = -0.16$, maximum $m_2 = +2.19$, spread $d = m_2 - m_1 = 2.35$.

This game is classified as **tense** (one-sided). A tense, one-sided game. White maintained the advantage throughout.



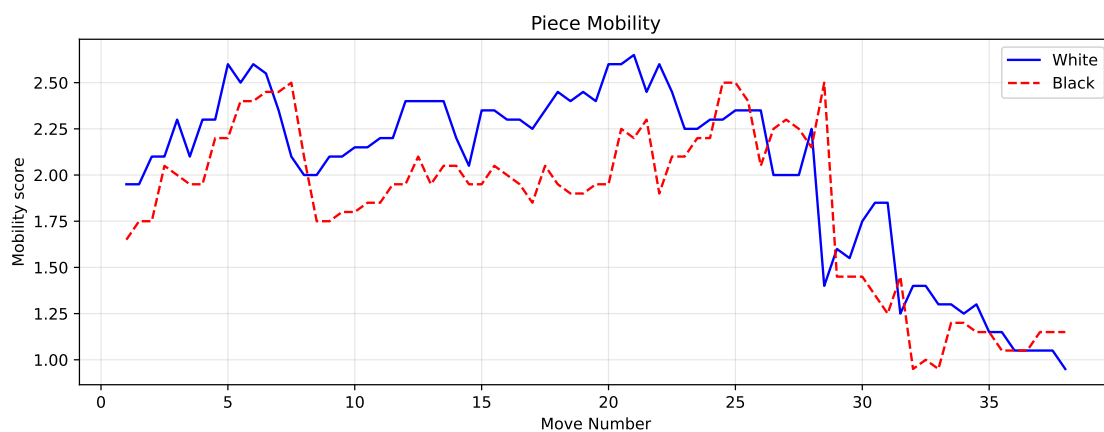
8.3.2 Space Control

- White average: 0.29 (range: 0.10 to 0.40)
- Black average: 0.27 (range: 0.10 to 0.50)



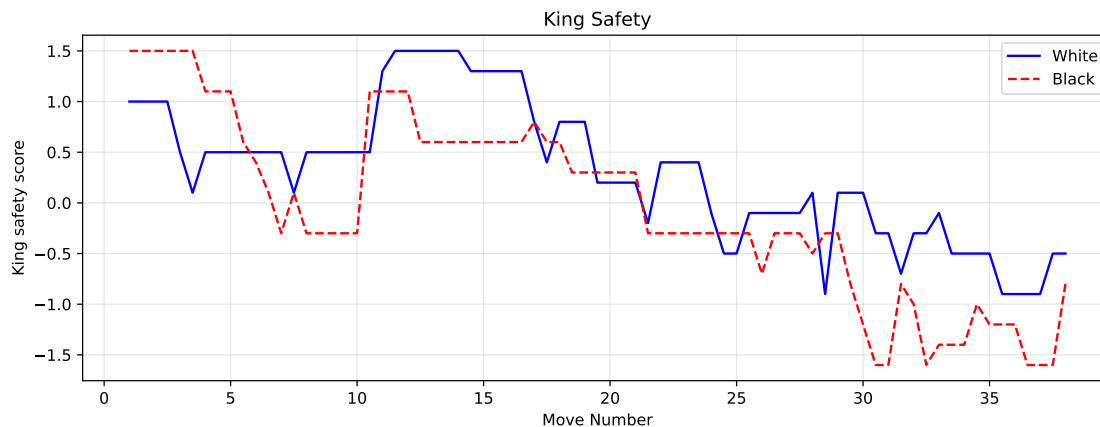
8.3.3 Mobility (Piece Activity)

- White average: 2.03 (range: 0.95 to 2.65)
- Black average: 1.85 (range: 0.95 to 2.50)



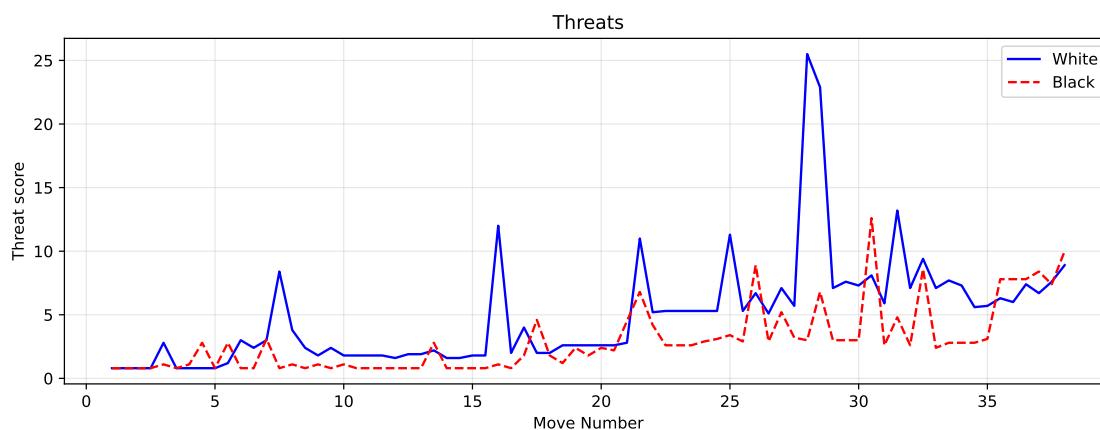
8.3.4 King Safety

- White average: 0.31
- Black average: -0.04



8.3.5 Threats

- White average: 4.95 (range: 0.80 to 25.50)
- Black average: 2.97 (range: 0.80 to 12.60)
- Average advantage: 1.98 (White maintained more threats)



8.4 Annotated Game

1. e4 c5 2. Nf3 Nc6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 e5 6. Ndb5 d6 7. Nd5! Nxd5! 8. exd5! Nb8 9. a4! Be7! 10. Be2! O-O 11. O-O Nd7! 12. Bd2! f5 13. a5! a6! 14. Na3! e4! 15. Nc4! Ne5 16. Nb6! Rb8 17. f4 exf3! 18. Bxf3! g5? *Black loses 69cp. Consider instead: Bf6, f4, Nd7.*

19. c4! f4 20. Bc3 Bf5?! 21. c5! Nxf3+?! 22. Qxf3 dxc5 23. Rad1 Bd6! 24. h3?? *White loses 168cp. Consider instead: Qh5, Nc4.*

24...Qe8! 25. Nc4! Qg6! 26. Nxd6! Qxd6! 27. h4 gxh4! 28. Qxf4! Qxf4 29. Rxf4 h5! 30. Re1! Bg4! 31. Rf6! Rxf6 32. Bxf6! Kf7! 33. Bxh4! Re8! 34. Rf1+ Kg8! 35. Rf6! Re2! 36. Rg6+! Kf8 37. d6 Rd2 38. Rg5!

1/2-1/2

8.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

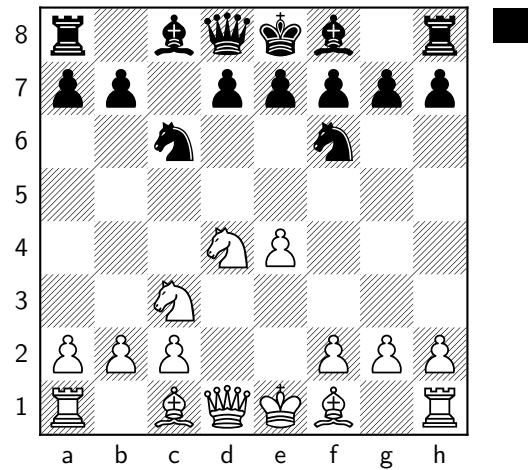
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 5. Nc3 ★

Biggest evaluation swing(s) (16cp)

Evaluation: +0.18



Metric	White	Black
Space	0.40	0.40
Mobility (MG)	5.20	4.40
King Safety	0.50	1.10
Threats	0.80	0.80

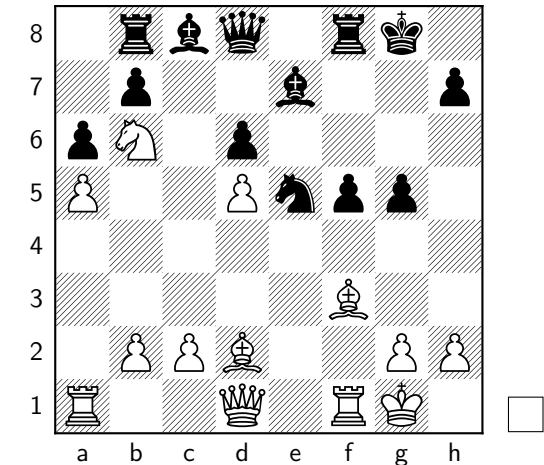
Instead of Nc3: Nxc6, Qd3

Best continuation: e5 Ndb5 d6

Position 2: After 18...g5 ★

Biggest evaluation swing(s) (61cp)

Evaluation: +0.96



Metric	White	Black
Space	0.30	0.30
Mobility (MG)	4.80	3.80
King Safety	0.80	0.30
Threats	2.60	1.20

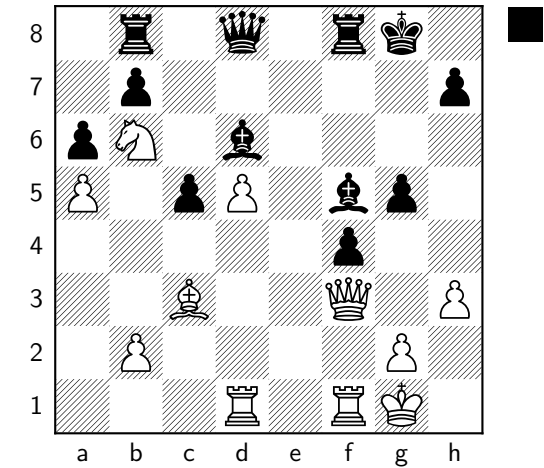
Instead of g5: Bf6, f4, Nd7

Best continuation: c4 Bf6 Bb4

Position 3: After 24. h3

White blunders (-180cp)

Evaluation: +0.37



Metric	White	Black
Space	0.20	0.10
Mobility (MG)	4.60	4.40
King Safety	-0.10	-0.30
Threats	5.30	2.90

Instead of h3: Qh5, Nc4

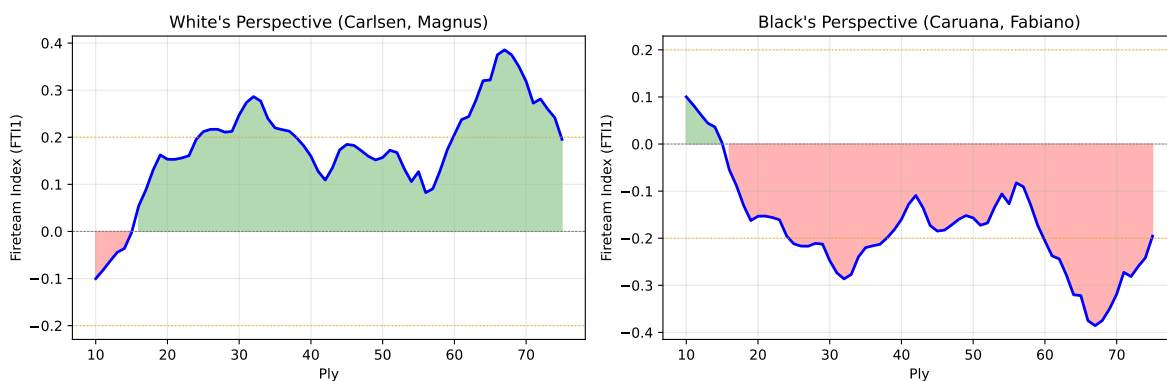
Best continuation: Qe8 Nc4 Qg6

8.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

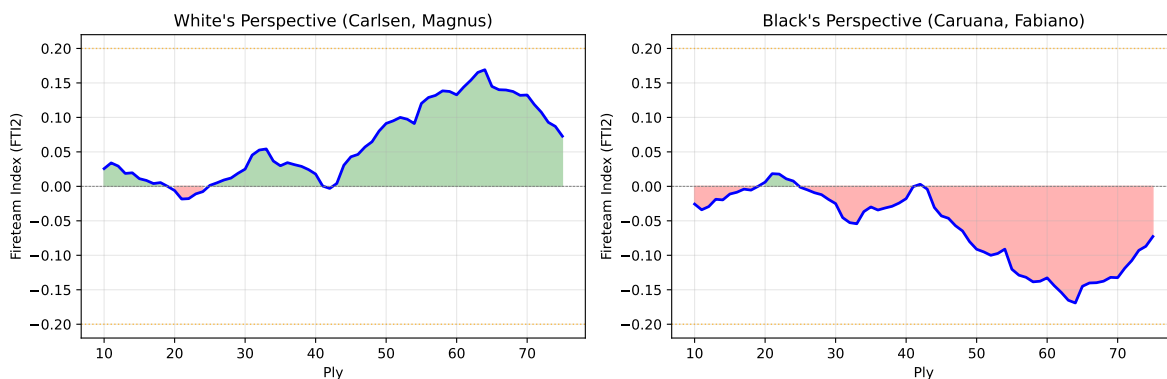
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	9 plies
Carlsen, Magnus	Windowed	WIN	15 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



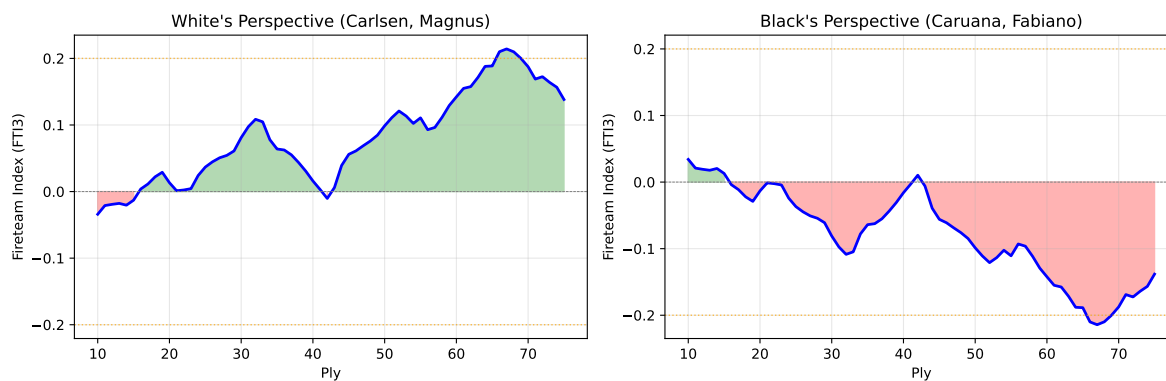
FTI2: Tactical (0.6, 0.1, 0, 0.3)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

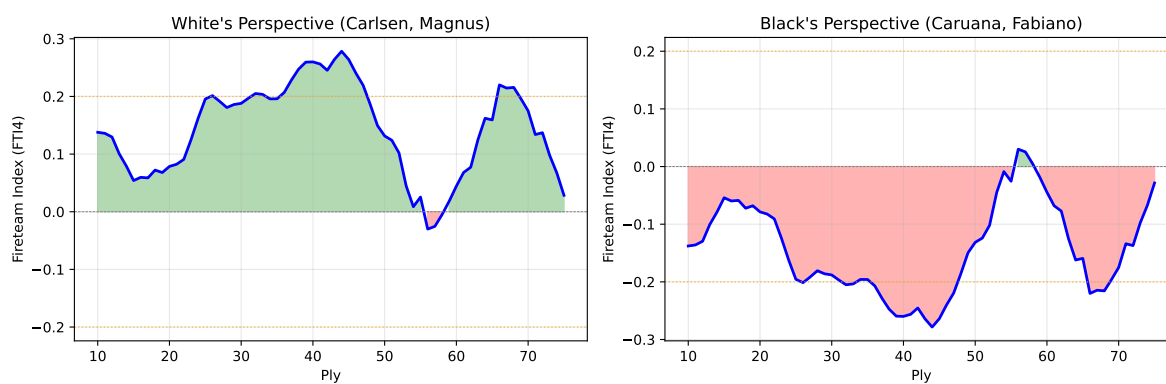


FTI3: Strategic (0.7, 0.1, 0.1, 0.1)

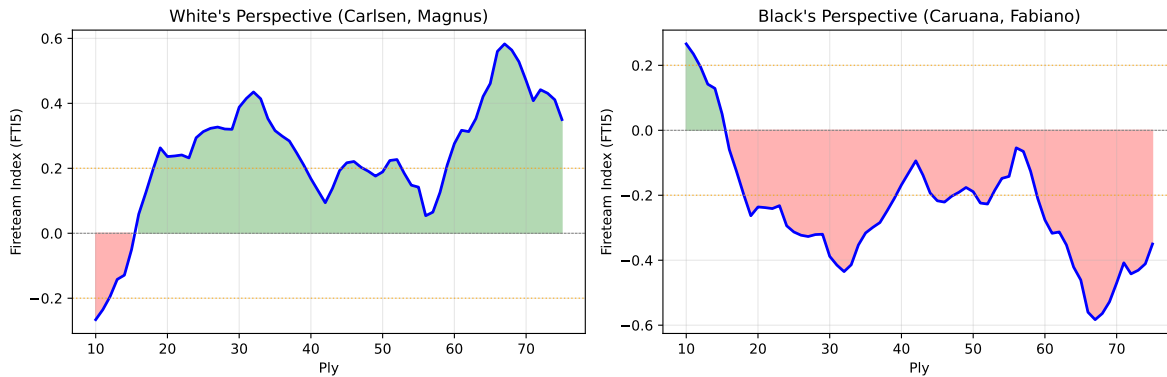
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	4 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI4: Dynamic** (0.2, 0.6, 0, 0.2)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	5 plies
Carlsen, Magnus	Windowed	WIN	12 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI5: Solid** (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	11 plies
Carlsen, Magnus	Windowed	WIN	21 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



Interpretation. White (Carlsen, Magnus) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting Black successfully defended or White failed to convert the advantage.

8.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 519.6 seconds

Chapter 9

Carlsen, Magnus vs Caruana, Fabiano

9.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.21
Round	9
White	Carlsen, Magnus (2847)
Black	Caruana, Fabiano (2823)
Result	1/2-1/2
Opening	A29 – English Opening, Four Knights, Kingside Fianchetto

9.2 Player Statistics

9.2.1 Carlsen, Magnus (White)

- Total moves: 56
- Accuracy: 98.0%
- Average centipawn loss: 3.9
- Best/Excellent moves: 32 / 15
- Good moves: 8
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 0

9.2.2 Caruana, Fabiano (Black)

- Total moves: 56
- Accuracy: 98.3%
- Average centipawn loss: 3.5

- Best/Excellent moves: 35 / 11
- Good moves: 10
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

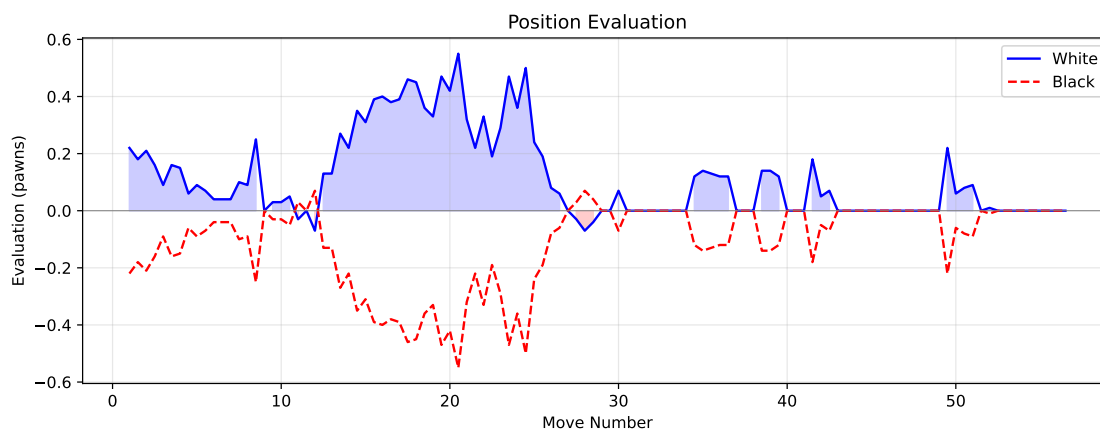
9.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

9.3.1 Position Evaluation Over Time

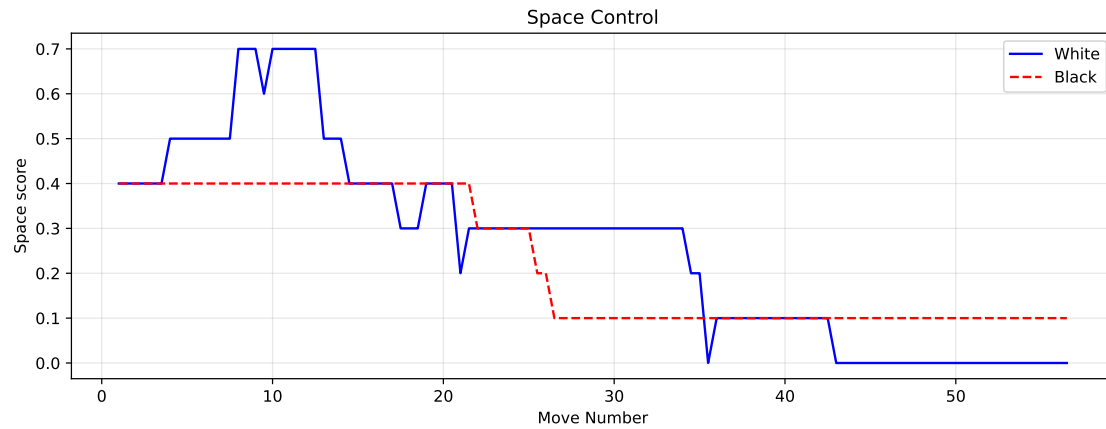
Game Character Classification. Evaluation range: minimum $m_1 = -0.07$, maximum $m_2 = +0.55$, spread $d = m_2 - m_1 = 0.62$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



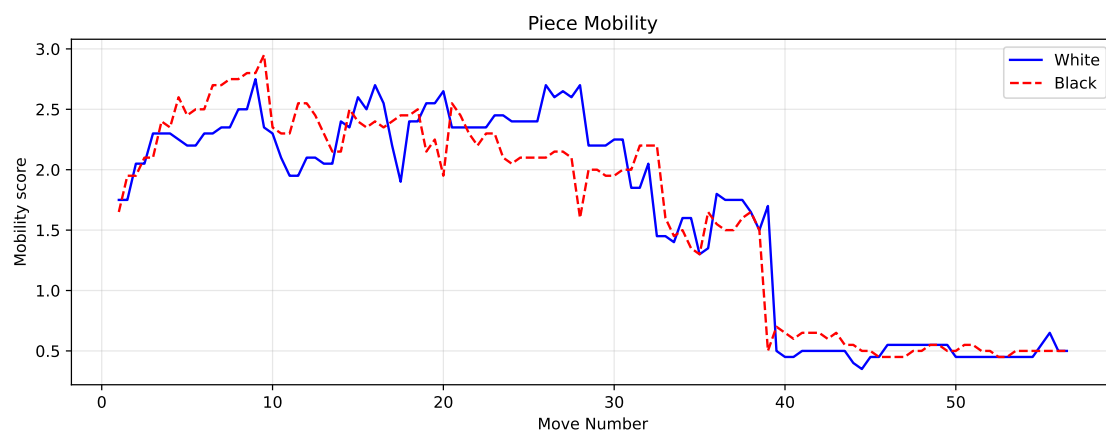
9.3.2 Space Control

- White average: 0.26 (range: 0.00 to 0.70)
- Black average: 0.23 (range: 0.10 to 0.40)



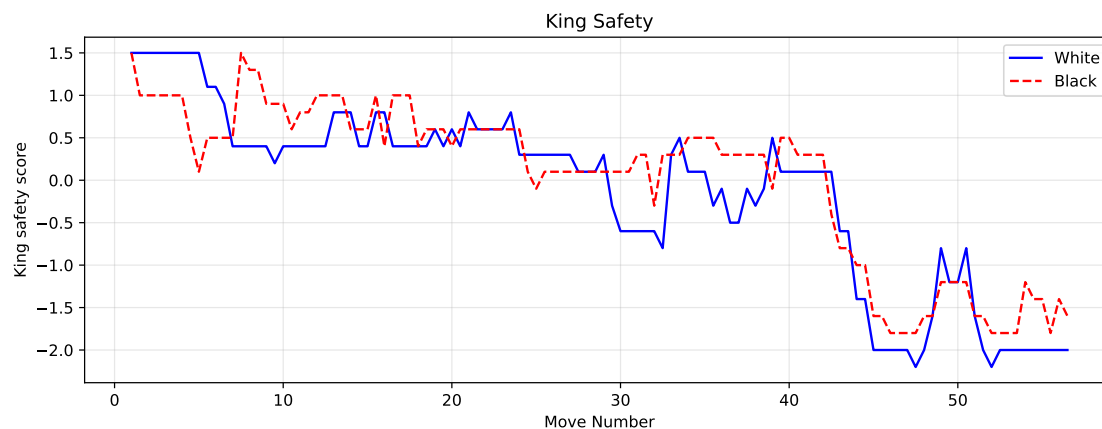
9.3.3 Mobility (Piece Activity)

- White average: 1.65 (range: 0.35 to 2.75)
- Black average: 1.64 (range: 0.45 to 2.95)



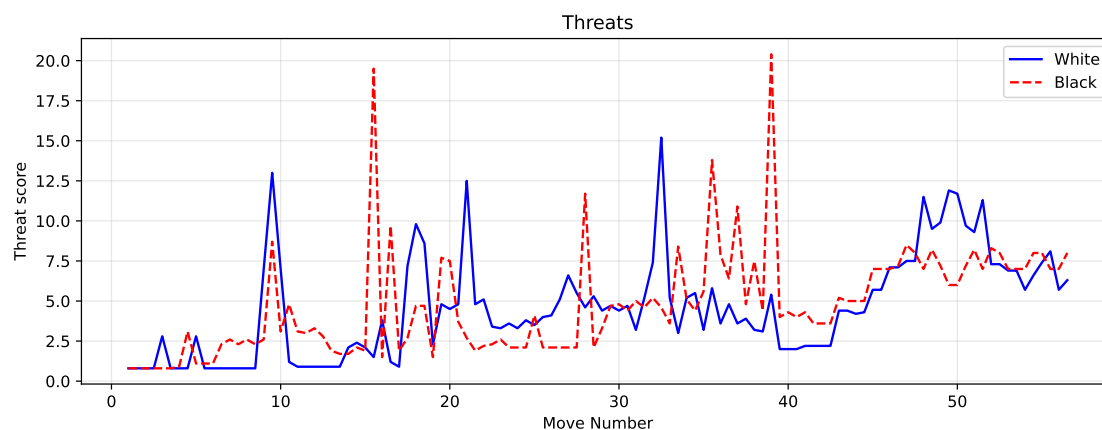
9.3.4 King Safety

- White average: -0.13
- Black average: 0.03



9.3.5 Threats

- White average: 4.60 (range: 0.80 to 15.20)
- Black average: 4.83 (range: 0.80 to 20.40)
- Average advantage: -0.23 (Black maintained more threats)



9.4 Annotated Game

1. c4 e5 2. Nc3 Nf6 3. Nf3 Nc6 4. g3 d5 5. cxd5 Nxd5 6. Bg2 Bc5 7. O-O! O-O 8. d3 Re8 9. Bg5 Nxc3! 10. bxc3! f6! 11. Bc1 Be6! 12. Bb2 Bb6 13. d4! Bd5 14. Qc2 exd4 15. cxd4! Be4! 16. Qb3+ Bd5 17. Qd1! Bxf3 18. Qb3+ Kh8! 19. Bxf3 Nxd4! 20. Bxd4! Qxd4 21. e3 Qe5! 22. Bxb7! Rad8 23. Rad1 Qe7 24. h4 g6! 25. h5?! gxh5! 26. Qc4 f5! 27. Bf3 h4! 28. Rxd8 Rxd8! 29. gxh4! Rg8+ 30. Kh1! Qf6 31. Qf4 Bc5! 32. Rg1! Rxg1+ 33. Kxg1! Bd6! 34. Qa4 f4 35. Qxa7! fxe3 36. Qxe3 Qxh4 37. a4 Qf6 38. Bd1! Qe5 39. Qxe5+! Bxe5! 40. a5 Kg7 41. a6! Bd4 42. Kg2 Kf6 43. f4! Bb6 44. Kf3! h6! 45. Ke4! Ba7 46. Bg4 Bg1! 47. Kd5 Bb6! 48. Kc6 Be3 49. Kb7 Bb6! 50. Bh3 Be3 51. Kc6 Bb6 52. Kd5! Ba7 53. Ke4! Bb6 54. Bf1 Ke6 55. Bc4+ Kf6! 56. Bd3 Ke6

1/2-1/2

9.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

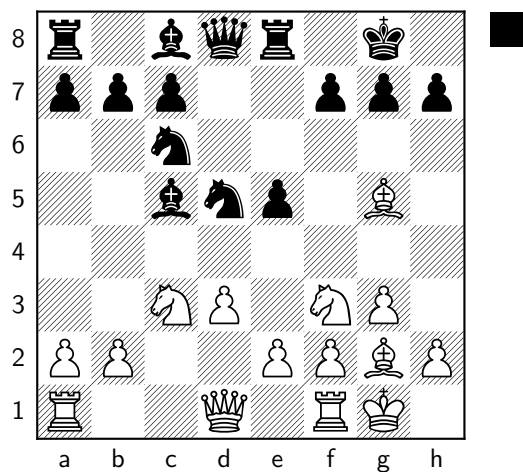
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 9. Bg5 ★

Biggest evaluation swing(s) (25cp)

Evaluation: +0.00



Metric	White	Black
Space	0.70	0.40
Mobility (MG)	5.50	5.60
King Safety	0.40	0.90
Threats	7.00	2.60

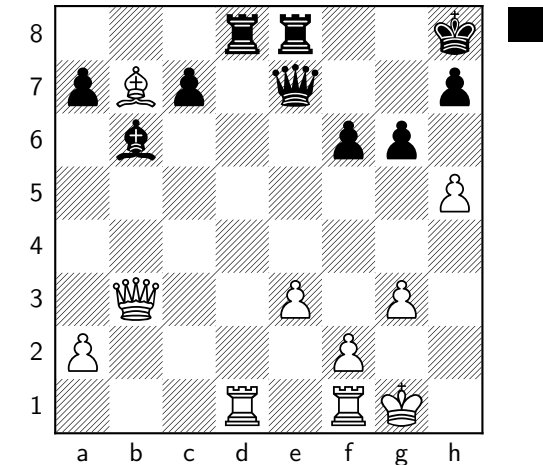
Instead of Bg5: Nxd5, Bd2

Best continuation: Nxc3 bxc3 f6

Position 2: After 25. h5 ★

Biggest evaluation swing(s) (26cp)

Evaluation: +0.24



Metric	White	Black
Space	0.30	0.30
Mobility (MG)	4.80	4.20
King Safety	0.30	-0.10
Threats	3.50	4.10

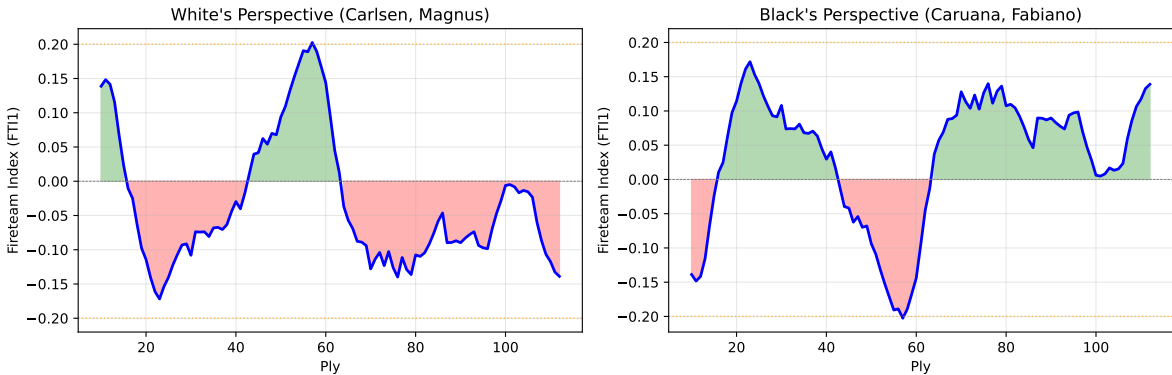
Instead of h5: Qc2, Bc6 Best continuation: gxh5 Bc6 Rg8

9.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

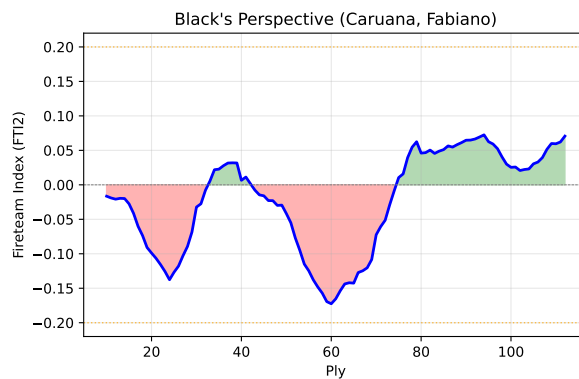
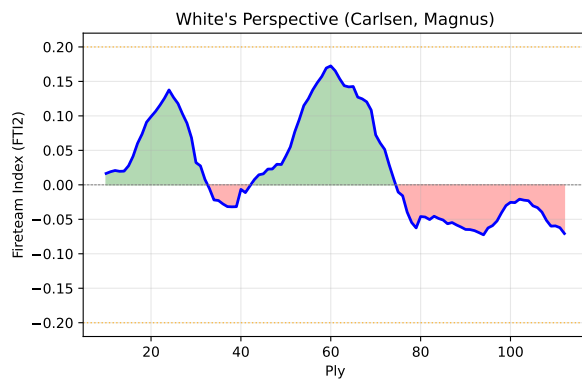
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	5 plies
Carlsen, Magnus	Windowed	DRAW	1 plies
Caruana, Fabiano	Per-ply	DRAW	3 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

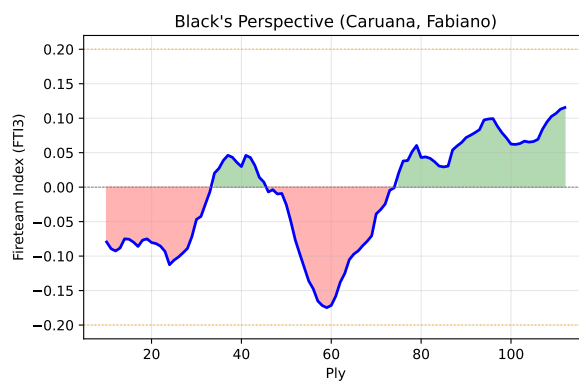
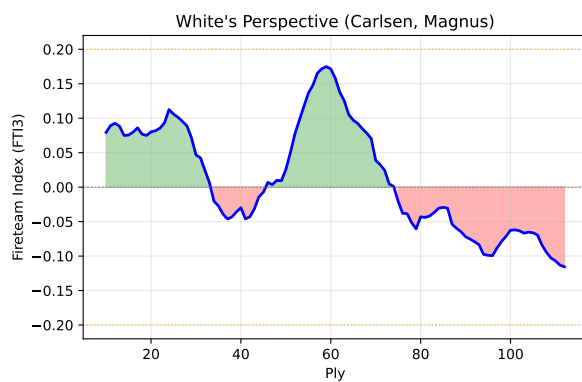


FTI2: Tactical (0.6, 0.1, 0, 0.3)

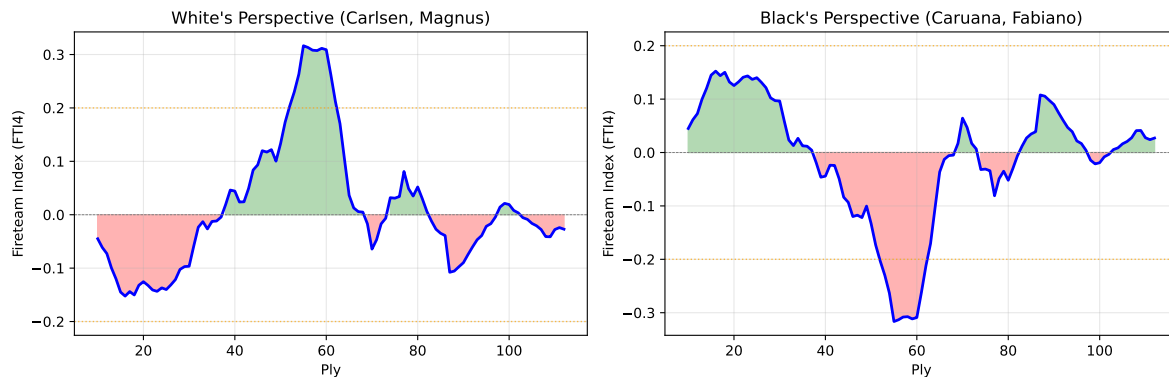
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI3: Strategic** (0.7, 0.1, 0.1, 0.1)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	4 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

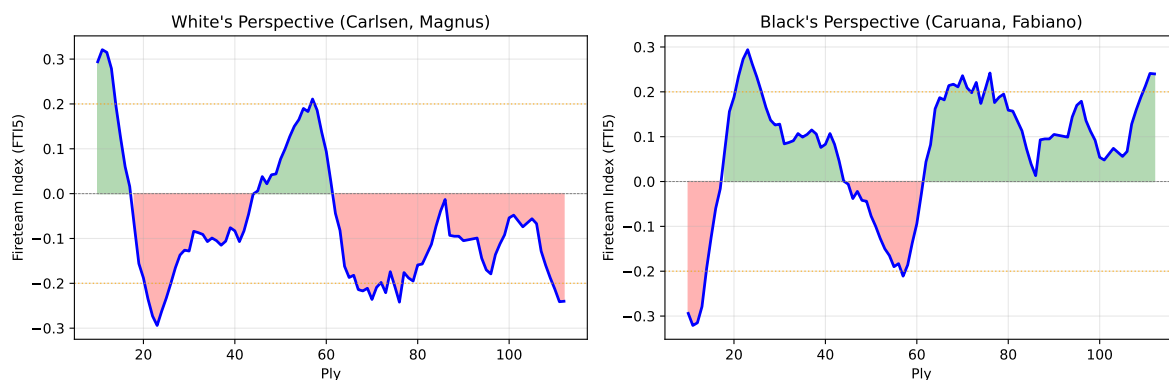
**FTI4: Dynamic** (0.2, 0.6, 0, 0.2)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	6 plies
Carlsen, Magnus	Windowed	WIN	11 plies
Caruana, Fabiano	Per-ply	DRAW	2 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



FTI5: Solid (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	3 plies
Carlsen, Magnus	Windowed	DRAW	1 plies
Caruana, Fabiano	Per-ply	DRAW	4 plies
Caruana, Fabiano	Windowed	DRAW	5 plies



Interpretation. Neither player achieved sustained positional dominance according to FTI1. Both algorithms predicted DRAW for both sides, which matches the actual result. This suggests a balanced game where neither side established lasting control.

9.7 Analysis Information

- Engine: Stockfish 17.1

- Depth: 23
- Analysis time: 435.8 seconds

Chapter 10

Carlsen, Magnus vs Caruana, Fabiano

10.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.22
Round	10
White	Carlsen, Magnus (2847)
Black	Caruana, Fabiano (2823)
Result	1/2-1/2
Opening	B33 – Sicilian Defence, Sveshnikov Variation

10.2 Player Statistics

10.2.1 Carlsen, Magnus (White)

- Total moves: 54
- Accuracy: 96.3%
- Average centipawn loss: 7.6
- Best/Excellent moves: 33 / 9
- Good moves: 9
- Inaccuracies: 2
- Mistakes: 0
- Blunders: 1

10.2.2 Caruana, Fabiano (Black)

- Total moves: 53
- Accuracy: 96.0%
- Average centipawn loss: 8.2

- Best/Excellent moves: 26 / 10
- Good moves: 15
- Inaccuracies: 1
- Mistakes: 1
- Blunders: 0

10.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

10.3.1 Position Evaluation Over Time

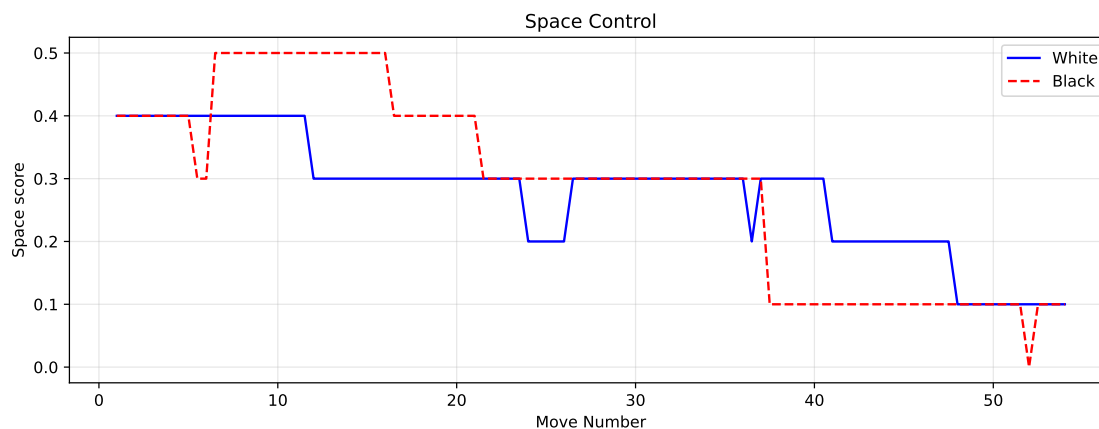
Game Character Classification. Evaluation range: minimum $m_1 = -0.60$, maximum $m_2 = +0.91$, spread $d = m_2 - m_1 = 1.51$.

This game is classified as **tense** (normal). A tense game with moderate swings.



10.3.2 Space Control

- White average: 0.28 (range: 0.10 to 0.40)
- Black average: 0.29 (range: 0.00 to 0.50)



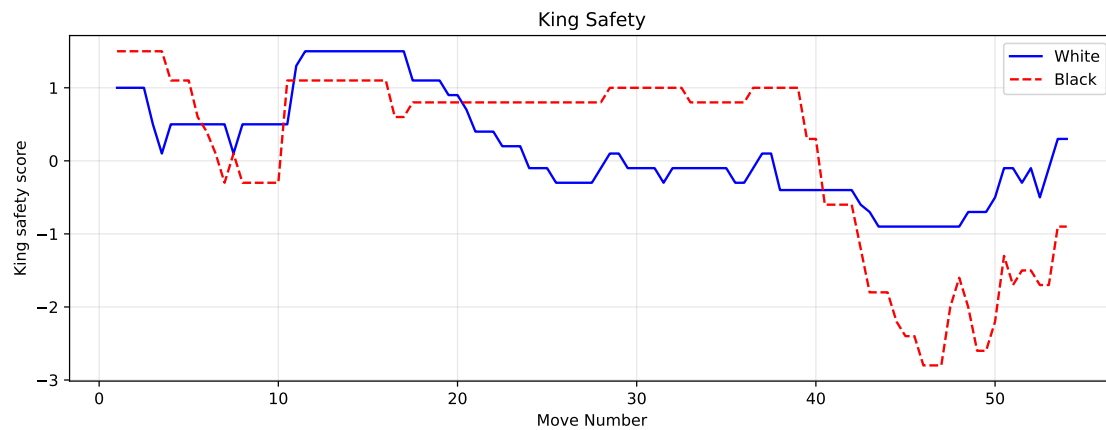
10.3.3 Mobility (Piece Activity)

- White average: 1.75 (range: 0.85 to 2.60)
- Black average: 1.79 (range: 0.75 to 2.60)



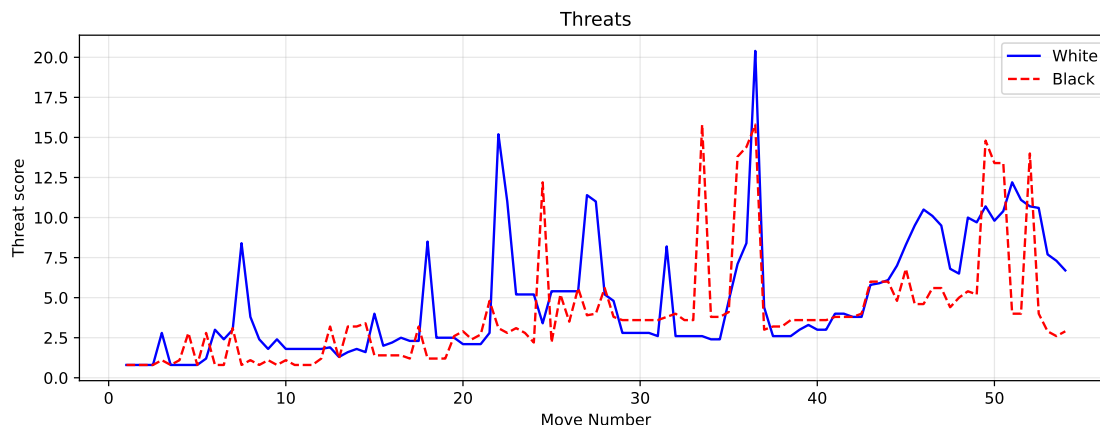
10.3.4 King Safety

- White average: 0.17
- Black average: 0.15



10.3.5 Threats

- White average: 4.82 (range: 0.80 to 20.40)
- Black average: 3.95 (range: 0.80 to 15.80)
- Average advantage: 0.86 (White maintained more threats)



10.4 Annotated Game

1. e4 c5 2. Nf3 Nc6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 e5 6. Ndb5 d6 7. Nd5! Nxd5! 8. exd5! Nb8 9. a4! Be7 10. Be2 O-O! 11. O-O Nd7! 12. b4! a6! 13. Na3! a5?! 14. bxa5 Rxa5! 15. Nc4! Ra8! 16. Be3! f5 17. a5 f4 18. Bb6! Qe8 19. Ra3?! Qg6! 20. Bc7 e4! 21. Kh1! b5 22. Nb6! Nxb6 23. Bxb6 Qg5? *Black loses 106cp. Consider instead: b4, Bd7, Bf6.*

24. g3?? *White loses 145cp. Consider instead: Bxb5.*

24...b4 25. Rb3! Bh3 26. Rg1! f3! 27. Bf1! Bxf1! 28. Qxf1! Qxd5 29. Rxb4! Qe6! 30. Rb5 Bd8 31. Qe1! Bxb6! 32. axb6! Rab8! 33. Qe3 Qc4 34. Rb2! Rb7 35. Rd1?! Qe2 36. Re1 Qxe3! 37. Rxe3! d5 38. h4! Rc8! 39. Ra3 Kf7 40. Kh2 Ke6! 41. g4! Rc6 42. Ra6! Ke5 43. Kg3! h6! 44. h5 Kd4 45. Rb5! Rd6! 46. Ra4+! Ke5 47. Rab4 Ke6! 48. c4 dxc4! 49. Rxc4 Rdx6! 50. Rxe4+! Kf7! 51. Rf5+ Rf6! 52. Rxf6+ Kxf6 53. Kxf3! Kf7 54. Kg3

1/2-1/2

10.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

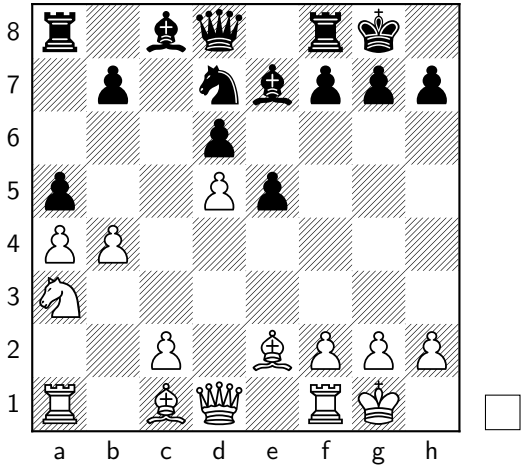
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 13...a5 ★

Biggest evaluation swing(s) (44cp)

Evaluation: +0.67



Metric	White	Black
Space	0.30	0.50
Mobility (MG)	4.20	3.70
King Safety	1.50	1.10
Threats	1.60	3.20

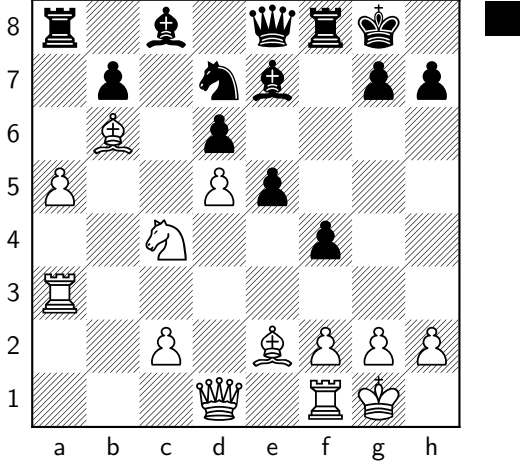
Instead of a5: b6, f5, e4

Best continuation: bxa5 Qxa5 Nc4

Position 2: After 19. Ra3 ★

Biggest evaluation swing(s) (82cp)

Evaluation: -0.02



Metric	White	Black
Space	0.30	0.40
Mobility (MG)	4.70	4.00
King Safety	1.10	0.80
Threats	2.50	1.20

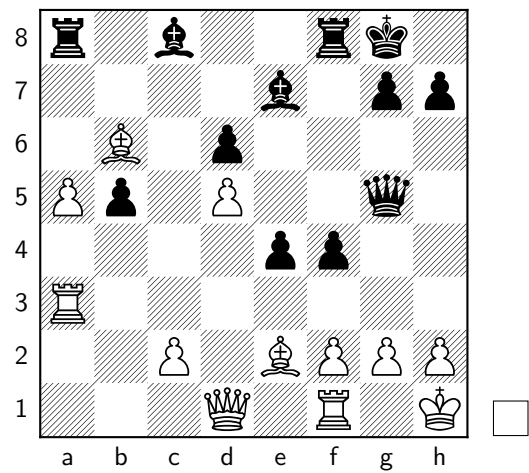
Instead of Ra3: Re1, Bh5, Bc7

Best continuation: Qg6 Kh1 Bd8

Position 3: After 23...Qg5

Critical moment

Evaluation: +0.91



Metric	White	Black
Space	0.30	0.30
Mobility (MG)	3.70	4.30
King Safety	0.20	0.80
Threats	5.20	2.80

Instead of Qg5: b4, Bd7, Bf6

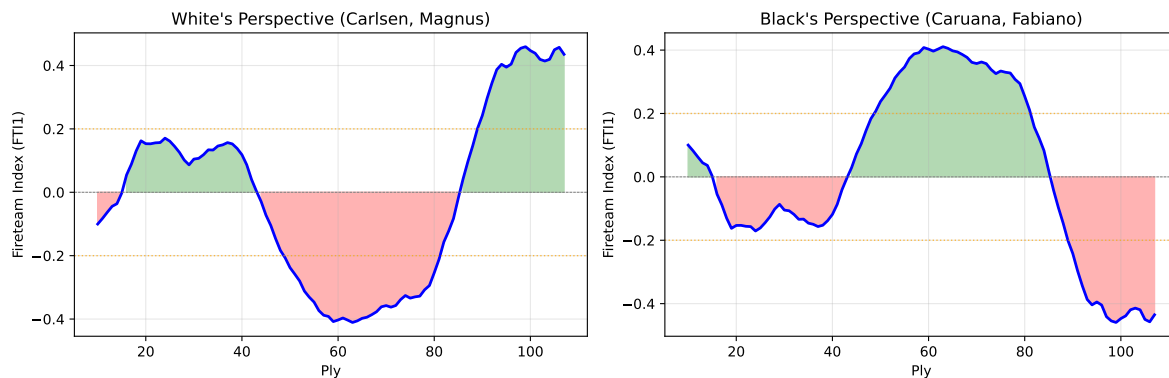
Best continuation: Bxb5 Rf6 Re1

10.6 Fireteam Index Prediction

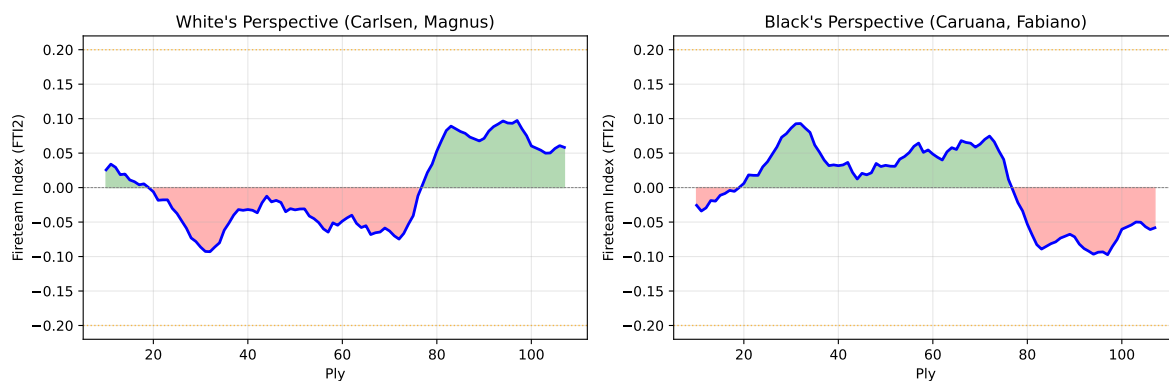
This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

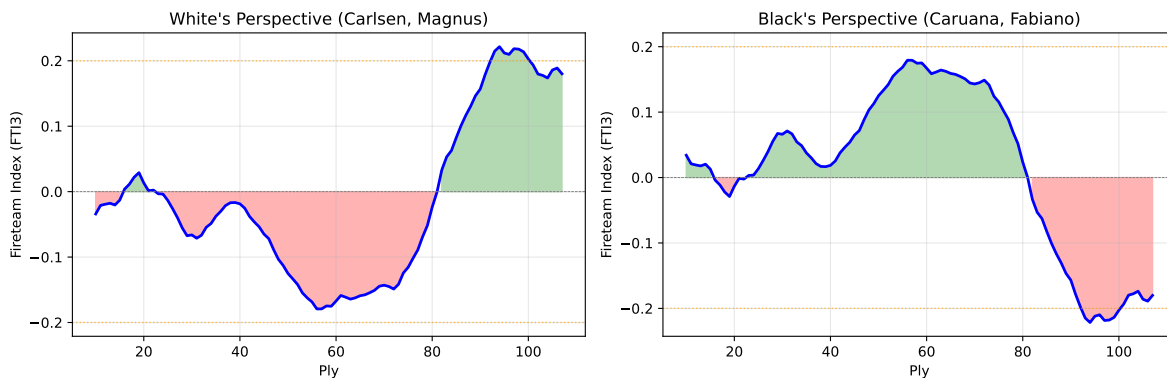
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	12 plies
Carlsen, Magnus	Windowed	WIN	19 plies
Caruana, Fabiano	Per-ply	WIN	31 plies
Caruana, Fabiano	Windowed	WIN	33 plies

**FTI2: Tactical** (0.6, 0.1, 0, 0.3)

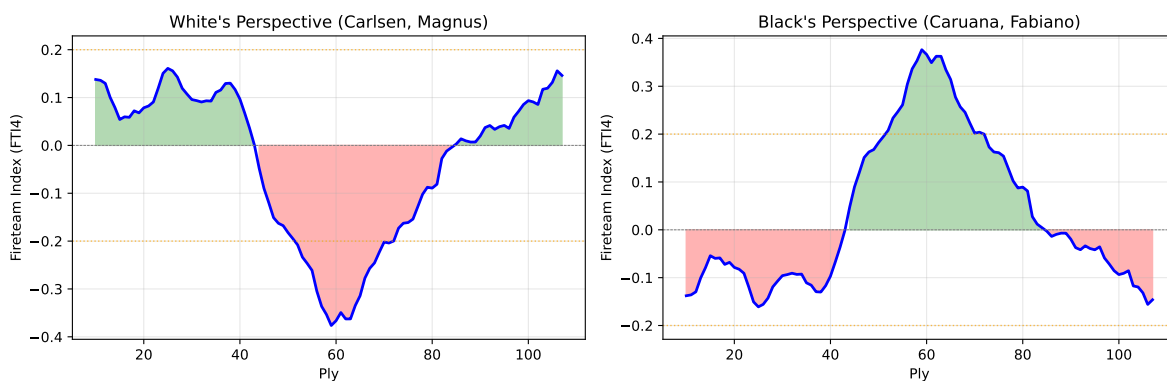
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI3: Strategic** (0.7, 0.1, 0.1, 0.1)

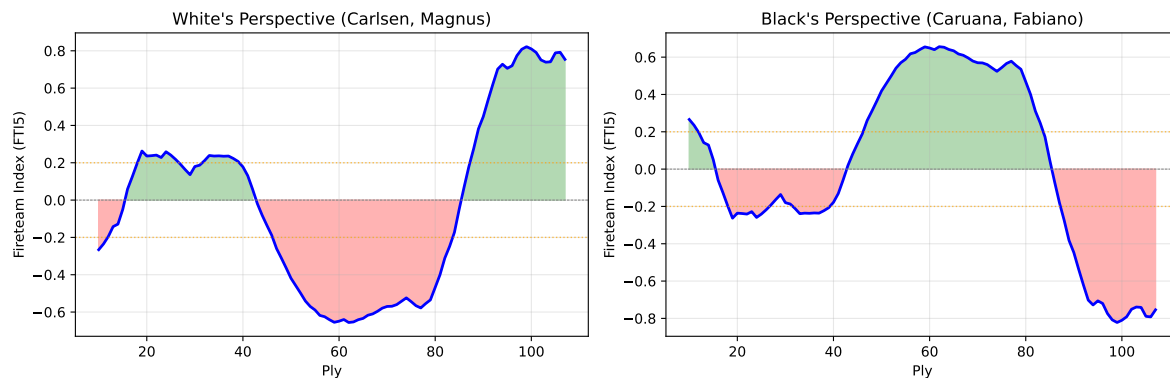
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	6 plies
Carlsen, Magnus	Windowed	DRAW	8 plies
Caruana, Fabiano	Per-ply	DRAW	2 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI4: Dynamic (0.2, 0.6, 0, 0.2)**

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	WIN	10 plies
Caruana, Fabiano	Windowed	WIN	20 plies

**FTI5: Solid (0.3, 0.2, 0.5, 0)**

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	24 plies
Carlsen, Magnus	Windowed	WIN	20 plies
Caruana, Fabiano	Per-ply	WIN	39 plies
Caruana, Fabiano	Windowed	WIN	37 plies



Interpretation. Both players achieved sustained positional dominance at different points in the game. The FTI1 Index predicted wins for both sides, reflecting a hard-fought battle where momentum shifted. The draw is a reasonable outcome given the mutual winning chances.

10.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 612.0 seconds

Chapter 11

Carlsen, Magnus vs Caruana, Fabiano

11.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.24
Round	11
White	Carlsen, Magnus (2847)
Black	Caruana, Fabiano (2823)
Result	1/2-1/2
Opening	C42 – Petrov’s Defence, Classical Variation

11.2 Player Statistics

11.2.1 Carlsen, Magnus (White)

- Total moves: 55
- Accuracy: 98.1%
- Average centipawn loss: 3.8
- Best/Excellent moves: 36 / 11
- Good moves: 8
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

11.2.2 Caruana, Fabiano (Black)

- Total moves: 55
- Accuracy: 98.4%
- Average centipawn loss: 3.1

- Best/Excellent moves: 38 / 7
- Good moves: 10
- Inaccuracies: 0
- Mistakes: 0
- Blunders: 0

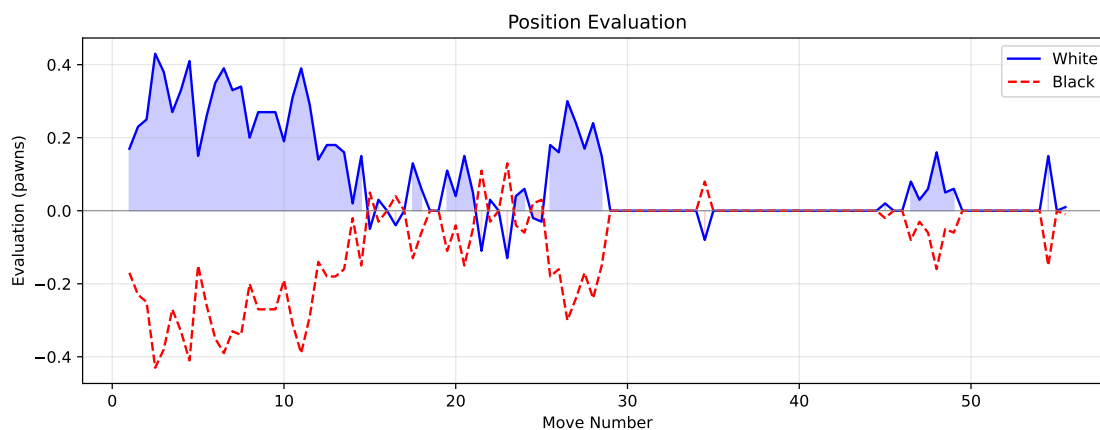
11.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

11.3.1 Position Evaluation Over Time

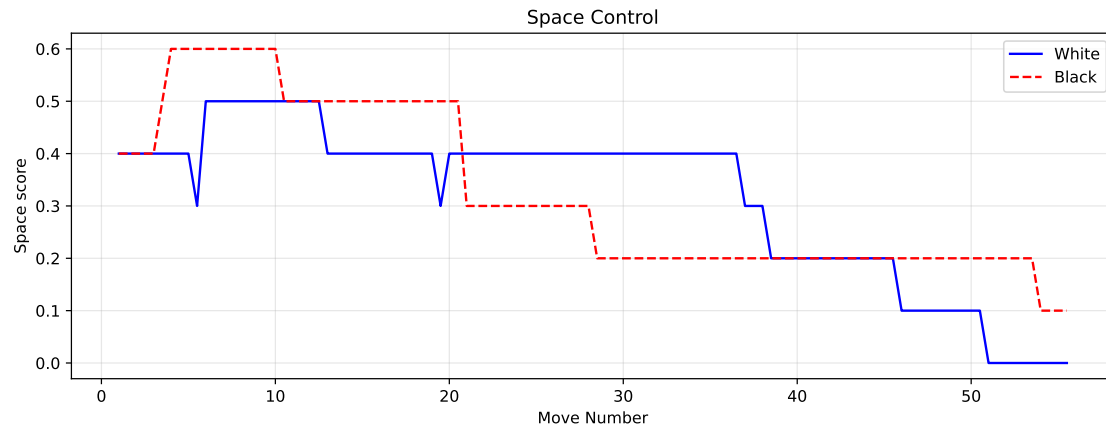
Game Character Classification. Evaluation range: minimum $m_1 = -0.13$, maximum $m_2 = +0.43$, spread $d = m_2 - m_1 = 0.56$.

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



11.3.2 Space Control

- White average: 0.32 (range: 0.00 to 0.50)
- Black average: 0.33 (range: 0.10 to 0.60)



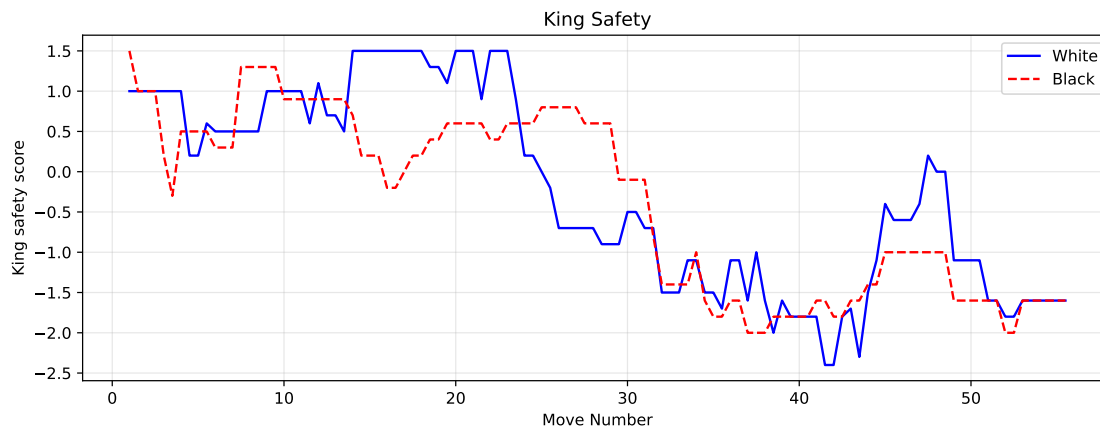
11.3.3 Mobility (Piece Activity)

- White average: 1.30 (range: 0.55 to 2.65)
- Black average: 1.24 (range: 0.40 to 2.60)



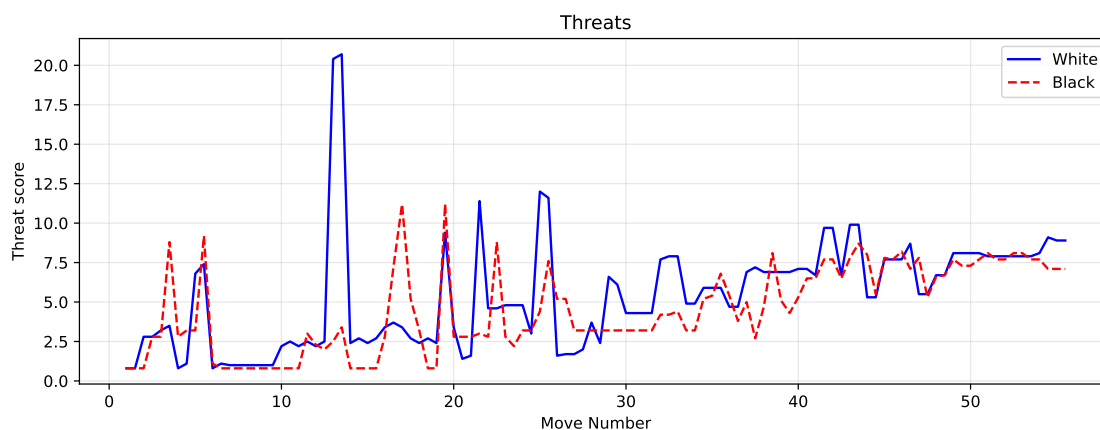
11.3.4 King Safety

- White average: -0.23
- Black average: -0.36



11.3.5 Threats

- White average: 5.41 (range: 0.80 to 20.70)
- Black average: 4.56 (range: 0.80 to 11.20)
- Average advantage: 0.85 (White maintained more threats)



11.4 Annotated Game

1. e4 e5 2. Nf3 Nf6 3. Nxe5 d6 4. Nf3 Nxe4 5. Nc3 Nxc3 6. dxc3 Be7 7. Be3! O-O 8. Qd2! Nd7! 9. O-O-O Nf6! 10. Bd3 c5 11. Rhe1 Be6! 12. Kb1 Qa5! 13. c4! Qxd2! 14. Bxd2 h6 15. Nh4 Rfe8 16. Ng6! Ng4! 17. Nxe7+! Rxe7 18. Re2! Ne5! 19. Bf4 Nxd3! 20. Rxd3! Rd7! 21. Rxd6! Rxd6! 22. Bxd6! Rd8 23. Rd2 Bxc4 24. Kc1 b6 25. Bf4! Rxd2 26. Kxd2 a6 27. a3 Kf8 28. Bc7! b5! 29. Bd6+ Ke8! 30. Bxc5! h5! 31. Ke3! Kd7! 32. Kd4 g6 33. g3 Be2! 34. Bf8 Kc6! 35. b3 Bd1 36. Kd3 Bg4 37. c4! Be6 38. Kd4 bxc4 39. bxc4! Bg4 40. c5 Be6 41. Bh6 Bd5 42. Be3 Be6 43. Ke5 Bd5 44. Kf4 Be6 45. Kg5 Bd5! 46. g4 hxg4 47. Kxg4! Ba2 48. Kg5 Bb3 49. Kf6 Ba2 50. h4 Bb3 51. f4! Ba2! 52. Ke7 Bb3 53. Kf6! Ba2 54. f5 Bb1! 55. Bf2 Bc2!

1/2-1/2

11.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

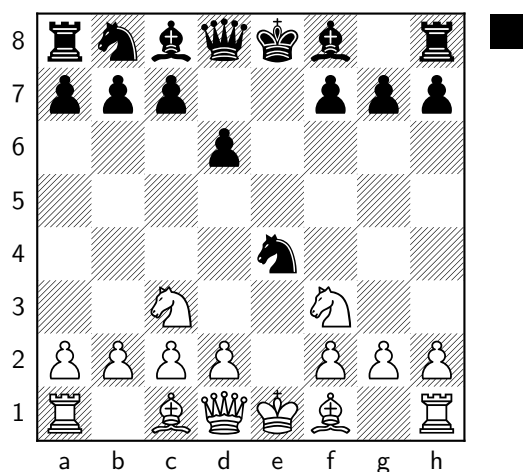
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 5. Nc3 ★

Biggest evaluation swing(s) (26cp)

Evaluation: +0.15



Metric	White	Black
Space	0.40	0.60
Mobility (MG)	4.60	4.30
King Safety	0.20	0.50
Threats	6.80	3.20

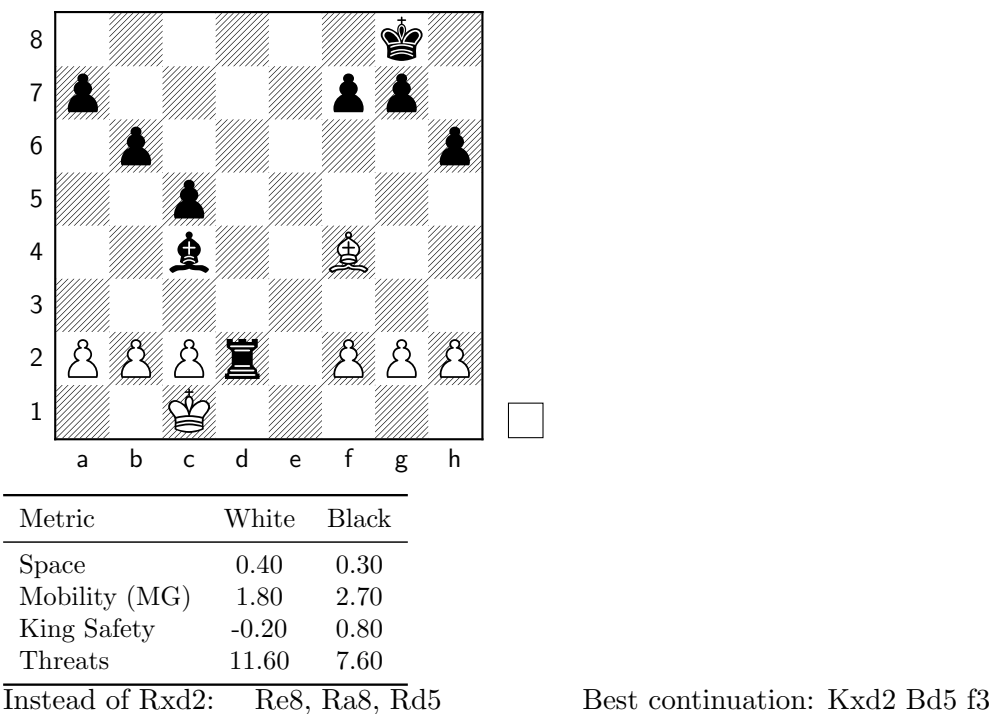
Instead of Nc3: d4, Bd3

Best continuation: Nxc3 dxc3 Be7

Position 2: After 25...Rxd2 ★

Biggest evaluation swing(s) (21cp)

Evaluation: +0.18

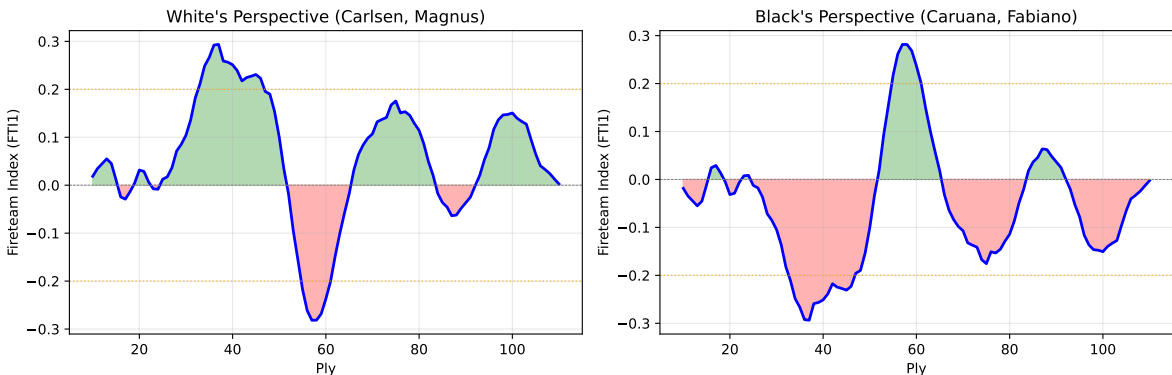


11.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

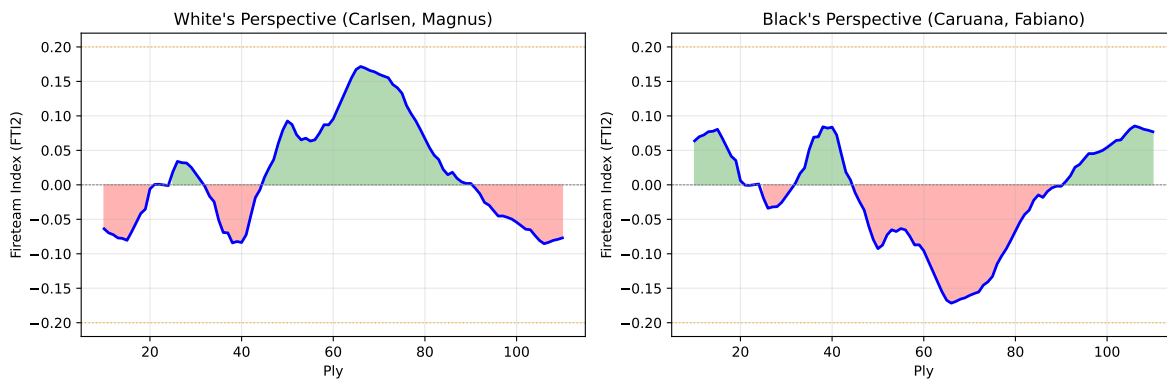
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	11 plies
Carlsen, Magnus	Windowed	WIN	14 plies
Caruana, Fabiano	Per-ply	DRAW	8 plies
Caruana, Fabiano	Windowed	DRAW	7 plies

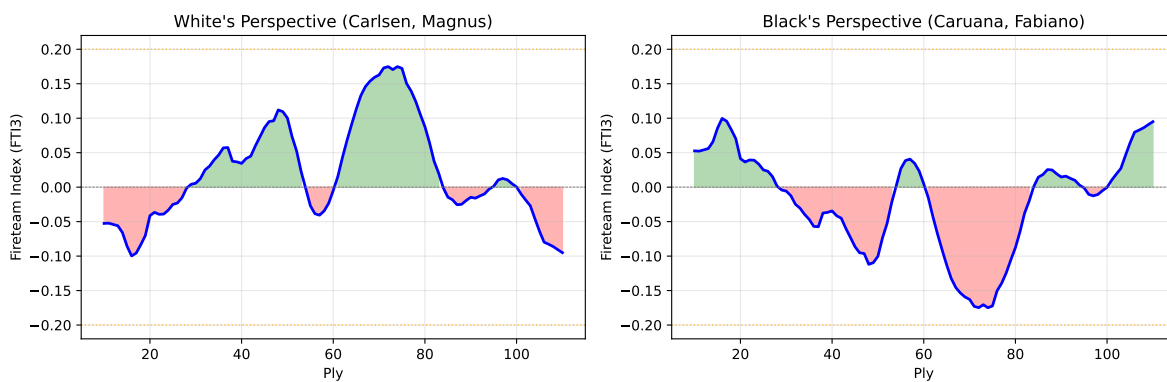


FTI2: Tactical (0.6, 0.1, 0, 0.3)

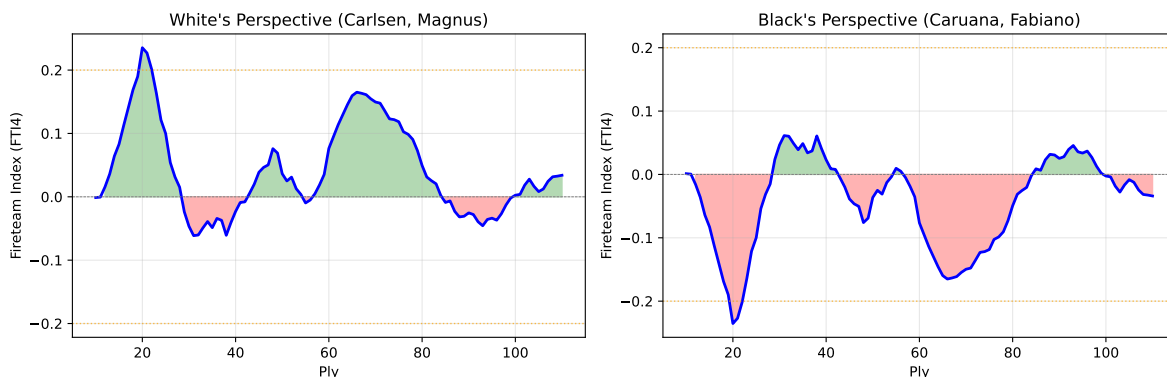
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI3: Strategic** (0.7, 0.1, 0.1, 0.1)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	0 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

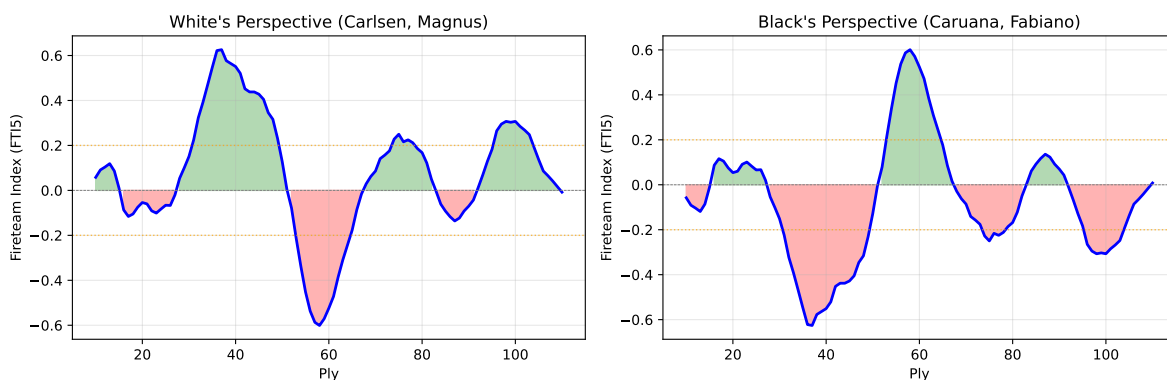
**FTI4: Dynamic** (0.2, 0.6, 0, 0.2)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	2 plies
Carlsen, Magnus	Windowed	DRAW	3 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



FTI5: Solid (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	WIN	11 plies
Carlsen, Magnus	Windowed	WIN	19 plies
Caruana, Fabiano	Per-ply	WIN	10 plies
Caruana, Fabiano	Windowed	WIN	12 plies



Interpretation. White (Carlsen, Magnus) achieved sustained positional dominance, with FTI1 predicting a win. However, the game ended in a draw, suggesting Black successfully defended or White failed to convert the advantage.

11.7 Analysis Information

- Engine: Stockfish 17.1

- Depth: 23
- Analysis time: 348.6 seconds

Chapter 12

Carlsen, Magnus vs Caruana, Fabiano

12.1 Game Information

Event	World Chess Championship 2018
Site	London
Date	2018.11.26
Round	12
White	Carlsen, Magnus (2847)
Black	Caruana, Fabiano (2823)
Result	1/2-1/2
Opening	B33 – Sicilian Defense, Sveshnikov Variation

12.2 Player Statistics

12.2.1 Carlsen, Magnus (White)

- Total moves: 31
- Accuracy: 92.9%
- Average centipawn loss: 14.7
- Best/Excellent moves: 11 / 9
- Good moves: 8
- Inaccuracies: 0
- Mistakes: 2
- Blunders: 1

12.2.2 Caruana, Fabiano (Black)

- Total moves: 31
- Accuracy: 95.0%
- Average centipawn loss: 10.4

- Best/Excellent moves: 14 / 6
- Good moves: 8
- Inaccuracies: 1
- Mistakes: 2
- Blunders: 0

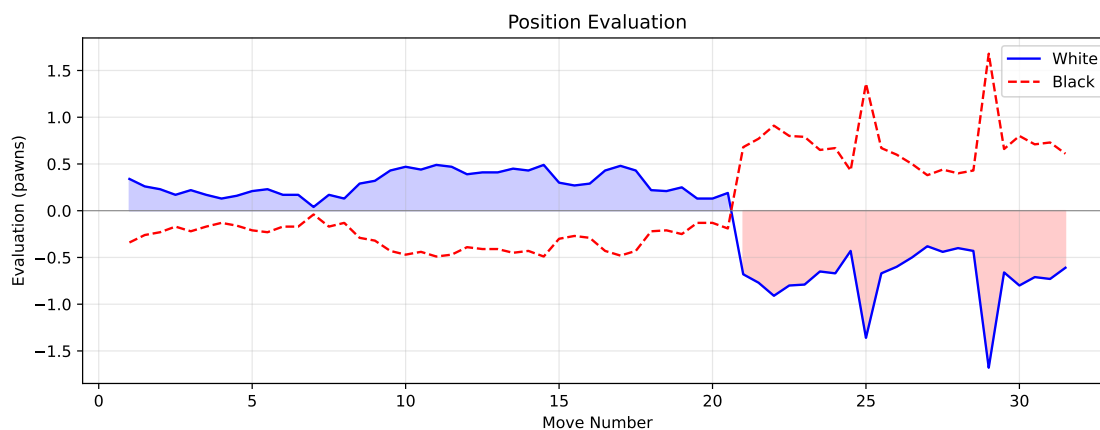
12.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

12.3.1 Position Evaluation Over Time

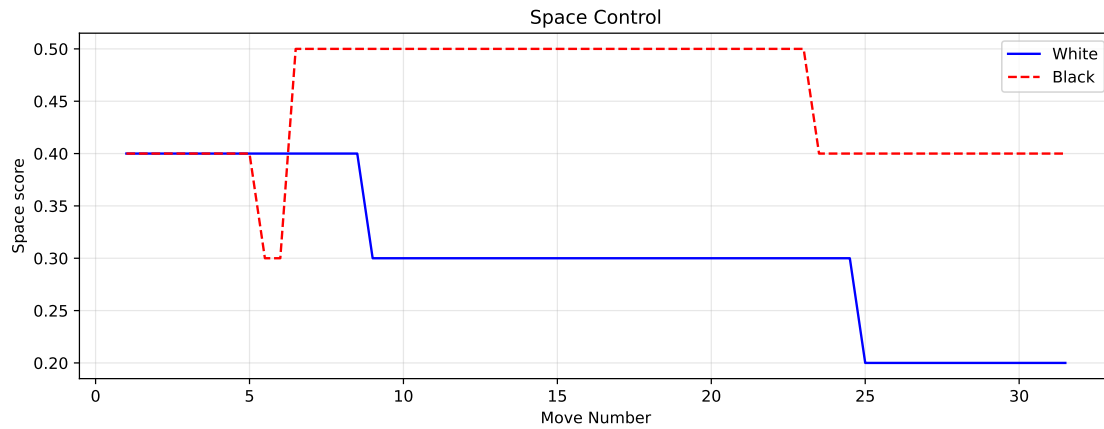
Game Character Classification. Evaluation range: minimum $m_1 = -1.68$, maximum $m_2 = +0.49$, spread $d = m_2 - m_1 = 2.17$.

This game is classified as **tense** (one-sided). A tense, one-sided game. Black maintained the advantage throughout.



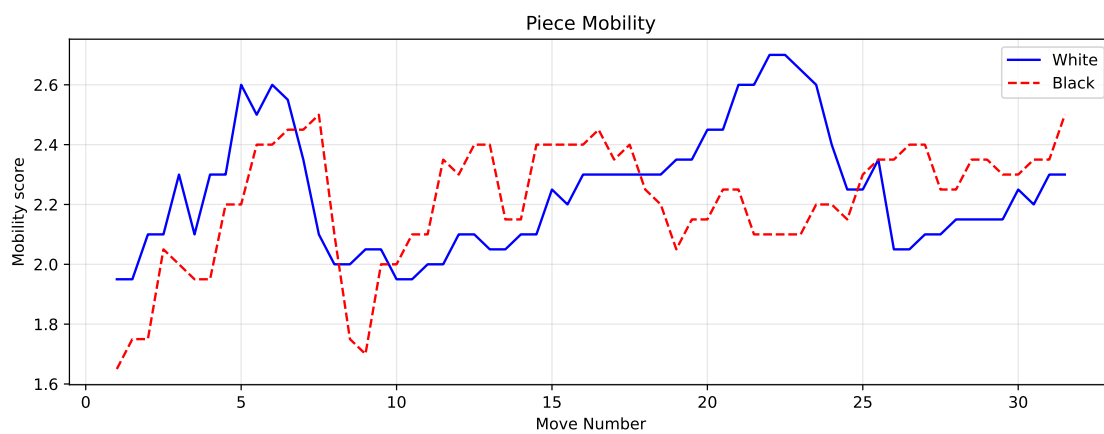
12.3.2 Space Control

- White average: 0.30 (range: 0.20 to 0.40)
- Black average: 0.45 (range: 0.30 to 0.50)



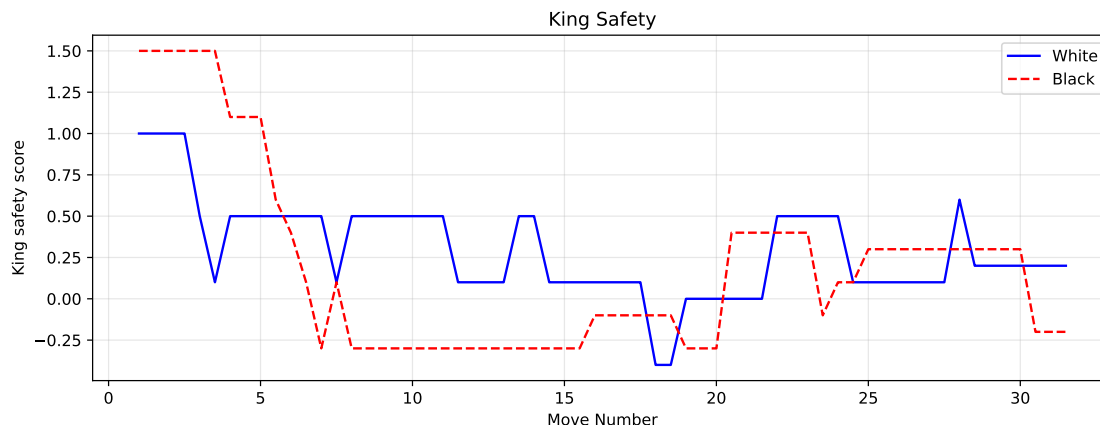
12.3.3 Mobility (Piece Activity)

- White average: 2.25 (range: 1.95 to 2.70)
- Black average: 2.21 (range: 1.65 to 2.50)



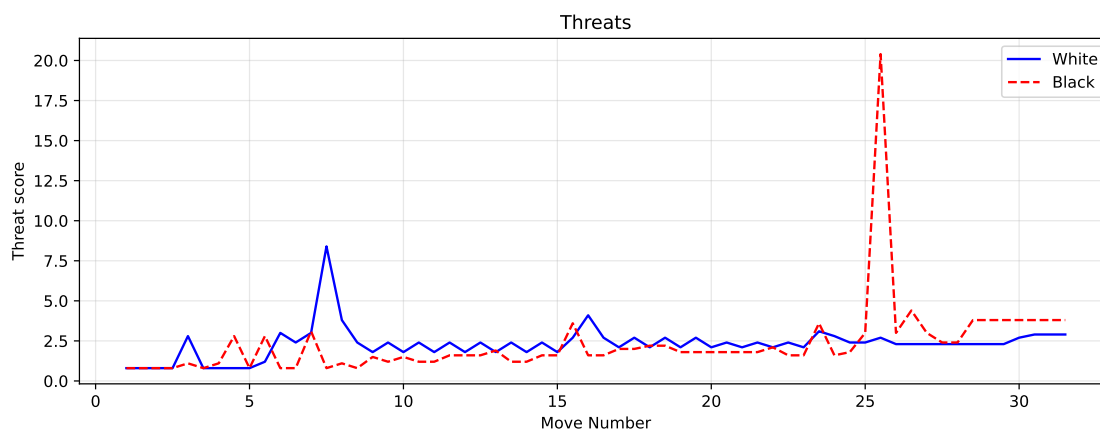
12.3.4 King Safety

- White average: 0.29
- Black average: 0.20



12.3.5 Threats

- White average: 2.31 (range: 0.80 to 8.40)
- Black average: 2.30 (range: 0.80 to 20.40)
- Average advantage: 0.01 (threats were balanced)



12.4 Annotated Game

1. e4 c5 2. Nf3 Nc6 3. d4 cxd4 4. Nxd4 Nf6 5. Nc3 e5 6. Ndb5 d6 7. Nd5 Nxd5 8. exd5 Ne7 9. c4 Ng6 10. Qa4 Bd7! 11. Qb4! Bf5! 12. h4 h5! 13. Qa4! Bd7! 14. Qb4! Bf5! 15. Be3 a6 16. Nc3! Qc7 17. g3! Be7 18. f3 Nf8! 19. Ne4 Nd7! 20. Bd3 O-O 21. Rh2? *White loses 91cp. Consider instead: O-O, Qd2, Rc1.*

21...Rac8! 22. O-O-O Bg6! 23. Rc2 f5! 24. Nf2! Nc5?! 25. f4? *White loses 94cp. Consider instead: Bxc5.*

25...a5? *Black loses 71cp. Consider instead: exf4, b5.*

26. Qd2! e4 27. Be2! Be8! 28. Kb1 Bf6 29. Re1?? *White loses 134cp. Consider instead: b3, Nh3, Rdc1.*

29...a4? *Black loses 102cp. Consider instead: Ba4.*

30. Qb4 g6 31. Rd1 Ra8
1/2-1/2

12.5 Critical Positions

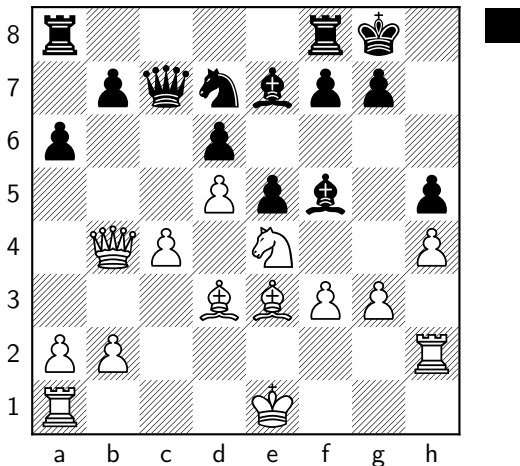
This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 21. Rh2 ★

Biggest evaluation swing(s) (87cp)
Evaluation: -0.68



Metric	White	Black
Space	0.30	0.50
Mobility (MG)	5.20	4.50
King Safety	0.00	0.40
Threats	2.10	1.80

Instead of Rh2:

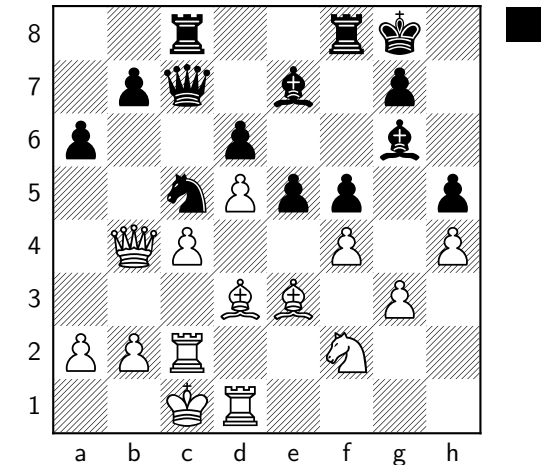
O-O, Qd2, Rc1

Best continuation:

Bg6 g4 b5

Position 2: After 25. f4 ★

Biggest evaluation swing(s) (93cp)
Evaluation: -1.36



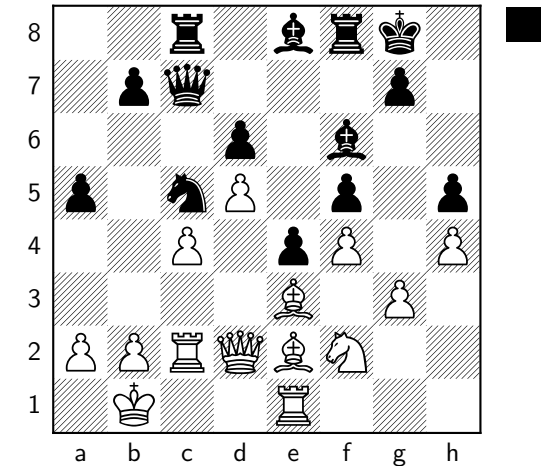
Metric	White	Black
Space	0.20	0.40
Mobility (MG)	4.50	4.60
King Safety	0.10	0.30
Threats	2.40	3.00

Instead of f4: Bxc5 Best continuation: exf4 gxf4 Bxh4

Position 3: After 29. Re1

White blunders (-125cp)

Evaluation: -1.68



Metric	White	Black
Space	0.20	0.40
Mobility (MG)	4.30	4.70
King Safety	0.20	0.30
Threats	2.30	3.80

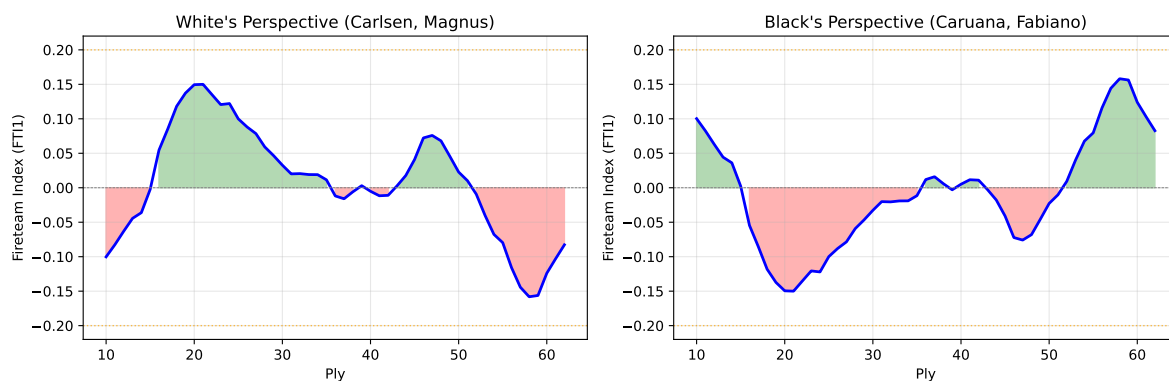
Instead of Re1: b3, Nh3, Rdc1 Best continuation: Ba4 Rcc1 b5

12.6 Fireteam Index Prediction

This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.

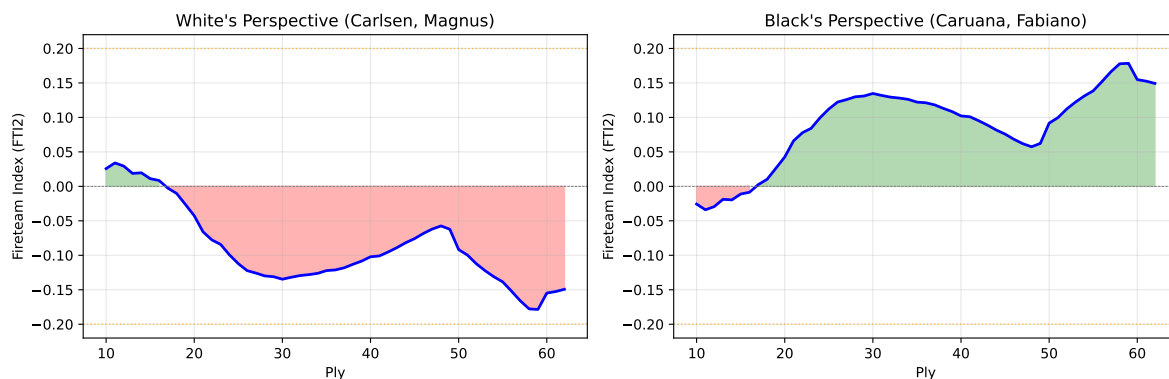
FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	1 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



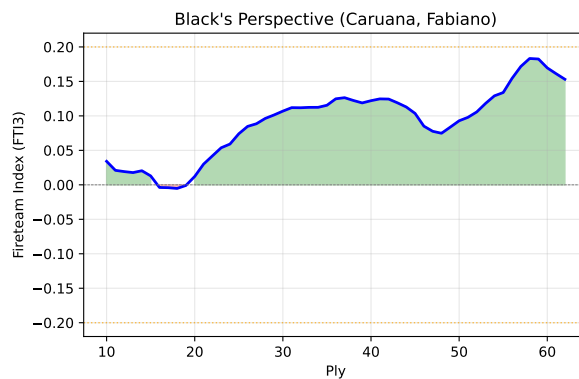
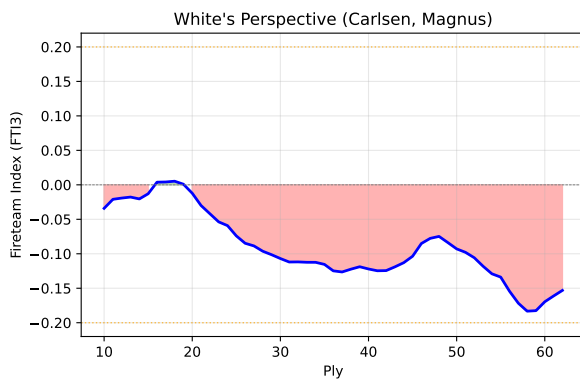
FTI2: Tactical (0.6, 0.1, 0, 0.3)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

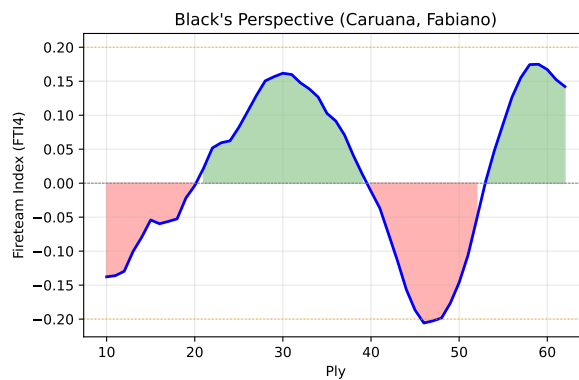
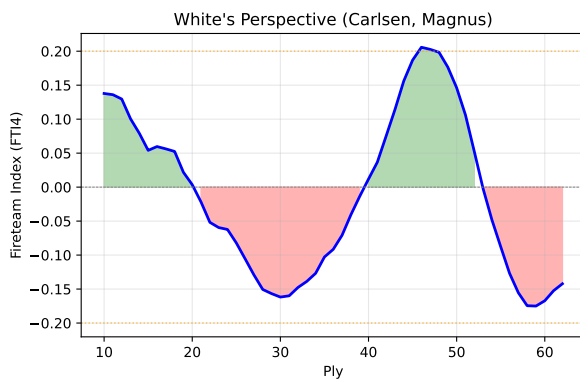


FTI3: Strategic (0.7, 0.1, 0.1, 0.1)

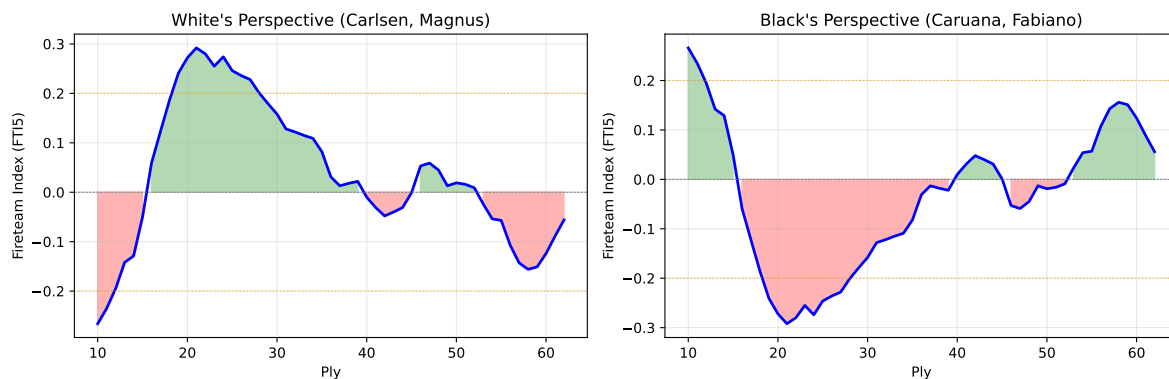
Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	0 plies
Carlsen, Magnus	Windowed	DRAW	0 plies
Caruana, Fabiano	Per-ply	DRAW	1 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI4: Dynamic** (0.2, 0.6, 0, 0.2)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	5 plies
Carlsen, Magnus	Windowed	DRAW	2 plies
Caruana, Fabiano	Per-ply	DRAW	4 plies
Caruana, Fabiano	Windowed	DRAW	0 plies

**FTI5: Solid** (0.3, 0.2, 0.5, 0)

Player	Algorithm	Prediction	Max Streak
Carlsen, Magnus	Per-ply	DRAW	5 plies
Carlsen, Magnus	Windowed	WIN	10 plies
Caruana, Fabiano	Per-ply	DRAW	3 plies
Caruana, Fabiano	Windowed	DRAW	0 plies



Interpretation. Neither player achieved sustained positional dominance according to FTI1. Both algorithms predicted DRAW for both sides, which matches the actual result. This suggests a balanced game where neither side established lasting control.

12.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 554.3 seconds

Chapter 13

Fireteam Index Prediction Summary

This chapter summarizes the accuracy of the Fireteam Index prediction algorithms across all games in this collection.

13.1 Overall Results

Metric	Value
Total games	12
Decisive games	0
Draws (both sides analyzed)	12
Games with prediction	12

13.1.1 FTI1: Harmonious ($\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$)

Algorithm	Correct	Accuracy
Per-ply (raw)	5/12	41.7%
Windowed (smoothed)	5/12	41.7%
Balance	0/12	0.0%

13.1.2 FTI2: Tactical (0.6, 0.1, 0, 0.3)

Algorithm	Correct	Accuracy
Per-ply (raw)	5/12	41.7%
Windowed (smoothed)	5/12	41.7%
Balance	0/12	0.0%

13.1.3 FTI3: Strategic (0.7, 0.1, 0.1, 0.1)

Algorithm	Correct	Accuracy
Per-ply (raw)	4/12	33.3%
Windowed (smoothed)	4/12	33.3%
Balance	0/12	0.0%

13.1.4 FTI4: Dynamic (0.2, 0.6, 0, 0.2)

Algorithm	Correct	Accuracy
Per-ply (raw)	4/12	33.3%
Windowed (smoothed)	4/12	33.3%
Balance	0/12	0.0%

13.1.5 FTI5: Solid (0.3, 0.2, 0.5, 0)

Algorithm	Correct	Accuracy
Per-ply (raw)	1/12	8.3%
Windowed (smoothed)	1/12	8.3%
Balance	0/12	0.0%

13.1.6 Comparison

FTI1 (Harmonious) (41.7%) and **FTI2 (Tactical)** (41.7%) show similar predictive accuracy (within 2%).

13.1.7 Empirically Optimal Weights

Using linear regression on the per-ply positional data from all games, the following weights best predict game outcomes (win=1, draw=0.5, loss=0):

$$\text{FTI}_{opt} = 0.00\Delta S + 0.00\Delta M + 0.00\Delta K + 1.00\Delta T' \quad (13.1)$$

Weights: $w_S = 0.001$, $w_M = 0.001$, $w_K = 0.001$, $w_T = 0.998$ ($R^2 = 0.000$)

Prediction Accuracy with Optimal Weights

Algorithm	Correct	Accuracy
Per-ply	12/12	100.0%
Windowed	11/12	91.7%
Balance	12/12	100.0%

For comparison, the predefined signatures are:

- FTI1 (Harmonious): (0.25, 0.25, 0.25, 0.25)
- FTI2 (Tactical): (0.60, 0.10, 0.00, 0.30)
- FTI3 (Strategic): (0.70, 0.10, 0.10, 0.10)
- FTI4 (Dynamic): (0.20, 0.60, 0.00, 0.20)
- FTI5 (Solid): (0.30, 0.20, 0.50, 0.00)

13.2 Interpretation

These predictions tracked the eventual winner in each decisive game. Three prediction algorithms are compared:

- **Per-ply:** Predicts WIN if $FTI > \epsilon$ for 10 consecutive plies (raw values)
- **Windowed:** Same logic but applied to a 10-ply rolling average
- **Balance:** Uses normalized balance B_n (plies ahead – plies behind) and volatility V (sign changes). Predicts DRAW if $|B_n| < 0.2$ or $V \geq 10$

The Balance algorithm explicitly models draws by detecting games where neither side maintains a consistent advantage, or where the advantage changes hands frequently.

The different FTI signatures may perform better or worse depending on the playing style of the players being analyzed. FTI2 (Tactical) tends to predict well for players who emphasize territorial control and tactical threats, while FTI3 (Strategic) may better capture the style of players who prioritize space and piece activity.

Appendix A

Methodology: How Metrics Are Computed

This appendix explains how the positional metrics are computed.

This section explains how Stockfish’s classical evaluation computes each positional factor. Note that modern Stockfish primarily uses NNUE (neural network) evaluation, but the classical evaluation terms provide interpretable metrics for educational purposes.

A.1 Space

Space evaluation counts squares in the center (files c-f, ranks 2-4 for White, 5-7 for Black) that are:

1. Attacked by at least one friendly pawn, AND
2. Not occupied by any enemy piece

The computation is weighted by the number of pieces behind the pawn chain—this rewards positions where you control territory AND have pieces to exploit that space. The space term is scaled by total piece count; it matters more in closed positions with many pieces.

A.2 Mobility

Mobility counts legal moves for each piece type, with different weights:

- **Knights:** Safe squares attacked (excluding squares defended by enemy pawns)
- **Bishops:** Safe squares on diagonals (bonus for long, unobstructed diagonals)
- **Rooks:** Squares on files/ranks (bonus for open files, 7th rank penetration)
- **Queens:** Combination of bishop and rook mobility patterns

“Safe squares” exclude those defended by enemy pawns or where exchanges would lose material. Mobility in the opponent’s territory is valued higher than mobility in your own.

A.3 King Safety

King safety combines several factors:

- **Pawn shield:** Pawns on 2nd/3rd rank directly in front of the castled king
- **King tropism:** Distance of enemy pieces to your king (closer = more dangerous)
- **Attack units:** Count of enemy pieces attacking squares adjacent to your king
- **Safe checks:** Number of safe checking squares available to the opponent

A.4 Threats

Threat evaluation considers:

- Hanging pieces (attacked but not adequately defended)
- Safe pawn advances that create threats
- Minor pieces attacking major pieces
- Weak squares in the pawn structure

A.5 Middlegame vs Endgame (MG/EG)

Each term has separate middlegame (MG) and endgame (EG) values. Stockfish blends these based on remaining material (the “phase”). With all pieces on the board, MG dominates; as pieces trade, EG values become more important. For example, space matters greatly in the middlegame but is nearly irrelevant in endgames.

A.6 Game Character Classification

Games are classified based on evaluation volatility and directionality. Let m_2 be the maximum evaluation for White during the game (in pawns) and m_1 be the minimum. The *spread* $d = m_2 - m_1$ measures the total evaluation range.

Spread Classifications. Based on the magnitude of d :

- **Balanced** ($d < 1$): Minimal advantage shifts; controlled, technical play with neither side gaining a significant edge.
- **Tense** ($1 \leq d < 3$): Normal competitive tension; typical of well-played games where small advantages are contested.
- **Tactical** ($3 \leq d < 6$): Significant evaluation swings; tactical complications, missed opportunities, or speculative play.
- **Chaotic** ($d \geq 6$): Wild swings in evaluation; likely includes major blunders, speculative sacrifices, or complex tactical melees.

Directionality Classifications. Based on whether the advantage changed hands:

- **One-sided:** Either $m_1 > -\frac{1}{2}$ (White always comfortable) or $m_2 < \frac{1}{2}$ (Black always comfortable). The leading side maintained control throughout.
- **Seesaw:** Both $m_1 < -1$ and $m_2 > 1$. The advantage genuinely changed hands, with each side holding a significant advantage at some point.
- **Normal:** Neither one-sided nor seesaw; moderate swings without complete reversals.

These classifications combine to describe the overall game character. For example, a “tactical seesaw battle” indicates significant evaluation swings ($3 \leq d < 6$) with the advantage changing hands multiple times.

A.7 Fireteam Index (FTI) and Win Prediction

The Fireteam Index is a composite metric inspired by the ancient mathematical board game *Rithmo-machia*, where a “progression victory” is achieved when a coordinated group of pieces (a “fireteam”) establishes control deep in enemy territory. Similarly, in chess, sustained positional dominance across multiple factors often precedes victory.

The FTI weights can be thought of as *signatures* that capture different playing styles. By adjusting which positional factors receive emphasis, the FTI can be tuned to better predict outcomes for players with different strategic preferences.

Threats Scaling. The raw Threats metric (T) produces values typically ranging from 5–20, while Space, Mobility, and King Safety range from approximately 0.1–0.5. To make these comparable, Threats is divided by a scaling factor of 20 before weighting. The scaled value $T' = T/20$ brings Threats into a similar range as the other metrics.

FTI1: Harmonious Formula. The harmonious Fireteam Index weights all four positional metrics equally (with Threats pre-scaled):

$$\text{FTI}_1 = \frac{1}{4}\Delta S + \frac{1}{4}\Delta M + \frac{1}{4}\Delta K + \frac{1}{4}\Delta T' \quad (\text{A.1})$$

where ΔS = Space, ΔM = Mobility, ΔK = King Safety, and $\Delta T' = T/20$ = scaled Threats, each computed as (Player’s value – Opponent’s value). Positive FTI indicates the player has overall positional superiority; negative indicates the opponent is better.

FTI2: Tactical. This variant emphasizes territorial control and tactical pressure through Space and Threats:

$$\text{FTI}_2 = 0.6\Delta S + 0.1\Delta M + 0.3\Delta T' \quad (\text{A.2})$$

FTI3: Strategic. This variant emphasizes long-term spatial dominance:

$$\text{FTI}_3 = 0.7\Delta S + 0.1\Delta M + 0.1\Delta K + 0.1\Delta T' \quad (\text{A.3})$$

FTI4: Dynamic. This variant emphasizes piece activity and maneuvering:

$$\text{FTI}_4 = 0.2\Delta S + 0.6\Delta M + 0.2\Delta T' \quad (\text{A.4})$$

FTI5: Solid. This variant emphasizes king safety and defensive play:

$$\text{FTI}_5 = 0.3\Delta S + 0.2\Delta M + 0.5\Delta K \quad (\text{A.5})$$

Prediction Algorithms. Three algorithms are used to predict game outcomes based on FTI (applied independently to each of the five variants):

1. **Per-Ply (Raw):** Examines the raw FTI value at each ply after the opening phase (ply 16, approximately 8 moves). If $\text{FTI} > \epsilon$ (default $\epsilon = 0.25$) for 10 consecutive plies (5 full moves), a WIN is predicted for that player.
2. **Windowed (Smoothed):** Computes a 10-ply rolling average of FTI before applying the same streak detection. This smooths out tactical noise and captures sustained trends rather than momentary fluctuations.
3. **Balance (Confidence-based):** Predicts outcomes using a confidence score that combines balance and volatility:

- *Normalized Balance* $B_n = \frac{(\text{plies with FTI} > \epsilon) - (\text{plies with FTI} < -\epsilon)}{\text{total plies}}$
- *Volatility* $V = \text{number of times FTI crosses zero (sign changes)}$
- *Volatility threshold* $V_{\max} = 0.2 \times \text{total plies}$ (scales with game length)
- *Confidence* $C = |B_n| \times \max(0, 1 - V/V_{\max})$

The prediction is: WIN if $C > 0.15$ and $B_n > 0$; LOSS if $C > 0.15$ and $B_n < 0$; DRAW otherwise. Low confidence (due to balanced play or high volatility) predicts a draw.

The Per-Ply and Windowed algorithms predict DRAW only when no streak is achieved, whereas the Balance algorithm explicitly detects drawish characteristics (balanced plies, back-and-forth play).

Interpretation. The FTI prediction measures *positional* dominance, which is a necessary but not sufficient condition for winning. A player may dominate positionally yet fail to convert due to:

- Tactical errors (blunders, missed combinations)
- Time pressure in rapid/blitz games
- Tenacious defense by the opponent
- Drawing tendencies in certain structures (opposite-colored bishops, fortresses)

Conversely, a player may win without ever achieving sustained FTI dominance through a single tactical strike or opponent error. The FTI metric is most predictive in games where positional understanding—rather than pure calculation—determines the outcome.

Choosing a Signature. Different FTI signatures may perform better for different players or playing styles. FTI2 (Tactical) tends to predict well for players who build up territorial control and convert through direct threats. FTI3 (Strategic) may better capture the style of players who emphasize piece activity and spatial domination. Comparing prediction accuracy across signatures can reveal which positional factors most strongly correlate with a player’s wins.

For Draws. When analyzing drawn games, the FTI is computed from both players’ perspectives. This reveals whether:

- Neither side achieved dominance (a “true draw” with balanced play)
- One side dominated but failed to convert (a “missed win”)
- Both sides dominated at different points (a “mutual chances draw”)